CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

DESIGN NOTE

RCC VOIDED SLAB (SPAN C/C EXP. JOINT = 25m)

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1.0 Design Report

The design note presents the detailed design calculations for RCC voided Slab 1x25.0m c/c exp. joint.

1.1 Introduction:-

Design is presented consistently in SI units; the following apply unless mentioned specifically otherwise:

Length	m
Force	kN
Stress	MPa
Hog Mom/Com Str	-ve
Sag Mom/Ten Str	+ve

1.2 Reference documents :-

1 IRC codes /guidelines/special publications
2 MORTH specification
3 Specialised literature as relevant

1.3 Assumptions:-

The following assumptions have been taken while designing the RCC Slab.

1 RCC Slab is designed for per metre width.
2 On top slab 75mm thick wearing coat is assumed for SIDL.
3 Deck width taken-8.4 m
4 Carriageway width- 7.5 m
5 Shear value is taken at dist. At effective depth from the face of the slab.
6 In case of load dispersion wearing coat thickness, fill thickness and top slab thickness is considered wherever applicable.
7

In case of design sheet under summary of moments, only magnitude of force has been considered.

1.4 Loads:-

The different types of loads used as per IRC 6 : 2017 are.

1 Dead load.
2 In SIDL fill, crash barrier, and wearing coat load is considered.
3 Live load - Class A & 70R Wheel load in case of top slab.
5 1.25 of Impact factor is considered.
6 Temperature loading for uniform rise and temperature gradient is considered.

1.5 Load combinations

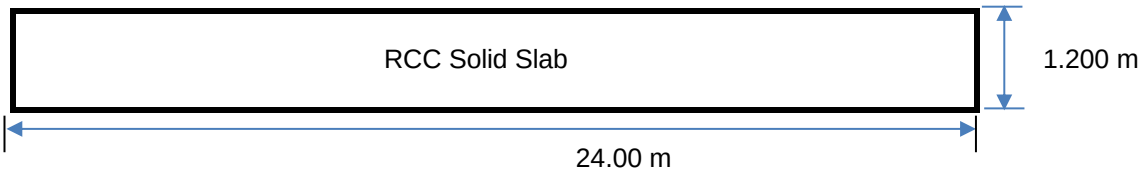
Load combinations as per IRC 6 2017 have been considered in staad load combination.

1.6 Material properties

1 Grade of Concrete **M35**
2 Grade of Steel Fe 500.

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RCC VOIDED SLAB (SPAN C/C EXP. JOINT = 25m)

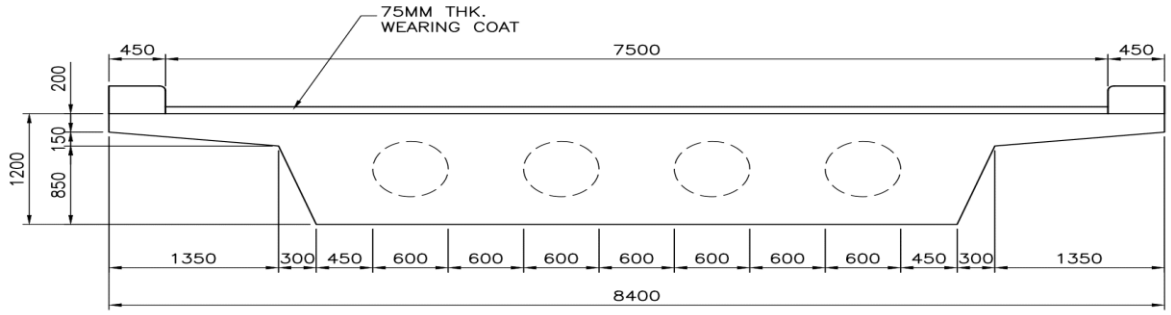


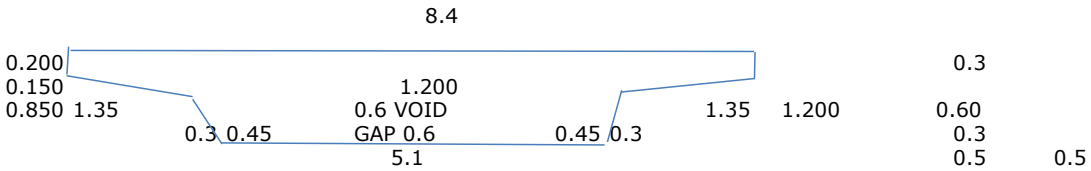
Longitudinal Elevation

2.1 Dimensions

No. of Span	=	1		
Span (c\c of bearing)	=	24.00 m	Span c/c exp. Joint	25
Slab thickness	=	1.200 m		
Total Deck width	=	8.40 m		
Carrriageway Width	=	7.50 m		
Footpath Dimensions	=	0.00 m		
Wearing Coat	=	75mm		
Crash barrier width	=	0.45 m		

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Section of slab

		Nos.	SIZE		AREA	Y	A*Y	$A*Y^2$	I_0
1	LT. CANTILEVER TAPERING	0.50	1.350	0.150	0.101	0.950	0.096	0.091	0.001
2	RT. CANTILEVER TAPERING	0.50	1.350	0.150	0.101	0.950	0.096	0.091	0.001
3	TOP SLAB	1.00	8.400	0.200	1.680	1.100	1.848	2.033	0.006
4	LT. CANTILEVER TAPERING	0.50	0.300	0.850	0.128	0.567	0.072	0.041	0.041
5	RT. CANTILEVER TAPERING	0.50	0.300	0.850	0.128	0.567	0.072	0.041	0.041
6	SOFFIT	1.00	5.100	1.000	5.100	0.500	2.550	1.275	0.425
7	VOIDS	-4.00	3.142	0.6 VOID	-1.131	0.6	-0.679	-0.407	-0.025
				TOTAL	6.1065		4.73	3.57	0.49

Percentage of voides = 1.131 / 6.1065 x 100 = 18.52 % < 40%

So, Normal calculation is adopted As per cl. 4.1.2 of IRC SP 64 : 2016.

3.2 Temperature load calculation

Effective Bridge Temperature

Maximum Air Shade temperature	=	47.5	/^{o}C (as per Fig 8 of IRC:6-2014)
Minimum Air Shade temperature	=	0	/^{o}C (as per Fig 9 of IRC:6-2014)
Mean of max and min temperature	=	23.75	/^{o}C (as per clause 215.2 of IRC:6-2014)
Bridge temperature to be assumed	=	33.75	
TEMPERATURE RISE		**33.75**	
TEMPERATURE FALL		**-33.75**	

Effect of temperature gradient

The box has been checked for temperature differential.

$F = E_c \, a Dt \, A$

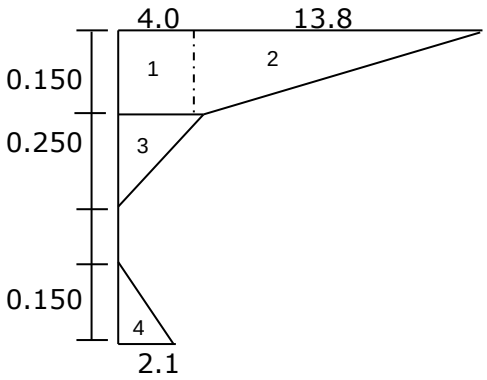
E_c = Modulus of Elasticity of Concrete	=	3.21E+06	t/m^2
a = Coefficient of Thermal expansion	=	1.20E-05	/^{o}C (as per IRC:6)

Dt = Temperature differential

A = X sectional Area of section where temperature differential is Dt

Average thickness of Deck slab = 1200 mm

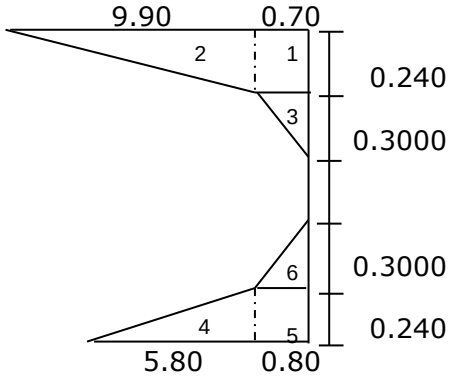
EFFECT OF TEMPERATURE RISE



Sr. No.	Dt	b	t	A = b x t	F (force)	Acting at	Eccentricity e*
1	4.0	1.0	0.150	0.150	23.13	0.075m from top	0.525
2	$\frac{13.8}{2}$	1.0	0.150	0.150	39.91	0.050m from top	0.550
3	$\frac{4.0}{2}$	1.0	0.250	0.250	19.28	0.233m from top	0.367
4	$\frac{2.1}{2}$	1.0	0.150	0.150	6.07	0.050m from bottom	-0.550
					SF = 88.39	M =	37.822

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EFFECT OF TEMPERATURE FALL



Sr. No.	Dt	b	t	A = b x t	F (force)	Acting at	Eccentricity e*
1	0.70	1.0	0.240	0.240	6.48	0.12 m from top	0.480
2	$\frac{9.90}{2}$	1.0	0.240	0.240	45.80	0.080m from top	0.520
3	$\frac{0.70}{2}$	1.0	0.3000	0.3000	4.05	0.340m from top	0.260
4	$\frac{5.80}{2}$	1.0	0.240	0.240	26.83	0.080m from bottom	-0.520
5	0.80	1.0	0.240	0.240	7.40	0.120m from bottom	-0.480
6	$\frac{0.80}{2}$	1.0	0.3000	0.3000	4.63	0.340m from bottom	-0.260
					SF = 95.19	M =	9.270

3.3 Staad input file

```
STAAD SPACE
INPUT WIDTH 79
UNIT METER KNS
JOINT COORDINATES
1 0 0 0
2 0.5 0 0
3 1.7 0 0
4 8 0 0
5 12.5 0 0
6 17 0 0
7 23.3 0 0
8 24.5 0 0
9 25 0 0
************************************
MEMBER INCIDENCES
1 1 2
2 2 3
3 3 4
4 4 5
5 5 6
6 6 7
7 7 8
8 8 9
****************************************
DEFINE MATERIAL START
ISOTROPIC CONCRETE1
E 32000000
POISSON 0.17
DENSITY 25
ALPHA 1.17e-005
DAMP 0.05
END DEFINE MATERIAL
MEMBER PROPERTY INDIAN
1 TO 8 PRIS YD 1.2 ZD 8.4 YB 0.925 ZB 5.4
*****************************************
CONSTANTS
MATERIAL CONCRETE1 ALL
SUPPORTS
2 8 PINNED
*****************************
UNIT METER MTON
DEFINE MOVING LOAD
*70R Wheeled (13.4m)
TYPE 1 LOAD 8 12 12 17 17 17 17
DIST 3.96 1.52 2.13 1.37 3.05 1.37
*Class A (18.8m)
TYPE 2 LOAD 2.7 2.7 11.4 11.4 6.8 6.8 6.8 6.8
DIST 1.1 3.2 1.2 4.3 3 3 3
UNIT METER KNS
***DL***
LOAD 1 DL
*SELFWEIGHT Y -1 ALL
MEMBER LOAD
1 2 7 8 UNI GY -152.663
```

3 TO 6 UNI GY -124.389
SIDL*
LOAD 2 SIDL CB
MEMBER LOAD
1 TO 8 UNI GY -16
**Surfacing (wearing coat)
LOAD 3 Surfacing (wearing coat)
MEMBER LOAD
1 TO 8 UNI GY -14.06
**********Temperature Loading*********
*********Uniform increase****
LOAD 4 Temperature Loading (Rise)
TEMPERATURE LOAD
2 TO 8 Temp 33.75
*********Uniform decrease****
LOAD 5 Temperature Loading (fall)
TEMPERATURE LOAD
2 TO 8 Temp 33.75
**********Temperature gradient
LOAD 6 Temperature gradient (Rise)
JOINT LOAD
2 FX -867.1
8 FX 867.1
2 MZ 371.03
8 MZ -371.03
LOAD 7 Temperature gradient (fall)
JOINT LOAD
2 FX 933.86
8 FX -933.86
2 MZ -90.94
8 MZ 90.94
**SYM. 70R Wheeled
LOAD GENERATION 1
TYPE 1 4.22 0 0 XINC 0.1
****
**1 Class A
LOAD GENERATION 1
TYPE 2 3.41 0 0 XINC 0.1
****

***Partial Safety for Verification of Structural Strength (Basic Combination)**

LOAD COMBINATION 10 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 8 1.725 4 0.9
LOAD COMBINATION 11 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 8 1.725 5 0.9
LOAD COMBINATION 12 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 8 1.725 6 0.9
LOAD COMBINATION 13 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 8 1.725 7 0.9
LOAD COMBINATION 14 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 9 1.725 4 0.9
LOAD COMBINATION 15 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 9 1.725 5 0.9
LOAD COMBINATION 16 BASIC COMBINATION
1 1.35 2 1.35 3 1.75 9 1.725 6 0.9
LOAD COMBINATION 17 BASIC COMBINATION

1 1.35 2 1.35 3 1.75 9 1.725 7 0.9

***Partial Safety for Verification of Serviceability Limit State (Rare Combination)**

LOAD COMBINATION 18 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 4 0.6
LOAD COMBINATION 19 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 5 0.6
LOAD COMBINATION 20 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 6 0.6
LOAD COMBINATION 21 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 7 0.6
LOAD COMBINATION 22 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 4 0.6
LOAD COMBINATION 23 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 5 0.6
LOAD COMBINATION 24 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 6 0.6
LOAD COMBINATION 25 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 7 0.6

***Partial Safety for Verification of Serviceability Limit State (Frequent Combination)**

LOAD COMBINATION 26 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 4 0.6
LOAD COMBINATION 27 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 5 0.6
LOAD COMBINATION 28 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 6 0.6
LOAD COMBINATION 29 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 7 0.6
LOAD COMBINATION 30 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 4 0.6
LOAD COMBINATION 31 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 5 0.6
LOAD COMBINATION 32 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 6 0.6
LOAD COMBINATION 33 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 7 0.6

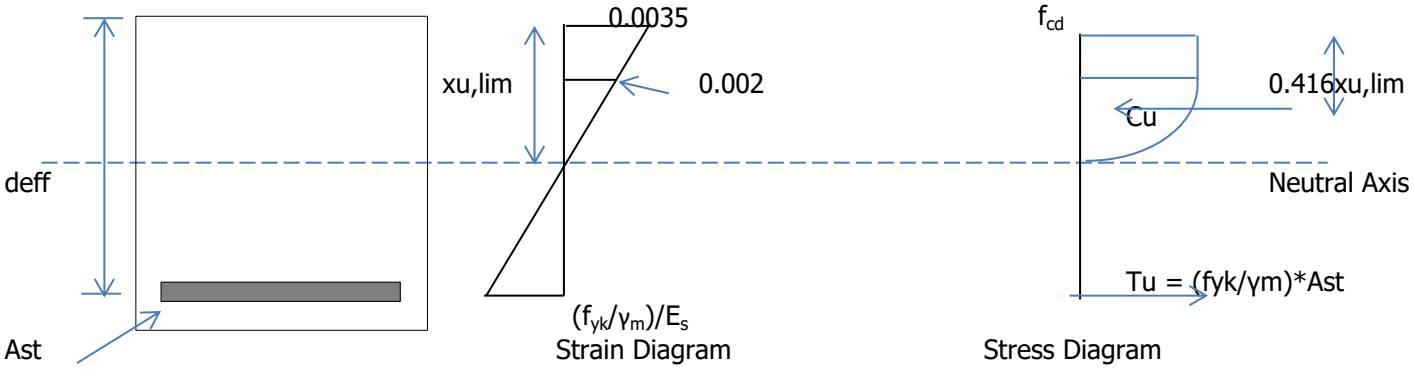
***Partial Safety for Verification of Serviceability Limit State (Quasi-permanent Combination)**

LOAD COMBINATION 34 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 4 0.6
LOAD COMBINATION 35 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 5 0.6
LOAD COMBINATION 36 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 6 0.6
LOAD COMBINATION 37 LL AS LEADING LOAD
1 1 2 1 3 1.2 8 1.15 7 0.6
LOAD COMBINATION 38 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 4 0.6
LOAD COMBINATION 39 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 5 0.6
LOAD COMBINATION 40 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 6 0.6
LOAD COMBINATION 41 LL AS LEADING LOAD
1 1 2 1 3 1.2 9 1.15 7 0.6
PERFORM ANALYSIS PRINT STATICS CHECK
FINISH

3.4 Summary of factored moments

	Solid Slab		
Load Case	Moment in Mid-Span 0.5L	Moment at Span 0.3L	Solid slab shear at deff
	kN-m	kN-m	kN
*Partial Safety for Verification of Structural Strength (Basic Combination) LC 10 TO 17	22890	19490	3159
*Partial Safety for Verification of Serviceability Limit State (Rare Combination) LC 18 TO 25	16304	13891	-
*Partial Safety for Verification of Serviceability Limit State (Frequent Combination) LC 26 TO 33	-	-	-
*Partial Safety for Verification of Serviceability Limit State (Quasi-permanent Combination) LC 34 TO 41	16304	13891	-

5.1.1 Verification of structural strength for RCC Solid slab



ULTIMATE LIMIT STATE

Grade of Concrete	f_{ck}	=	35	N/mm^2	
As per clause 6.4.2.8, IRC:112-2011					
	f_{cd}	=	15.63	N/mm^2	For Basic Combination
	f_{cd}	=	19.54	N/mm^2	For Accidental Combination
	f_{cd}	=	15.63	N/mm^2	For Seismic Combination
E_c		=	32000	MPa	
Grade of steel	f_Y	=	500	N/mm^2	
	f_{Yd}	=	435	N/mm^2	For Basic Combination
	f_{Yd}	=	500	N/mm^2	For Accidental Combination
	f_{Yd}	=	435	N/mm^2	For Seismic Combination

Refer Fig. 6.2 of IRC:112-2011

For steel reinforcement, simplified bilinear diagram is used

Minimum strain in steel reinforcement = 0.87 fy / Es

Es = 2.0E+05 MPa E_c = 32000 MPa

Refer Chaper 5 of Reinfroced Concrete Limit State Design by Ashok K. Jain

Cu = fcd*b*(3/7$xu_{,lim}$ + 2/3*4/7$xu_{,lim}$)

= 17/21*fcd*b*xu

= 0.8095 *fcd*b*xu

cg of compression block from top = 0.416 xu

Tu = fyd*Ast

$R_{lim} = M_{u,Lim}/bd^2$ = 0.8095fcd*(xu $_{lim}$/d)*(1-0.416*xu $_{lim}$/d)

	Basic Comb	Accidental Comb	Seismic Comb
$xu_{,lim}/d$	0.62	0.58	0.62
$R_{lim} = M_{u,Lim}/bd^2$	5.80	6.99	5.80

Here R_{lim} is in MPa

Calculation of Reinforcement

Width of section b	=	5100 mm
Depth of section D	=	1200 mm
Clear cover	=	40

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Moment on the section		**Slab Bottom Mid Span**
	Basic Comb 0.25L	Basic Comb 0.5L
Actual moment (KNm)	19490.0	22890.0
b	5100	5100
D	1200	1200
c	40	40
d	1060.0	1060.0
f_{cd}	15.63	15.63
f_{Yd}	435	435
$xu_{,lim}/d$	0.62	0.62
$R_{sls} = M_{u,sls}/bd^2$	5.80	5.80
$M_{u,Lim}$ (KNm)	33255	33255
	OK	**OK**
Ast Req.	48511	58811
Dia of bar (main tension) (mm)	32	32
Spacing (mm) Total 98 nos.	104	104
+ dia of bar (main tension) (mm)	32	32
Spacing (mm)	208	104
Ast provided (sq mm)	59159	78878
Dia of bar (main compresion) (mm)	16	16
Spacing (mm)	104	104
Area of main compresion (mm^2)	1933	1933
f_{ctm}	2.8	2.8
f_{yk}	435	435
cl. 16.6.1 (2) of IRC :112-2011		
$A_{S.min} = 0.26\ f_{ctm}\ b_t\ d\ /\ f_{yk} >= 0.0013\ b_t\ d$	9052	9052
A_{ct}	4087542	3410056
$f_{ct,eff}$	2.9	2.9
$k_c = 0.4\ \{\ 1 - \sigma_c / (\ k_1\ f_{ct,eff}\ h/h^*)\ \} <= 1$	0.4	0.4
For Bending or bending combined with axial force		
k	0.6500	0.6500
σ_s	435	435
cl. 12.3.6 (4) of IRC :112-2011		
$A_{S.min} = k_c\ k\ f_{ct,eff}\ A_{ct} / \sigma_s$	7089	5914
	OK	**OK**
As.max = 0.025 Ac (main tension)	153000	153000
cl. 16.5.1.1 (2) of IRC :112-2011	**OK**	**OK**
As.max = 0.04 Ac (tension + compresion)	244800	244800
x (mm)	399	531
x/d	0.376	0.501
	OK	**OK**
z (mm)	894	839
MR (KNm)	23000	28772
	OK	**OK**

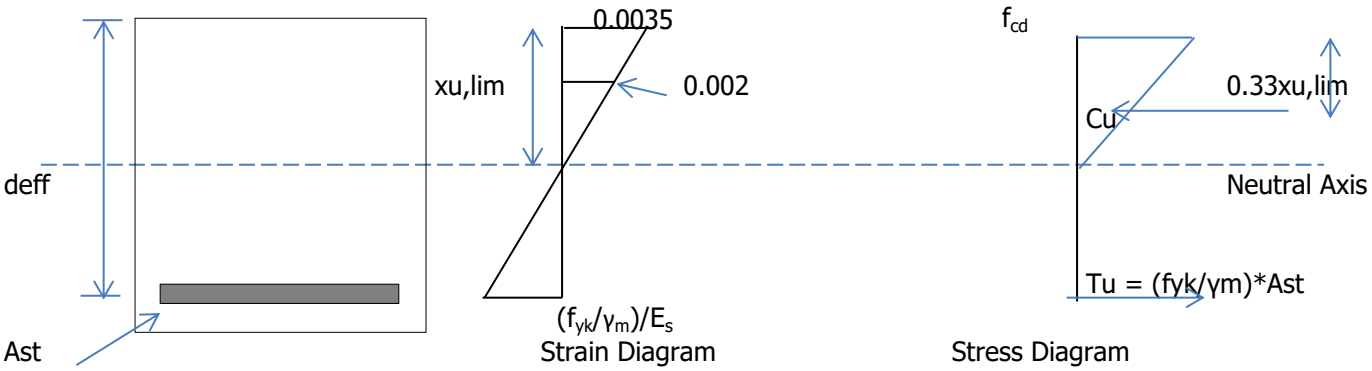
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Shear on the section	**Slab Top End support**		
Actual shear V_{Ed} (KN)	619.4		
Actual shear stress (N/mm2)	0.1		
Max shear capacity, 0.135 fck(1-fck/310)	4.2		
	OK.		
Min shear capacity, 0.0924 fck(1-fck/310)	2.9		
$\Theta = 0.5 \times \sin^{-1}$ (Applied shear stress / 0.135/fck/(1-fck/310))			
Min angle of inclination, Θ (deg)	21.8		
cl. 10.3.2(2) Eq. 10.2 of IRC :112-2010			
K = 1+Sqrt(200/d) <= 2.0	1.434		
cl. 10.3.2(2) Eq. 10.3 of IRC :112-2010			
$\nu_{min} = 0.031\ K^{3/2}\ fck^{1/2}$	0.315		
cl. 10.3.1 of IRC :112-2011			
$\rho 1 = A_{sl}/(b_w\ d) <= 0.02$	0.011		
	OK		
$0.12\ K\ (80\ \rho 1\ f_{ck})^{0.33}$	0.533		
Axial compressive force N_{Ed} (KN)	0		
$\sigma_{cp} = N_{Ed} / A_c <= 0.2\ f_{cd}$	0.0		
cl. 10.3.2(2) Eq. 10.1 of IRC :112-2010			
$V_{Rd.c} = [0.12K(80\rho 1\ f_{ck})^{0.33} + 0.15\sigma_{cp}]b_w d <= (\nu_{min} + 0.15\ \sigma_{cp})\ b_w\ d$ (KN)	1703		
	OK.		
Min shear stress	0.3		
Min shear force for providing reinf., V_E (N)	1532876.2		
No. of link for shear reinf.	**2**		
Dia. of bar for shear reinf.	**12**		
$S = Asw \times 0.9 \times d \times \cot \Theta \times fy / V_E$	153		
S (mm)	100		
A_{SW}	226		
cl. 16.5.2(7) Eq. 16.6 of IRC :112-2011			
$S_{l.max} = 0.75\ d$	795		
Spacing provided in Long. Direction (mm)	**100**		
cl. 16.5.2(9) Eq. 16.8 of IRC :112-2011			
$S_{t.max} = 0.75\ d <= 600mm$	600		
$S_{t.}$ mm	500		
	OK		
z (mm)	894		
f_{ywd}	435		
cl. 10.3.3.3 Eq. 10.17 of IRC :112-2010			
$V_{Ed} <= A_{SW}\ f_{ywd}$ (KN)	98		
cl. 10.3.3.3 (6) of IRC :112-2010			
$M_{Ed} / z + 0.5\ V_{Ed}$ (KN)	21845		
$M_{ed\ max} / z$ (KN)	25721		
	OK		

cl. 10.3.3.2 Eq. 10.7 of IRC :112-2011			
$V_{Rd.s} = A_{SW}\ z\ f_{ywd} / S$ (KN)	879		
	OK		
$\alpha_{cw} =$ ($\sigma_{cp} = N_{Ed} / A_c = 0$)	1.0		
ν_1	0.6		
cl. 10.3.3.2 Eq. 10.8 of IRC :112-2011			
$V_{Rd.max} = \alpha_{cw}\ b_w\ z\ \nu_1\ f_{cd}$ (KN)	42777		
	OK		
cl. 10.3.3.2 Eq. 10.10 of IRC :112-2011			
$A_{SW.max} <= 0.5\ \alpha_{cw}\ \nu_1\ f_{cd}\ b_w\ S / f_{ywd}$	5501		
	OK		

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5.1.2 Verification for serviceability limit state for RCC Solid slab



SERVICEABILITY LIMIT STATE

Grade of Concrete	f_{ck}	=	35	N/mm^2	
As per clause 12.2.1, IRC:112-2011					
	f_{cd}	=	16.80	N/mm^2	For Rare Combination
	f_{cd}	=	16.80	N/mm^2	For Frequent Combination
	f_{cd}	=	12.60	N/mm^2	For Quasi-Perma. Combination
As per clause 12.2.2, IRC:112-2011					
Grade of steel	f_Y	=	500	N/mm^2	
	f_{Yd}	=	300	N/mm^2	For Rare Combination
	f_{Yd}	=	300	N/mm^2	For Frequent Combination
	f_{Yd}	=	300	N/mm^2	For Quasi-Perma. Combination

Refer Fig. 6.2 of IRC:112-2011
For steel reinforcement, simplified bilinear diagram is used
Minimum strain in steel reinforcement = 0.87 fy / Es
Es = 2.0E+05 MPa E_c = 32000 MPa

Cu = 1/2*fcd*b*xu
= 0.5 *fcd*b*xu
cg of compression block from top = 0.33 xu
Tu = fyd*Ast
$R_{sls} = M_{u,sls}/bd^2$ = 0.5fcd*(xu/d)*(1-0.33*xu/d)

Refer Chaper 5 of Reinfroced Concrete Limit State Design by Ashok K. Jain

	Rare Comb	Frequent Comb	Quasi-Perma. Comb
xu,sls/d	0.70	0.70	0.70
$R_{sls} = M_{u,sls}/bd^2$	4.52	4.52	3.39

Here R_{sls} is in MPa

Calculation of Reinforcement

Width of section b	=	5100	mm
Depth of section d	=	1200	mm
Clear cover	=	40	

-	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	RCC VOIDED SLAB (SPAN C/C EXP. JOINT = 25m)	Date & Rev.	09-01-2024

Moment on the section	**Slab Top End support**			**Slab Bottom Mid Span**		
	Rare Comb 0.25L		Quasi-Perma. Comb 0.25L	Rare Comb 0.5L		Quasi-Perma. Comb 0.5L
Actual moment (KNm)	13891.0		13891.0	16304		16304
b	5100		5100	5100		5100
D	1200		1200	1200		1200
c	40		40	40		40
d	1060.0		1060.0	1060.0		1060.0
f_{cd}	16.80		12.60	16.80		12.60
f_{Yd}	300		300	300		300
xu,sls/d	0.70		0.70	0.70		0.70
$R_{sls} = M_{u,sls}/bd^2$	4.52		3.39	4.52		3.39
$M_{u,sls}$ (KNm)	25911		19433	25911		19433
	OK		**OK**	**OK**		**OK**
Ast Req.	48894		51346	58811		62697
Dia of bar (main tension) (mm)	32		32	32		32
Spacing (mm)	104		104	104		104
+ dia of bar (main tension) (mm)	32		32	32		32
Spacing (mm)	208		208	104		104
Ast provided (sq mm)	59159		59159	78878		78878
Dia of bar (main compresion) (mm)	16		16	16		16
Spacing (mm)	104		104	104		104
Area of main compresion (mm^2)	1933		1933	1933		1933
f_{ctm}	2.8		2.8	2.8		2.8
x (mm)	414.3		552.4	552.4		736.5
x/d	0.391		0.521	0.521		0.695
	OK		**OK**	**OK**		**OK**
z (mm)	923		878	878		817
MR_{sls} (KNm)	16386		15577	20770		19332
	OK		**OK**	**OK**		**OK**
$\sigma_{sc} = M/(A_s z)$	254		268	235		253
	OK		**OK**	**OK**		**OK**
$\sigma_{ca} = M/(0.8095 z b x_u)$	14.24		11.24	13.19		10.63
	OK		**OK**	**OK**		**OK**

-	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	RCC VOIDED SLAB (SPAN C/C EXP. JOINT = 25m)	Date & Rev.	09-01-2024

Calculation of crack width	**Slab Top End support**			**Slab Bottom Mid Span**		
n_1			10			10
n_2			5			10
$\phi_{eq} = (n_1\phi_1^2 + n_2\phi_2^2) / (n_1\phi_1+n_2\phi_2)$			32			32
cl. 12.3.4 (3) of IRC :112-2011						
c			40			40
k1			0.8			0.8
k2			0.50			0.50
For skew slab refer eq. 12.10 of IRC :112-2011						
$\rho_{p.eff} = A_s / A_{c,eff}$			0.083			0.110
$S_{r,max} = \{ 3.4\ c + (0.425\ k_1\ k_2\ \phi) /\ \rho_{p.eff} \}$			202			185
cl. 12.3.4 (3) of IRC :112-2011						
k_t			0.5			0.5
$f_{ct,eff}$			2.90			2.90
E_s			200000			200000
E_{cm}			32000			32000
$\alpha_e = E_s / E_{cm}$			6.25			6.25
$(\varepsilon_{sm}-\varepsilon_{cm})=(\sigma_{sc}-k_t f_{ct,eff}(1+\alpha_e \rho_{p,eff})/\rho_{p,eff})/E_s$ $>=0.6\sigma_{sc}/E_s$			0.0012			0.0012
cl. 12.3.4 (2) of IRC :112-2011						
$W_k = S_{r,max} (\varepsilon_{sm} - \varepsilon_{cm})$			0.24			0.21
cl. 12.3.4 (1) of IRC :112-2011						
			OK			**OK**
Calculation of deflection						
Span (mm)					24000	
span/800					30.0	
cl. 12.4.1 (2) of IRC :112-2011						
Short term elastic deflection from STAAD					0.331	
					OK	

6.0 Summary of provided Reinforcement

Top Slab

At top of Mid Span

deff.				=	1136.0 mm	
Area of Steel Required				=	1916 mm^2/m	
Area of Steel Provided				=	1933 mm^2/m	
16mm dia	@	104mmc/c	Top slab (Top main reinforcement)			OK

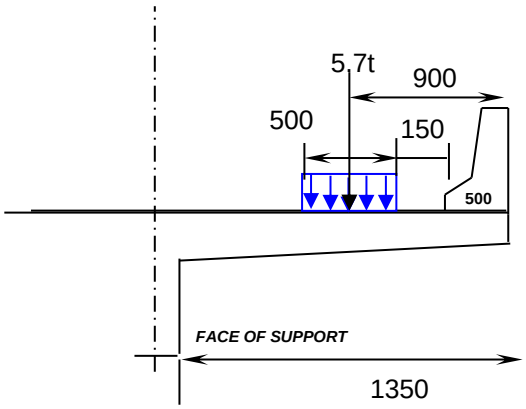
At bottom of Mid Span

deff.				=	1128.0 mm	
Area of Steel Required				=	11532 mm^2/m	
Area of Steel Provided				=	15466 mm^2/m	
98 nos. 32mm dia	@	104mmc/c	Top slab (Bottom main reinforcement)			
32mm dia	@	104mmc/c	Top slab (Bottom extra reinforcement)			OK

Distribution Reinforcement As per cl. 16.6.1.1 (3) of IRC :112-2011

Top Slab

Req. Reinforcement				=	2306 mm^2/m	
Provided Reinforcement				=		
	16mm dia	@	125mmc/c	3217.0	mm^2/m	OK
	16mm dia	@	125mmc/c			
Shear Link 2	12mm dia	@	100mmc/c	Longitudinal dir.		At support
		@	500mmc/c	Transverse dir.		



Calculation of cantilever reinforcement

DATA

Length of cantilever slab beyond end girder	=	1.350	m
Depth of deck slab (Rectangular part)	=	0.200	m
Depth of deck slab (triangular part)	=	0.150	m
C.G of load from end of crash Barrier	=	0.900	m
Depth of wearing course (Asphaltic)	=	0.075	m
Cross sectional area of crash Barrier	=	0.3125	m^2
Width of crash Barrier	=	0.500	m
Density of Concrete	=	2.55	T/m^3
Clear cover for main reinforcement	=	40	mm
Diameter of bars	=	16	mm
Grade of concrete	M 35		
Grade of steel	Fe 500		

Dead Load

Intensity of Dead Load (Rect. Part)	=	0.688	T/m
Intensity of Dead Load (Tri. Part)	=	0.258	T/m
Bending moment due to dead load	=	**0.581**	T-m/m

Super imposed Dead Load

Intensity due to Crash Barrier	=	0.796	T/m
Bending Moment due to SIDL	=	**0.876**	T-m/m

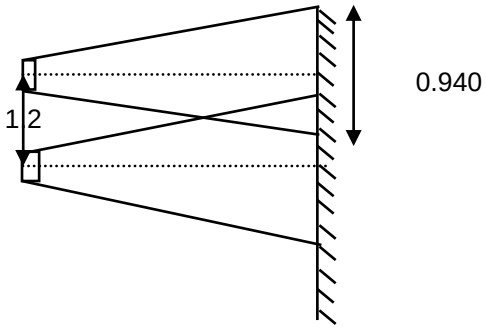
Due to Live Load

Due to the specified minimum clearance , Class AA / 70R loading will not operate on the cantilever slab . Class A loading is to be considered and the loading will be as shown in the figure.

Tyre Width	=	0.250	m
Total live load on cantilever portion	=	5.700	T

Effective Width of dispersion be = 1.2x + bw

x	=	0.450	m
bw	=	0.400	m
be	=	**0.940**	m



Live load per meter width including impact	=	9.096 T/m
Maximum moment due to Live load	=	4.093 T-m/m

Total Max. Bending Moment = **5.550** T-m/m

	Moment	PSF	IF	Moment
DL	0.58	1.1		6.26
Crash barrier	0.88	1.1		9.45
Due to wearing coat	0.17	1.35		2.31
LL	5.55	1.5	1.5	122.50
				140.52

Data			
Grade of concrete	M	35	N/sqmm
Fy	Fe	500	N/sqmm
Depth		350	mm
Width		1000	mm
Clear Cover		40	mm
Diameter of main bar		16	mm
effective depth		302	mm
Moment		141	Nm
fcd		15.633	N/sqmm
b1		0.80952	
b2		0.41597	
x		38.84	mm
xlim	0.6168d	186.274	OK
z		285.842	
As		1130	sqmm

Provide 12 mm dia @ 150 +
16 mm dia @ 150 2094 sqmm

AT top of slab OK

DESIGN OF ELASTOMERIC BEARING :-

DESIGN DATA

Span c/c of Expansion Joint	=	25.00	m	Seismic Zone	II	0.1
Expansion gap	=	0.040	m	Importance factor		1.2
Overall span	=	24.96	m	Sa/g		2.5
Deck cantilever	=	**0.48**	m			
c/c of brg.	=	24.00	m	seismic coeficient		0.15
Nos. of bearing proposed	=	**3**	Nos. / support			
Total width of deck	=	8.40	m			

Rise and Fall of temperature degree +/- 25

Temp. coeff (As per cl. 215.4 of IRC:6-2010) 0.000012

Movement of Superstructure due to Temp.,Shrinkage and Creep 7.2 mm

Reaction due to Perm loads over supports (ULS Factored)			
Brg. Typ.	DL	SIDL	
	tonne	tonne	Seismic load
L1	101	34.47	20.3
L2	88	-14.32	11.0
L3	101	34.47	20.3
R1	101	34.47	20.3
R2	88	-14.32	11.0
R3	101	34.47	20.3

Reaction at Left & Right support due to Live Load without impact (from grillage model of structure)

S.No.	**Bearing number**		**L1**	**L2**	**L3**	**L4**	**L5**	**R1**	**R2**	**R3**	**R4**
	Loads (in Tonnes)	Max	61	59	26	0	0	32	14	14	0
		Min	57	0	-2	0	0	15	1	-1	0

Impact Factor :

Effective Span = 24.00 m

Impact factor = 0.150

Summary of Maximum & Minimum Vertical Loads over Bearings

Support No.	DL (in T)	SIDL (in T)	LL (with impact) (in T)		Total (in T)		
			Max	Min	Max	Min	
L1	101	34	70	65	206	201	
L2	88	-14	68	0	141	73	
L3	101	34	30	-2	165	134	
R1	101	34	36	17	172	153	
R2	88	-14	16	1	89	74	
R3	101	34	16	-1	152	135	
Design Load (kN)	Vertical		Max.		2019		
			Perm.		1330		
			Min.		719		
	Transverse				20	Due to centrifugal	
	Longitudinal				33		
Translation (mm)	Transverse				0		
	Longitudinal				7		
Rotation (rad.)	Longitudinal				0.0001		

Design parameters :-

Design vertical, Vmax	**=**	**2019.11 kN**	Design displacement, e_l(long.)	=	7 mm
Design horizontal load, Hmax (long.)	=	32.5532 kN	Design displacement, e_l(lat.)	=	0 mm
Design horizontal load, Hmax (lat.)	=	20.3349 kN	Bulk modulus, E_b	=	2000 N/mm2
Maximum design strain, εtd (Cl. 5.2.3)	=	7.0	Yield stress of steel, f_y	=	250 N/mm2
Shear modulus, G	=	0.9 N/mm2	Partial safety factor, γ_m	=	1.0
Stress correction factor, Kp	=	1.3	Factor for tensile stresses, K_h	=	1.0
Design rotation, α	=	0.0001	Type loading factor, K_L	=	1.0
Design vertical load, Vmin	**=**	**719.229 kN**			

Bearing parameters :-

Length, a	=	450 mm	No. of elastomer layer, n	=	3.0
Width, b	=	350 mm	No. of steel laminate, n+1	=	4.0
Total height, T	**=**	**52 mm**	Side cover, c	=	5 mm
Thickness of individual elastomer layer, t_i or t_e	=	10 mm	Edge cover	=	5 mm
Thickness of steel laminate, t_s	=	3 mm	Effective area, Ae = (a-2c)x(b-2c)	=	149600 mm2
			Reduced area, Ar	=	143864 mm2
Total elastomer thickness, T_e	=	40 mm	Ae*(1-(v_{xd}/(a-2c))-(v_{yd}/(b-2c)))		
			K_{rd}	=	3.0

Design Check :- (Cl No. 5.2.3.2)

Compressive strain

Shape factor, S=0.5x((a-2c)x(b-2c))/((a-2c)+(b-2c))xti	=	**9.59**				
Compressive strain, $\varepsilon_{c,d}$ =1.5 x F_{zd} / G x A_r x S	=	2.44				

Shear Strain :- (Cl No.5.2.3.5)

Shear displacement, long $V_{x,d}$	=	16.9	{ (H*T_e)/(A_e*G)} + el		
Shear displacement, lat $V_{y,d}$	=	0.0			
Shear displacement ,$V_{xy,d}$	=	16.9			
Strain,εqd=vxyd/Te	=	0.32	< 1.00	**HENCE OK**	

Strain due to angular rotation :- (Cl No. 5.2.3.4)

Strain,εαd=(α*((a-2c)^2+(b-2c)^2)*ti)/(2*Σti^3)	=	0.05			

Total Design strain (Cl No.5.2.3)

Total strain, εtd=K_L(εcd+εqd+εαd)	=	2.82	< 7.00	**HENCE OK**	

Thickness of steel laminate (ts) (Cl.No.5.2.3.5)

Allowable thickness of steel laminate	>	2 mm		
Required thickness, ts=(K_p*V_{max}*(t_1+t_2)*K_h x γ_m)/(f_yxA_r)	=	1.46 mm	**HENCE OK**	
Thickness provided	=	3 mm	**Thickness given > Required thickness**	

Buckling Stability :- (Cl. No.5.2.3.6)

Buckling Stability, 2*(a-2c)*GxS/3*Te	=	63.3 N/mm2	**HENCE OK**	
Max Compressive stress, σ_{max}=(V_{max}/A_r)	=	14.0 N/mm2	**Buckling Stability > σmax**	

Vertical Deflection (Cl. No.5.2.3.7)

Total Vertical defl., ΣV_{zd}=σ_{max}x$t_i$xnx(1/5xGxS_2+1/E_b)	=	1.23 mm		

Rotational deflection

Rotational deflection, ((a-2C)xα)+(b-2C)xα))/K_{rd}	=	0.03 mm	<	1.23	**HENCE OK**	(Cl. No.5.2.3.6)

Friction Check

Min compressive stress, σ_{min}=(V_{min}/A_r)	=	5.00	>	3.00	**HENCE OK**	(Cl. No.5.2.3.6)
Coefficient of friction, μ=(0.1+1.5 x k_f/σ_{min})	=	0.28				
Resultant Horizontal force, Hr	=	32.6 kN				
Friction force,Fr = μxVmin	=	201.4 kN	>	32.55	**HENCE OK**	

STAAD FLOOR
START JOB INFORMATION
ENGINEER DATE 23-May-19
END JOB INFORMATION
*Staad for Bearing design load
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 1.875; 2 0.5 0 1.875; 3 1.7 0 1.875; 4 4.4 0 1.875; 5 7.1 0 1.875;
6 9.8 0 1.875; 7 12.5 0 1.875; 8 15.2 0 1.875; 9 17.9 0 1.875; 10 20.6 0 1.875;
11 23.3 0 1.875; 12 24.5 0 1.875; 13 25 0 1.875; 14 0 0 4.20033;
15 0.5 0 4.20033; 16 1.7 0 4.20033; 17 4.4 0 4.20033; 18 7.1 0 4.20033;
19 9.8 0 4.20033; 20 12.5 0 4.20033; 21 15.2 0 4.20033; 22 17.9 0 4.20033;
23 20.6 0 4.20033; 24 23.3 0 4.20033; 25 24.5 0 4.20033; 26 25 0 4.20033;
40 0 0 6.52538; 41 0.5 0 6.52538; 42 1.7 0 6.52538; 43 4.4 0 6.52538;
44 7.1 0 6.52538; 45 9.8 0 6.52538; 46 12.5 0 6.52538; 47 15.2 0 6.52538;
48 17.9 0 6.52538; 49 20.6 0 6.52538; 50 23.3 0 6.52538; 51 24.5 0 6.52538;
52 25 0 6.52538; 53 0 0 0; 54 0.5 0 0; 55 1.7 0 0; 56 4.4 0 0; 57 7.1 0 0;
58 9.8 0 0; 59 12.5 0 0; 60 15.2 0 0; 61 17.9 0 0; 62 20.6 0 0; 63 23.3 0 0;
64 24.5 0 0; 65 25 0 0; 66 0 0 8.40038; 67 0.5 0 8.40038; 68 1.7 0 8.40038;
69 4.4 0 8.40038; 70 7.1 0 8.40038; 71 9.8 0 8.40038; 72 12.5 0 8.40038;
73 15.2 0 8.40038; 74 17.9 0 8.40038; 75 20.6 0 8.40038; 76 23.3 0 8.40038;
77 24.5 0 8.40038; 78 25 0 8.40038; 79 0 0 7.05038; 80 0.5 0 7.05038;
81 1.7 0 7.05038; 82 4.4 0 7.05038; 83 7.1 0 7.05038; 84 9.8 0 7.05038;
85 12.5 0 7.05038; 86 15.2 0 7.05038; 87 17.9 0 7.05038; 88 20.6 0 7.05038;
89 23.3 0 7.05038; 90 24.5 0 7.05038; 91 25 0 7.05038; 92 0 0 1.35;
93 0.5 0 1.35; 94 1.7 0 1.35; 95 4.4 0 1.35; 96 7.1 0 1.35; 97 9.8 0 1.35;
98 12.5 0 1.35; 99 15.2 0 1.35; 100 17.9 0 1.35; 101 20.6 0 1.35;
102 23.3 0 1.35; 103 24.5 0 1.35; 104 25 0 1.35;
MEMBER INCIDENCES
1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 6 7; 7 7 8; 8 8 9; 9 9 10; 10 10 11;
11 11 12; 12 12 13; 13 14 15; 14 15 16; 15 16 17; 16 17 18; 17 18 19; 18 19 20;
19 20 21; 20 21 22; 21 22 23; 22 23 24; 23 24 25; 24 25 26; 37 40 41; 38 41 42;
39 42 43; 40 43 44; 41 44 45; 42 45 46; 43 46 47; 44 47 48; 45 48 49; 46 49 50;
47 50 51; 48 51 52; 49 66 67; 50 67 68; 51 68 69; 52 69 70; 53 70 71; 54 71 72;
55 72 73; 56 73 74; 57 74 75; 58 75 76; 59 76 77; 60 77 78; 61 53 54; 62 54 55;
63 55 56; 64 56 57; 65 57 58; 66 58 59; 67 59 60; 68 60 61; 69 61 62; 70 62 63;
71 63 64; 72 64 65; 73 1 92; 74 2 93; 75 3 94; 76 4 95; 77 5 96; 78 6 97;
79 7 98; 80 8 99; 81 9 100; 82 10 101; 83 11 102; 84 12 103; 85 13 104;
86 1 14; 87 2 15; 88 3 16; 89 4 17; 90 5 18; 91 6 19; 92 7 20; 93 8 21;
94 9 22; 95 10 23; 96 11 24; 97 12 25; 98 13 26; 125 40 79; 126 41 80;
127 42 81; 128 43 82; 129 44 83; 130 45 84; 131 46 85; 132 47 86; 133 48 87;
134 49 88; 135 50 89; 136 51 90; 137 52 91; 138 14 40; 139 15 41; 140 16 42;
141 17 43; 142 18 44; 143 19 45; 144 20 46; 145 21 47; 146 22 48; 147 23 49;
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155 80 81; 156 82 69; 157 81 82; 158 83 70; 159 82 83; 160 84 71; 161 83 84;
162 85 72; 163 84 85; 164 86 73; 165 85 86; 166 87 74; 167 86 87; 168 88 75;
169 87 88; 170 89 76; 171 88 89; 172 90 77; 173 89 90; 174 91 78; 175 90 91;
176 92 53; 177 93 54; 178 92 93; 179 94 55; 180 93 94; 181 95 56; 182 94 95;
183 96 57; 184 95 96; 185 97 58; 186 96 97; 187 98 59; 188 97 98; 189 99 60;
190 98 99; 191 100 61; 192 99 100; 193 101 62; 194 100 101; 195 102 63;
196 101 102; 197 103 64; 198 102 103; 199 104 65; 200 103 104;
*
MEMBER PROPERTY AMERICAN
13 TO 24 PRIS YD 0.925 ZD 2.325

```
1 TO 12 37 TO 48 PRIS YD 0.925 ZD 1.425
153 155 157 159 161 163 165 167 169 171 173 175 178 180 182 184 186 188 190 -
192 194 196 198 200 PRIS YD 0.925 ZD 0.9375
49 TO 72 PRIS YD 0.275 ZD 0.675
75 TO 83 88 TO 96 127 TO 135 140 TO 148 154 156 158 160 162 164 166 168 170 -
179 181 183 185 187 189 191 193 195 PRIS YD 0.275 ZD 2.7
74 84 87 97 126 136 139 149 PRIS YD 1.7 ZD 1
152 172 177 197 PRIS YD 0.275 ZD 0.85
73 85 86 98 125 137 138 150 151 174 176 199 PRIS YD 0.25 ZD 0.25
*********
DEFINE MATERIAL START
ISOTROPIC CONCRETE-1
E 3.1e+007
POISSON 0.17
DENSITY 25
ALPHA 1.17e-005
DAMP 0.05
END DEFINE MATERIAL
***
CONSTANTS
MATERIAL CONCRETE-1 ALL
SUPPORTS
12 51 PINNED
25 PINNED
2 41 PINNED
15 PINNED
UNIT METER MTON
DEFINE MOVING LOAD
*70R Wheeled (13.4m)
*TYPE 1 LOAD 4 6 6 8.5 8.5 8.5 8.5
*DIST 3.96 1.52 2.13 1.37 3.05 1.37 WID 1.93
TYPE 1 LOAD 8.5 8.5 8.5 8.5 6 6 4
DIST 1.37 3.05 1.37 2.13 1.52 3.96 WID 1.93
*Class A (18.8m)
TYPE 2 LOAD 1.35 1.35 5.7 5.7 3.4 3.4 3.4 3.4
DIST 1.1 3.2 1.2 4.3 3 3 3 WID 1.8
UNIT METER KN
LOAD 1 DL GIRDER
FLOOR LOAD
YRANGE 0 0 FLOAD -26.3 XRANGE 0 25  ZRANGE 1.35 7.05  GY
YRANGE 0 0 FLOAD -6.875 XRANGE 0 25  ZRANGE 0 1.35  GY
YRANGE 0 0 FLOAD -6.875 XRANGE 0 25  ZRANGE 7.05 8.4  GY
LOAD 2 DL DECK SLAB & CROSS GIRDER
FLOOR LOAD
YRANGE 0 0 FLOAD -0 XRANGE 0 15  ZRANGE 0 8.4  GY
LOAD 3 SIDL CRASH BARRIER & RAILING
MEMBER LOAD
49 TO 72 UNI GY -8
LOAD 4 SIDL SURFACING(WEARING COAT)
FLOOR LOAD
YRANGE 0 0 FLOAD -1.875 XRANGE 0 25  ZRANGE 0 8.4  GY
LOAD 5 FPLL
FLOOR LOAD
YRANGE 0 0 FLOAD -0 XRANGE 0 19.5  ZRANGE 0 12  GY
************************************
```

```
LOAD GENERATION 1
TYPE 2 -4.3 0 2.65 XINC 0.2
*ECC.  2 Class A CB side (18.8m)
LOAD GENERATION 1
TYPE 2 -4.3 0 2.65 XINC 0.2
TYPE 2 -4.3 0 6.15 XINC 0.2
*ECC. 70R Wheeled CB side (13.4m)
LOAD GENERATION 1
TYPE 1 0 0 4.06 XINC 0.2
*ECC.  3 Class A (18.8m)
LOAD GENERATION 1
TYPE 2 0 0 2.65 XINC 0.2
TYPE 2 0 0 6.15 XINC 0.2
*ECC. 70R Wheeled + 1 Class A
LOAD GENERATION 1
TYPE 1 0 0 4.06 XINC 0.2
*TYPE 2 0 0 7.74 XINC 0.2
PERFORM ANALYSIS
FINISH
```

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	**DESIGN OF ABUTMENT A1 & A2**	Date & Rev.	09-01-2024

Introduction

This design note present the detailed calculations for abutment having elastomaric bearings. Design note contains the typical design of abutment cap, abutment, pile cap and pile. In the analysis, the forces have been worked out at the pile cap bottom level for pile cap design and at the top of pile cap for abutment shaft design.

This design is applicable for abutments

abutment no.	Road Top Level (M)	GROUND LEVEL (M)	Pile cap bottom (M)	DIFF (M)
A1	92.575	88.910	86.810	5.765

Soffit RL **90.800** LOWEST

Design Data

Span Arrangement C/C of exp. Joint	=	**25**	m
Structural system	=	simply supported	
construction type	=	RCC	
Effective Span	=	**24.00**	m
Expansion gap in abutment	=	**40**	mm
No of Lane considered for design	=	**2**	
Bearing type	=	**Elastomeric Bearing**	
Skew Angle	=	**0**	Degree

Salient Reduced Levels (RL's)

Deck Level (Crash barrier Edge)	=	92.575
Soffit Level (at high end)	=	90.800
abutment Cap Top Level	=	90.300
abutment Cap Bottom Level	=	89.800
HFL	=	90.000
Ground Level (LBL)	=	88.910
Pile cap bottom Level	=	86.810
Scour Level	=	74.800

Details of superstructure and other parameters

Type of structure		**RCC**
Depth of Superstructure	=	**1.700**
Thickness of wearing coat	=	**0.075**
Camber in both direction	=	**2.5%**
Expansion joint Type	=	**Strip seal**
Impact Factor	=	**1.150**

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Material Parameters

Concrete **Refer Table 6.5, IRC:112-2011**

Grade of concrete for sub structure and foundation		=	**M30**	
Cube strength of concrete at 28 days	f_{ck}	=	**30**	MPa
Design value of concrete compressive strength	f_{cd}	=	$\alpha f_{ck}/\gamma_m$	
		α =	0.67	
For Basic Combination	f_{cd}	=	13.40	MPa
For Accidental Combination	f_{cd}	=	16.75	MPa
For Seismic Combination	f_{cd}	=	13.40	MPa
Modulus of Elasticity	E_c	=	31000	MPa
Mean value of axial tensile strength of concrete	f_{ctm}	=	2.5	MPa
Wet Density **2.00** t/m^3	Dry Density	=	**2.50**	t/m^3

Reinforcing Steel

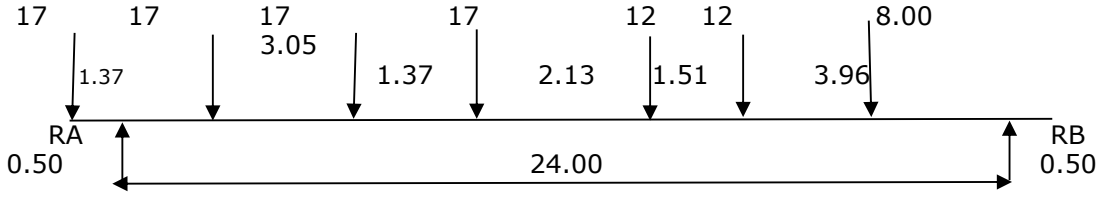
Grade		=	**Fe500**	
Characteristics yield strength	f_{yk}	=	500	MPa
Design yield strength	f_{yd}	=	f_{yk}/γ_m	
For Basic Combination	f_{yd}	=	434.8	MPa
For Accidental Combination	f_{yd}	=	500.0	MPa
For Seismic Combination	f_{yd}	=	434.8	MPa
Modulus of Elasticity	E_s	=	**2.0E+05**	MPa
Density		=	**7.85**	t/m^3
Soil density		=	**2.00**	t/m^3

(As per STAAD Output)

25 m span Reactions (KN)				**25 m span Reactions (KN)**			
Bearing No.	**DL**	**SIDL1 (WC)**	**SIDL2(CB)**	**Bearing No.**	**DL**	**SIDL1 (WC)**	**SIDL2(CB)**
1	735	85	166	1	735	85	166
2	637	28	-132	2	637	28	-132
3	735	85	166	3	735	85	166
	2106	**197**	**200**		**2106**	**196.884**	**200**

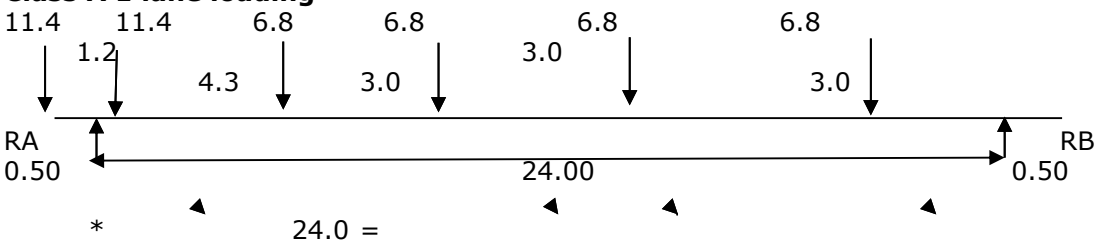
Live load

a) 70 R 1lane loading



Ra	*		24.00				
	4.00	*	17.00	*	21.61	=	1469.1
	2.00	*	12.00	*	15.83	=	379.80
	1.00	*	8.00	*	11.11	=	88.88
						Total :	1937.8
Ra =		1937.82	/	24.00		=	**80.7** **t**
		Impact	1.150				**92.9** **t**
RB =							19.26 **t**

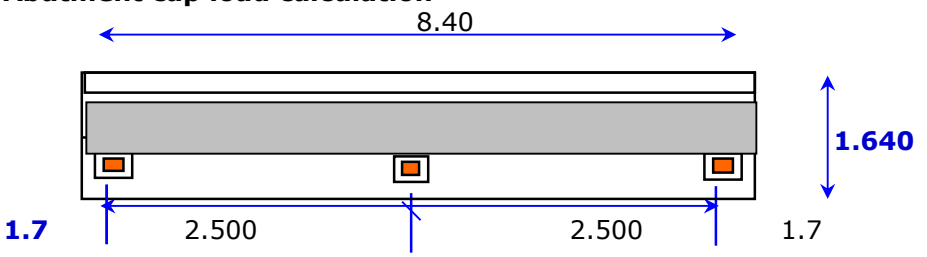
b) Class A 1 lane loading

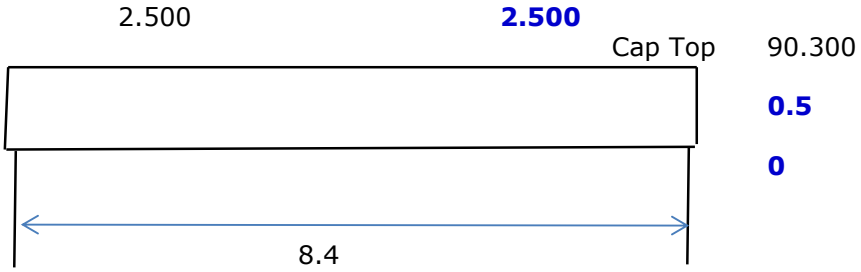


Ra		*	24.0 =				
	4	*	6.8	*	14.50	=	394
	2	*	11.4	*	23.9	=	545
						Total :	939
Ra =						=	**39.1** **t**
				Impact	1.150		**45.0** **t**
RB =							10.9 **t**

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Abutment cap load calculation





Abutment cap is fully resting on abutment wall, so provide min. reinforcement.
cl. 16.5.1.1 (1) of IRC :112-2011
$A_{S.min}$ = 0.26 f_{ctm} b_t d / f_{yk} >= 0.0013 b_t d

f_{ctm}	2.5	
b_t	1640	
d	500	
f_{yk}	500	
0.26 f_{ctm} b_t d / f_{yk}	1066	mm^2
0.0013 b_t d	1066	mm^2

Provide

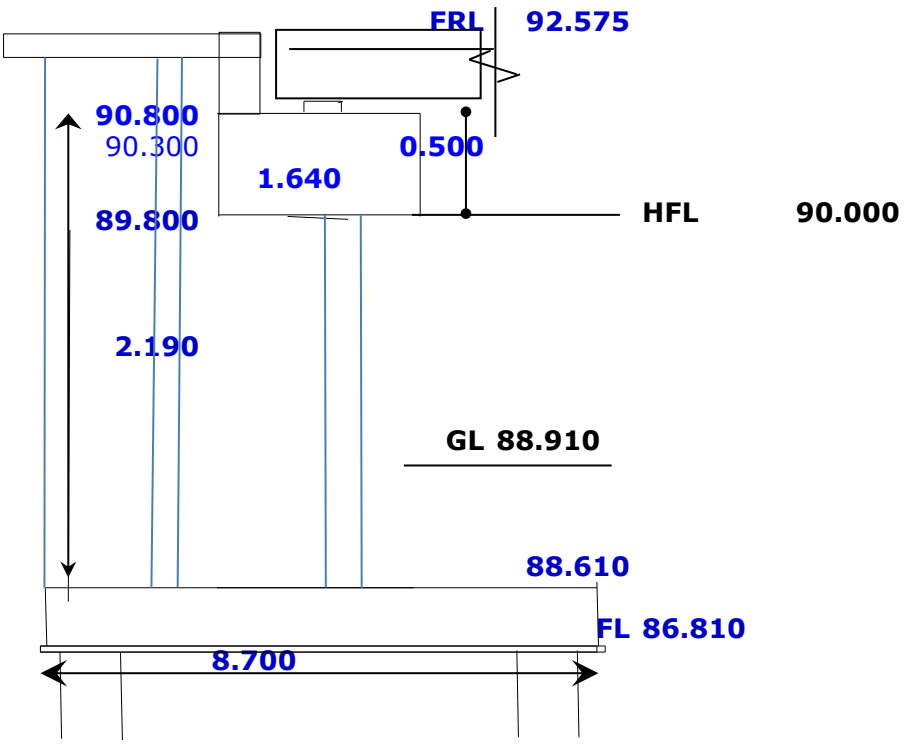
12 ϕ @	**150 mm c/c**	**754 OK**	**mm^2**	**at top and bottom**
12 ϕ @	**150 mm c/c**	**754 OK**	**mm^2**	**as distribution in the form of stirrup**

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Load calculation for abutment & pile cap

1. Salient Reduced Levels (RL's)

Deck Level	=	92.575	m
Bearing Top Level	=	90.800	m
abutment Cap Top Level	=	90.300	m
abutment Bottom Level	=	88.610	m
Depth of Pile cap	=	1.800	
GL	=	88.910	m
Pile cap bottom Level		86.810	m
HFL	=	90.000	m
LBL	=	88.910	m
Expansion Gap	=	0.02	m
Width of abutment	=	0.90	m
Length of the abut	=	8.40	m
Length of Pile cap	=	8.70	m
Width of Pile cap	=	8.70	m
Thickness of Wearing coat (W/c)	=	0.075	m
Height of Deck+Girder	=	1.700	m
Thickness of bearingBearing	=	0.052	m
Height of Pedestal	=	0.448	m
Total Height from w/c to Cap top	=	2.275	m
Span EJ to EJ	=	25.00	m



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Load Calculation

Taking Moments about founding level

		LHS		L (Long)	L (Trans)	ML	Mt
		t		m	m	t-m	t-m
Dead Load		214.68		1.16	0.00	249.03	0.00
SIDL1 w/c		20.07		1.16	0.00	23.28	0.00
SIDL 2 exl w/c		20.39		1.16	0.00	23.65	0.00

LL2 (Max Moment)							
Class A		39.14		1.16	2.350	45.40	91.98
Class 2 A		78.28		1.16	0.600	90.80	46.97
70 R		80.74		1.16	1.055	93.66	85.18

Horizontal Force Due to braking at				Elastomeric Bearing				
			(as per IRC 6:2010)	Fh/2+Vr ltc	**@foundation**		**abut Bottom**	
	Fh	Rg	Rq	**H**	**Lev**	**ML**	**Lev**	**ML**
	t	**t**	**t**	**t**	**m**	**t-m**	**m**	**t-m**
Class 2 A	13.85	214.68	78.28	18.27	3.99	72.88	2.19	40.00
Class 70 R	20.00	214.68	80.74	21.34	3.99	85.15	2.19	46.73

DUE TO BEARING

TEMPERATURE CORRECTIONS L a T

L	=	SPAN	24.00	M
a	=	EXPANSION COEFFICIENT	1.20E-05	$/^0C$
T	=	TEMPERATURE DIFFERENCE	50.00	^0C

TEMPERATURE CORRECTIONS L α T 0.014 M

ON EACH SIDE OF THE abut 0.0144/2 *1000 7.20 mm

DUT TO SHRINKAGE

SHRINKAGE 0.0002 M

TOTAL DEFORMATION 0.0002*24*1000/2+7.2 9.60 mm

NO OF BEARINGS one side 3.00 NOS

F = G*A* d /T

WHERE

F	=	FORCE IN BEARING				
G	=	SHEAR MODULUS	10.00	KG/CM^2		
d	=	DEFORMATION	0.96	CM		
A	=	AREA OF BEARING (450*350)	1575.00	CM2		
T	=	THICKNESS OF BEARING	4.00	CM	5.20	1.20

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F = 3.78 T

TOTAL FORCE ON BEARINGS 11.34 T

WIND FORCES ON SUPER STRUCTURE

REF: IRC 6-2010 CL NO:209.2

Basic wind speed	=	39	m/s	Plain terrain				1
Design wind speed	=	32.85	m/s	at height	10.00	m	27.80	
Design wind pressure P_Z		647.65	N/m^2					

$F_T = P_Z A_1 G C_D$	317	Kg/m^2	G =	**2.00**	**C_D =**	**2.40**	
F_L =	79.22	Kg/m^2					
$F_V = P_Z A_3 G C_L$	99	Kg/m^2	G =	**2.00**	**C_L =**	**0.75**	

Up ward or downward vertical wind load Fv

F_V =	12.50	x	**8.40**	x	99.03	=	**10.40**	**t**

WIND FORCES ON LIVE LOAD

Basic wind speed	=	39	m/s	Plain terrain				1
Design wind speed	=	32.85	m/s	at height	10.00	m	27.80	
Design wind pressure P_Z		647.65	N/m^2					

$F_T = P_Z A_1 G C_D$	158	Kg/m^2	G =	**2.00**	**C_D =**	**1.20**	
F_L =	39.61	Kg/m^2					
$F_V = P_Z A_3 G C_L$	99	Kg/m^2	G =	**2.00**	**C_L =**	**0.75**	

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WIND FORCES ON SUB STRUCTURE

Basic wind speed	=	39	m/s	Plain terrain			1
Design wind speed	=	32.85	m/s	at height	10.00	m	27.80
Design wind pressure P_Z		647.65	N/m^2				

REF: IRC 6-2010 CL NO:209.4

$F_T = P_Z A_1 G C_D$	66	Kg/m^2	G =	**2.00**	**C_D =**	**0.50**
F_L =	16.50	Kg/m^2				
$F_V = P_Z A_3 G C_L$	99	Kg/m^2	G =	**2.00**	**C_L =**	**0.75**

WIND PRESSURE 317 KG/M^2

WIND PRESSURE ON MOVING LOAD 475 KG/M

TRANSVERSE DIRECTION		**Moments about FL**			**abutment Bottom**	
DESCRITPTION		**H**	**Lev**	**Mt**	**Lev**	**Mt**
		t	**M**	**t-m**	**M**	**t-m**
WIND ON SUPER STR ACT @	91.69	7.03	4.88	34.29	3.08	21.64
WIND ON MOVING LOAD @	94.08	5.94	7.27	43.17	5.47	32.47
WIND ON abutment CAP @	90.05	0.05	3.24	0.18	1.44	0.08
	TOTAL	**13.08**		**77.64**		**54.19**
		7.14		**34.47**		**21.72**

LONGITUDIANL DIRECTION		**Moments about FL**			**abutment Bottom**	
DESCRITPTION		**H**	**Lev**	**Mt**	**Lev**	**Mt**
		t	**M**	**t-m**	**M**	**t-m**
WIND ON SUPER STR ACT @	91.69	1.18	4.88	5.76	3.08	3.64
WIND ON MOVING LOAD @	94.08	1.00	7.27	7.25	5.47	5.46
WIND ON abutment CAP @	90.05	0.00	3.24	0.00	1.44	0.00
	TOTAL	**2.18**		**13.01**		**9.09**
		1.18		**7.25**		**5.46**

Dead wt of abutment & Foundation
As per abutment calculation sheet

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	870.10	305.03		846.31	296.69
	565.07				-549.6

Earth pressure		At foundation		At abut. Bottom	
		LWL case	HFL case	LWL case	HFL case
Height	m	5.765	5.77	3.97	3.97
Dry unit wt. of soil	t/m3	2.00		2.00	
wet unit wt. of soil	t/m3		1.00		1.00
Coulomb's theory Ka (Normal case)					
ϕ Angle of friction		30.00			
α wall inclination to horizontal		90.00			
α wall inclination to vertical		0.00			
β soil surcharge inclination to horizontal		0.00			
δ		20.00			
Ka (Normal case)		0.279			
Coefficient of active earth pressure		0.279	0.279	0.279	0.279
Live load surcharge	m	1.20	1.20	1.20	1.20
Active earth pressure		3.22	1.61	2.21	1.11
Pressure due to live load surcharge		0.67	0.33	0.67	0.33
Force due to live load surcharge	t/m	32	16	22	11
Force due to active earth pressure	t/m	78	39	37	18
Moment due to live load surcharge	t.m/m	93	47	44	22
Moment due to active earth pressure	t.m/m	189	94	61	31

Seismic Force Calculations	**For Foundation**					
	Horizontal	**Vertical (2/3 of Horizontal)**				
Long Seismic Coefficient	0.060	0.040	***Refer Seismic Calculations Appendex 1***			
Trans Seismic Coefficient	0.060	0.040				

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Seismic

						@ Foundation		
	V (t)	**H Comp**	**V Comp**	**Acting At**	**Ecc (For H)**	**ML Due to H**	**Moment Due to V**	
							ML	**MT**
DL	214.7	12.9	8.6	90.8	**4.0**	51.4	14.9	0.0
SIDL1	20.1	1.2	0.8	90.8	**4.0**	4.8	1.4	0.0
SIDL2	20.4	1.2	0.8	90.8	**4.0**	4.9	1.4	0.0
: Cap+dirt wall	0.0	0.0	0.0	0.0	**-86.8**	0.0		
abut	870.1	52.2	34.8	89.7	**2.9**	151.1		
return wall	0.0	0.0	0.0	90.6	**3.8**	0.0		
Soil above f	0.0	0.0	0.0	90.6	**3.8**	0.0		
LL1	16.1	1.0	0.6	90.8	**4.0**	3.9	3.7	0.7
LL2	16.1	1.0	0.6	90.8	**4.0**	3.9	3.7	0.7
LL3	7.8	0.5	0.3	90.8	**4.0**	1.9	1.8	0.7

Longitudinal seismic

	V (t)	ML (t.m)	MT (t.m)
DL	43.39	217.47	0.00
SIDL1	0.80	4.80	0.00
SIDL2	0.82	4.88	0.00
LL1	0.65	7.61	0.68
LL2	0.65	7.61	0.68
LL3	0.31	3.69	0.74

Tranverse seismic

	V (t)	ML (t.m)	MT (t.m)
DL	43.39	14.94	202.53
SIDL1	0.80	1.40	4.80
SIDL2	0.82	1.42	4.88
LL1	0.65	3.75	4.55
LL2	0.65	3.75	4.55
LL3	0.31	3.69	2.61

Combination of Orthogonal Seismic Force

		V (t)	ML (t.m)	MT (t.m)	V (t)	ML (t.m)	MT (t.m)
DL		43.39	278.23	0.00	**43.4**	**278.2**	**0.0**
SIDL1		0.80	6.25	0.00	**0.8**	**6.2**	**0.0**
SIDL2		0.82	6.34	0.00	**0.8**	**6.3**	**0.0**
LL1		0.65	8.98	0.68	**0.6**	**9.0**	**0.7**
LL2		0.65	8.98	0.68	**0.6**	**9.0**	**0.7**
LL3		0.31	4.47	0.74	**0.3**	**4.5**	**0.7**
DL		43.39	14.94	267.77	**43.4**	**14.9**	**267.8**
SIDL1		0.80	1.40	6.25	**0.8**	**1.4**	**6.2**
SIDL2		0.82	1.42	6.34	**0.8**	**1.4**	**6.3**
LL1		0.65	3.75	6.83	**0.6**	**3.7**	**6.8**
LL2		0.65	3.75	6.83	**0.6**	**3.7**	**6.8**
LL3		0.31	1.82	3.72	**0.3**	**1.8**	**3.7**

	V (t)	ML (t.m)	MT (t.m)
DL+SIDL1+SIDL2+LL1	45.66	299.80	0.68
DL+SIDL1+SIDL2+LL2	45.66	299.80	0.68
DL+SIDL1+SIDL2+LL3	45.32	295.30	0.74
DL+SIDL1+SIDL2+LL1	45.66	21.50	287.19
DL+SIDL1+SIDL2+LL2	45.66	21.50	287.19
DL+SIDL1+SIDL2+LL3	45.32	19.57	284.08
DL+SIDL1+SIDL2+LL1	**46**	**300**	**1**
DL+SIDL1+SIDL2+LL2	**46**	**300**	**1**
DL+SIDL1+SIDL2+LL3	**46**	**22**	**287**

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Combination for design of foundation & base pressure

Forces and moments are calculated at Pile cap bottom level

Following load combinations are con As per Annex B of IRC:112-2011

1 Combination 1

2 Combination 2

3 Seismic Combination

4 Accidental Combination Skew Angle 0 cosθ 1.000 sinθ 0.000

5 Combination for base pressure check

		P (t)	ML (tm)	MT (tm)	HL(t)	HT(t)
1	DL Dead Load (due to superstructure+substructure+foundation)	1085	-597	0	0	0
2	DL_SUP	215	249	0	0	0
3	ABUT	870	-846	0	0	0
4	ABUT + RETURN WALL	0	0	0	0	0
5	FDN	0	0	0	0	0
6	SOIL ABOVE FDN	0	0	0	0	0
7	SIDL1 Super-imposed Dead Load due to wearing course	20	23		0	0
8	SIDL2 Super-imposed Dead Load other than wearing course	20	24		0	0
9	LL1 Live Load Maximum Reaction Case	81	94	85	0	0
10	LL2 Live Load Maximum Longitudinal Moment Case	81	94	85	0	0
11	LL3 Live Load Maximum Transverse Moment Case	39	45	92	0	0
12	LL1B Live Load Moment due to braking	2	85	0	21.34	0
13	LL2B Live Load Moment due to braking	2	85	0	21.34	0
14	LL3B Live Load Moment due to braking	0	0	0	0.00	0
15	W1 Wind vertically down without LL	10	7	34	1.18	7.14
16	W2 Wind vertically up without LL	-10	7	34	1.18	7.14
17	W3 Wind vertically down with LL	10	13	78	2.18	13.08
18	W4 Wind vertically up with LL	-10	13	78	2.18	13.08
19	S1 Seimic	46	300	1	68.5	0
20	S2	46	22	287	0.0	68.5
21	S3	-46	300	1	68.5	0
22	S4	-46	22	287	0.0	68.5
25	HFL	-565	-550	0.00	0.00	0.00
26	Active earth pressure LWL	0	189	0	77.9	0.0
27	LL surcharge LWL	0	93	0	32.4	0.0
28	Active earth pressure HFL	0	94	0	38.9	0.0
29	LL surcharge HFL	0	47	0	16.2	0.0

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Note:- In combination 1 & 2 wind is taken as accompanying load and LL is taken as the leading load.

	Combination 1				Combination 2				Seismic Combination				Combination for base pressure check		
	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)	P(t)	ML(tm)	MT(tm)
DL	1.35	1464	-806	0	1.00	1085	-597	0	1.25	1356	-747	0	1085	-597	0
SIDL1	1.75	35	41	0	1.00	20	23	0	1.75	35	41	0	20	23	0
SIDL2	1.35	28	32	0	1.00	20	24	0	1.25	25	30	0	20	24	0
LL1	1.50	121	140	128	1.30	105	122	111	0.75	61	70	64	81	94	85
LL2	1.50	121	140	128	1.30	105	122	111	0.75	61	70	64	81	94	85
LL3	1.50	59	68	138	1.30	51	59	120	0.75	29	34	69	39	45	92
LL1B	1.50	4	128	0	1.30	3	111	0	0.75	2	64	0	2	85	0
LL2B	1.50	4	128	0	1.30	3	111	0	0.75	2	64	0	2	85	0
LL3B	1.50	0	0	0	1.30	0	0	0	0.75	0	0	0	0	0	0
W1	0.90	9	7	31	0.80	8	6	28	0.00	0	0	0	10	7	34
W2	0.90	-9	7	31	0.80	-8	6	28	0.00	0	0	0	-10	7	34
W3	0.90	9	12	70	0.80	8	10	62	0.00	0	0	0	10	13	78
W4	0.90	-9	12	70	0.80	-8	10	62	0.00	0	0	0	-10	13	78
S1	0.00	0	0	0	0.00	0	0	0	1.50	68	450	1	46	300	1
S2	0.00	0	0	0	0.00	0	0	0	1.50	68	32	431	46	22	287
S3	0.00	0	0	0	0.00	0	0	0	1.50	-68	450	1	-46	300	1
S4	0.00	0	0	0	0.00	0	0	0	1.50	-68	32	431	-46	22	287
HFL	1.00	-565	-550	0	1.00	-565	-550	0	1.00	-565	-550	0	-565	-550	0
Active ear	1.00	0	189	0	0.85	0	160	0	1.00	0	189	0	0	189	0
LL surcha	1.20	0	112	0	1.00	0	93	0	0.20	0	19	0	0	93	0
Active ear	1.00	0	94	0	0.85	0	80	0	1.00	0	94	0	0	94	0
LL surcha	1.20	0	56	0	1.00	0	47	0	0.20	0	9	0	0	47	0

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
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	Job	DESIGN OF ABUTMENT A1 & A2	Date & Rev.	09-01-2024

1 Combination 1

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2	77.9	0.0	1527	-545	0
Case 2	DL+SIDL1+SIDL2+LL1	148.8	0.0	1652	-165	128
Case 3	DL+SIDL1+SIDL2+LL2	142.3	0.0	1652	-165	128
Case 4	DL+SIDL1+SIDL2+LL3	116.8	0.0	1586	-365	138
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	79.0	6.4	1536	-539	31
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	150.8	11.8	1661	-153	198
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	144.3	11.8	1661	-153	198
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	118.8	11.8	1595	-353	208
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	79.0	6.4	1518	-539	31
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	150.8	11.8	1643	-153	198
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	144.3	11.8	1643	-153	198
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	118.8	11.8	1576	-353	208
HFL case						
Case 13	DL+SIDL1+SIDL2 + HFL	38.9	0.0	962	-1189	0
Case 14	DL+SIDL1+SIDL2+LL1+ HFL	90.4	0.0	1087	-865	128
Case 15	DL+SIDL1+SIDL2+LL2+ HFL	90.4	0.0	1087	-865	128
Case 16	DL+SIDL1+SIDL2+LL3+ HFL	58.4	0.0	1021	-1065	138
Downward Wind Condition						
Case 17	DL+SIDL1+SIDL2 + Wind + HFL	40.0	6.4	971	-1182	31
Case 18	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	92.4	11.8	1096	-853	198
Case 19	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	92.4	11.8	1096	-853	198
Case 20	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	60.4	11.8	1030	-1053	208
Upward Wind Condition						
Case 21	DL+SIDL1+SIDL2 + Wind + HFL	40.0	6.4	953	-1182	31
Case 22	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	92.4	11.8	1078	-853	198
Case 23	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	92.4	11.8	1078	-853	198
Case 24	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	60.4	11.8	1011	-1053	208

151	12	1661	-153	198
151	12	1661	-153	198
		1595	-353	208

2 Combination 2

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2	66.2	0.0	1125	-390	0
Case 2	DL+SIDL1+SIDL2+LL1	126.4	0.0	1233	-64	111
Case 3	DL+SIDL1+SIDL2+LL2	126.4	0.0	1233	-64	111
Case 4	DL+SIDL1+SIDL2+LL3	98.6	0.0	1176	-238	120
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	67.2	5.7	1125	-297	0
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	128.1	1.7	1242	-54	173
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	128.1	1.7	1242	-54	173
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	100.4	1.7	1184	-227	182
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	67.2	5.7	1117	-384	28
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	128.1	10.5	1225	-54	173
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	128.1	10.5	1225	-54	173
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	100.4	10.5	1168	-227	182
HFL case						
Case 13	DL+SIDL1+SIDL2 + HFL	33.1	0.0	560	-1020	0
Case 14	DL+SIDL1+SIDL2+LL1+ HFL	77.1	0.0	668	-741	111
Case 15	DL+SIDL1+SIDL2+LL2+ HFL	77.1	0.0	668	-741	111
Case 16	DL+SIDL1+SIDL2+LL3+ HFL	49.3	0.0	611	-914	120
Downward Wind Condition						
Case 17	DL+SIDL1+SIDL2 + Wind + HFL	34.0	5.7	560	-846	0
Case 18	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	78.8	1.7	677	-730	173
Case 19	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	78.8	1.7	677	-730	173
Case 20	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	51.1	1.7	619	-904	182
Upward Wind Condition						
Case 21	DL+SIDL1+SIDL2 + Wind + HFL	34.0	5.7	552	-1014	28
Case 22	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	78.8	10.5	660	-730	173
Case 23	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	78.8	10.5	660	-730	173
Case 24	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	51.1	10.5	603	-904	182

128	2	1242	-54	173
128	10	1242	-54	173
		1184	-227	182

Case 3 Seismic Combination

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Downward seismic in longitudinal direction						
Case 1	DL+SIDL1+SIDL2+ seismic	180.6	0.0	1485	-38	1
Case 2	DL+SIDL1+SIDL2+LL1+ seismic	203.1	0.0	1547	115	65
Case 3	DL+SIDL1+SIDL2+LL2+ seismic	203.1	0.0	1547	115	65
Case 4	DL+SIDL1+SIDL2+LL3+ seismic	187.1	0.0	1514	15	70
Downward seismic in transverse direction						
Case 5	DL+SIDL1+SIDL2+ seismic	77.9	102.7	1485	-455	431

Case 6	DL+SIDL1+SIDL2+LL1+ seismic	100.4	102.7	1547	-303	495
Case 7	DL+SIDL1+SIDL2+LL2+ seismic	100.4	102.7	1547	-303	495
Case 8	DL+SIDL1+SIDL2+LL3+ seismic	84.4	102.7	1514	-403	500
Upward seismic in longitudinal direction						
Case 9	DL+SIDL1+SIDL2+ seismic	180.6	0.0	1348	-38	1
Case 10	DL+SIDL1+SIDL2+LL1+ seismic	203.1	0.0	1411	115	65
Case 11	DL+SIDL1+SIDL2+LL2+ seismic	203.1	0.0	1411	115	65
Case 12	DL+SIDL1+SIDL2+LL3+ seismic	187.1	0.0	1377	15	70
Upward seismic in transverse direction						
Case 13	DL+SIDL1+SIDL2+ seismic	77.9	102.7	1348	-644	431
Case 14	DL+SIDL1+SIDL2+LL1+ seismic	100.4	102.7	1411	-491	495
Case 15	DL+SIDL1+SIDL2+LL2+ seismic	100.4	102.7	1411	-491	495
Case 16	DL+SIDL1+SIDL2+LL3+ seismic	84.4	102.7	1377	-591	500
HFL case						
Downward seismic in longitudinal direction						
Case 17	DL+SIDL1+SIDL2+HFL	141.7	0.0	920	-682	1
Case 18	DL+SIDL1+SIDL2+LL1+ seismic+HFL	160.9	0.0	982	-538	65
Case 19	DL+SIDL1+SIDL2+LL2+ seismic+HFL	160.9	0.0	982	-538	65
Case 20	DL+SIDL1+SIDL2+LL3+ seismic+HFL	144.9	0.0	949	-639	70
Downward seismic in transverse direction						
Case 21	DL+SIDL1+SIDL2+HFL	38.9	102.7	920	-1099	431
Case 22	DL+SIDL1+SIDL2+LL1+ seismic+HFL	58.2	102.7	982	-956	495
Case 23	DL+SIDL1+SIDL2+LL2+ seismic+HFL	58.2	102.7	982	-956	495
Case 24	DL+SIDL1+SIDL2+LL3+ seismic+HFL	42.2	102.7	949	-1056	500
Upward seismic in longitudinal direction						
Case 25	DL+SIDL1+SIDL2+HFL	141.7	0.0	783	-682	1
Case 26	DL+SIDL1+SIDL2+LL1+ seismic+HFL	160.9	0.0	845	-538	65
Case 27	DL+SIDL1+SIDL2+LL2+ seismic+HFL	160.9	0.0	845	-538	65
Case 28	DL+SIDL1+SIDL2+LL3+ seismic+HFL	144.9	0.0	812	-639	70
Upward seismic in transverse direction						
Case 29	DL+SIDL1+SIDL2+HFL	38.9	102.7	783	-1099	431
Case 30	DL+SIDL1+SIDL2+LL1+ seismic+HFL	58.2	102.7	845	-956	495
Case 31	DL+SIDL1+SIDL2+LL2+ seismic+HFL	58.2	102.7	845	-956	495
Case 32	DL+SIDL1+SIDL2+LL3+ seismic+HFL	42.2	102.7	812	-1056	500

203	0	1547	115	65
78	103	1547	115	65
		1514	-403	500

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4 Combination for base pressure check

Load Case		Factored Forces/Moments		
		V	ML	MT
		t	t-m	t-m
Normal				
Case 1	DL+SIDL1+SIDL2	1125	-362	0
Case 2	DL+SIDL1+SIDL2+LL1	1208	-89	85
Case 3	DL+SIDL1+SIDL2+LL2	1208	-89	85
Case 4	DL+SIDL1+SIDL2+LL3	1164	-223	92
Downward Wind Condition				
Case 5	DL+SIDL1+SIDL2 + Wind	1136	-355	34
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	1219	-76	163
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	1219	-76	163
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	1175	-210	170
Upward Wind Condition				
Case 9	DL+SIDL1+SIDL2 + Wind	1115	-355	34
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	1198	-76	163
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	1198	-76	163
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	1154	-210	170
Downward seismic in longitudinal direction				
Case 13	DL+SIDL1+SIDL2+ seismic	1171	-62	1
Case 14	DL+SIDL1+SIDL2+LL1+ seismic	1254	210	86
Case 15	DL+SIDL1+SIDL2+LL2+ seismic	1254	210	86
Case 16	DL+SIDL1+SIDL2+LL3+ seismic	1210	77	93
Downward seismic in transverse direction				
Case 17	DL+SIDL1+SIDL2+ seismic	1171	-340	287
Case 18	DL+SIDL1+SIDL2+LL1+ seismic	1254	-68	372
Case 19	DL+SIDL1+SIDL2+LL2+ seismic	1254	-68	372
Case 20	DL+SIDL1+SIDL2+LL3+ seismic	1210	-201	379
Upward seismic in longitudinal direction				
Case 21	DL+SIDL1+SIDL2+ seismic	1080	-62	1
Case 22	DL+SIDL1+SIDL2+LL1+ seismic	1163	210	86
Case 23	DL+SIDL1+SIDL2+LL2+ seismic	1163	210	86
Case 24	DL+SIDL1+SIDL2+LL3+ seismic	1119	77	93
Upward seismic in transverse direction				
Case 25	DL+SIDL1+SIDL2+ seismic	1080	-340	287
Case 26	DL+SIDL1+SIDL2+LL1+ seismic	1163	-68	372
Case 27	DL+SIDL1+SIDL2+LL2+ seismic	1163	-68	372
Case 28	DL+SIDL1+SIDL2+LL3+ seismic	1119	-201	379
	HFL condition			
Case 29	DL+SIDL1+SIDL2+HFL	560	-1006	0
Case 30	DL+SIDL1+SIDL2+LL1+HFL	643	-780	85
Case 31	DL+SIDL1+SIDL2+LL2+HFL	643	-780	85
Case 32	DL+SIDL1+SIDL2+LL3+HFL	599	-914	92
Downward Wind Condition				
Case 33	DL+SIDL1+SIDL2 + Wind +HFL	571	-998	34
Case 34	DL+SIDL1+SIDL2+LL1+ Wind+HFL	654	-767	163
Case 35	DL+SIDL1+SIDL2+LL2+ Wind+HFL	654	-767	163
Case 36	DL+SIDL1+SIDL2+LL3+ Wind+HFL	610	-901	170
Upward Wind Condition				
Case 37	DL+SIDL1+SIDL2 + Wind +HFL	550	-998	34
Case 38	DL+SIDL1+SIDL2+LL1+ Wind+HFL	633	-767	163
Case 39	DL+SIDL1+SIDL2+LL2+ Wind+HFL	633	-767	163
Case 40	DL+SIDL1+SIDL2+LL3+ Wind+HFL	589	-901	170
Downward seismic in longitudinal direction				
Case 41	DL+SIDL1+SIDL2+HFL	606	-706	1
Case 42	DL+SIDL1+SIDL2+LL1+ seismic+HFL	689	-480	86
Case 43	DL+SIDL1+SIDL2+LL2+ seismic+HFL	689	-480	86
Case 44	DL+SIDL1+SIDL2+LL3+ seismic+HFL	645	-614	93
Downward seismic in transverse direction				

Case 45	DL+SIDL1+SIDL2+HFL	606	-984	287
Case 46	DL+SIDL1+SIDL2+LL1+ seismic+HFL	689	-759	372
Case 47	DL+SIDL1+SIDL2+LL2+ seismic+HFL	689	-759	372
Case 48	DL+SIDL1+SIDL2+LL3+ seismic+HFL	645	-892	379
Upward seismic in longitudinal direction				
Case 49	DL+SIDL1+SIDL2+HFL	515	-706	1
Case 50	DL+SIDL1+SIDL2+LL1+ seismic+HFL	598	-480	86
Case 51	DL+SIDL1+SIDL2+LL2+ seismic+HFL	598	-480	86
Case 52	DL+SIDL1+SIDL2+LL3+ seismic+HFL	554	-614	93
Upward seismic in transverse direction				
Case 53	DL+SIDL1+SIDL2+HFL	515	-984	287
Case 54	DL+SIDL1+SIDL2+LL1+ seismic+HFL	598	-759	372
Case 55	DL+SIDL1+SIDL2+LL2+ seismic+HFL	598	-759	372
Case 56	DL+SIDL1+SIDL2+LL3+ seismic+HFL	554	-892	379

1254	210	86
1254	210	86
1210	-201	379

1 Normal			
Load Case	Factored Forces/Moments		
	V	ML	MT
	t	t-m	t-m
Case 1 Normal max. Vertical load	1208	-89	85
Case 2 Normal max. longitudinal moment	1208	-89	85
Case 3 Normal max. transverse moment	1164	-223	92
2 Seismic / wind			
Load Case	Factored Forces/Moments		
	V	ML	MT
	t	t-m	t-m
Case 1 Seismic/wind max. Vertical load	1254	210	86
Case 2 Seismic/wind max. longitudinal moment	1254	210	86
Case 3 Seismic/wind max. transverse moment	1210	-201	379

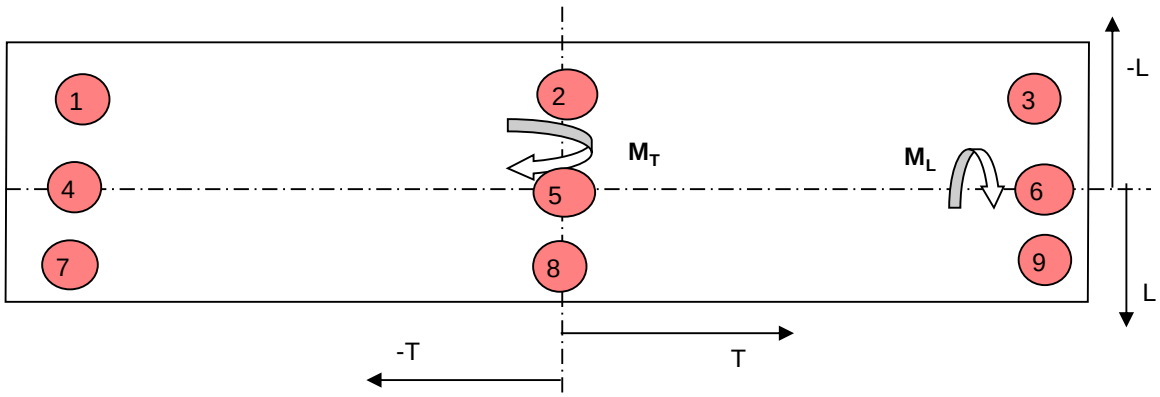
	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
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Load on pile for base pressure check

		V	ML	MT
		t	t-m	t-m
DL+SIDL1+SIDL2	Case 1	1208	89	85
DL+SIDL1+SIDL2+LL1	Case 2	1208	89	85
DL+SIDL1+SIDL2+LL2	Case 3	1164	223	92
DL+SIDL1+SIDL2+Seismic	Case 4	1254	210	86
DL+SIDL1+SIDL2+LL1+Seismic	Case 5	1254	210	86
DL+SIDL1+SIDL2+LL2+Seismic	Case 6	1210	201	379

Diameter of Pile	=	1.200	m
Shift	=	0.000	m
Vertical capacity of pile required	=	145.000	T
Lateral capacity of pile required	=	15.000	T

SKETCH OF PILE CAP



No. of piles = 9

Calculations

Pile No.	Dist.(L)	Dist. (T)	L^2	T^2	Z_L	Z_T
1	-3.600	-3.600	12.960	12.960	-21.600	-21.600
2	-3.600	0.000	12.960	0.000	-21.600	0.000
3	-3.600	3.600	12.960	12.960	-21.600	21.600
4	0.000	-3.600	0.000	12.960	0.000	-21.600
5	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	3.600	0.000	12.960	0.000	21.600
7	3.600	-3.600	12.960	12.960	21.600	-21.600
8	3.600	0.000	12.960	0.000	21.600	0.000
9	3.600	3.600	12.960	12.960	21.600	21.600

I_{LL} = **77.76** sqm

I_{TT} = **77.760** sqm

Normal case :- Lateral force

Case 1 **Horizontal force = 131.656 t Force per Pile = 14.6 t** **OK**

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total	Inference
No.				M_L/Z_L	M_T/Z_T	Load	
	(T)	(T)	(T)	(T)	(T)	(T)	
1	134.27	-4.14	-3.94	0.00	0.00	126.19	O.K
2	134.27	-4.14	0.00	0.00	0.00	130.13	O.K
3	134.27	-4.14	3.94	0.00	0.00	134.07	O.K
4	134.27	0.00	-3.94	0.00	0.00	130.33	O.K
5	134.27	0.00	0.00	0.00	0.00	134.27	O.K
6	134.27	0.00	3.94	0.00	0.00	138.22	O.K
7	134.27	4.14	-3.94	0.00	0.00	134.47	O.K
8	134.27	4.14	0.00	0.00	0.00	138.42	O.K
9	134.27	4.14	3.94	0.00	0.00	142.36	O.K

Case 2

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total	Inference
No.				M_L/Z_L	M_T/Z_T	Load	
	(T)	(T)	(T)	(T)	(T)	(T)	
1	134.27	-4.14	-3.94	0.00	0.00	126.19	O.K
2	134.27	-4.14	0.00	0.00	0.00	130.13	O.K
3	134.27	-4.14	3.94	0.00	0.00	134.07	O.K
4	134.27	0.00	-3.94	0.00	0.00	130.33	O.K
5	134.27	0.00	0.00	0.00	0.00	134.27	O.K
6	134.27	0.00	3.94	0.00	0.00	138.22	O.K
7	134.27	4.14	-3.94	0.00	0.00	134.47	O.K
8	134.27	4.14	0.00	0.00	0.00	138.42	O.K
9	134.27	4.14	3.94	0.00	0.00	142.36	O.K

Case 3

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total	Inference
No.				M_L/Z_L	M_T/Z_T	Load	
	(T)	(T)	(T)	(T)	(T)	(T)	
1	129.37	-10.32	-4.26	0.00	0.00	114.80	O.K
2	129.37	-10.32	0.00	0.00	0.00	119.06	O.K
3	129.37	-10.32	4.26	0.00	0.00	123.31	O.K
4	129.37	0.00	-4.26	0.00	0.00	125.12	O.K
5	129.37	0.00	0.00	0.00	0.00	129.37	O.K
6	129.37	0.00	4.26	0.00	0.00	133.63	O.K
7	129.37	10.32	-4.26	0.00	0.00	135.44	O.K
8	129.37	10.32	0.00	0.00	0.00	139.69	O.K
9	129.37	10.32	4.26	0.00	0.00	143.95	O.K

Seismic Case :- Lateral force

Case 4 **Horizontal force = 157.126 t Force per Pile = 17.5 t OK**

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total	Inference
No.				M_L/Z_L	M_T/Z_T	Load	
	(T)	(T)	(T)	(T)	(T)	(T)	
1	139.35	-9.74	-3.98	0.00	0.00	125.63	O.K
2	139.35	-9.74	0.00	0.00	0.00	129.61	O.K
3	139.35	-9.74	3.98	0.00	0.00	133.58	O.K
4	139.35	0.00	-3.98	0.00	0.00	135.37	O.K
5	139.35	0.00	0.00	0.00	0.00	139.35	O.K
6	139.35	0.00	3.98	0.00	0.00	143.32	O.K
7	139.35	9.74	-3.98	0.00	0.00	145.11	O.K
8	139.35	9.74	0.00	0.00	0.00	149.08	O.K
9	139.35	9.74	3.98	0.00	0.00	153.06	O.K

Case 5

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total	Inference
No.				M_L/Z_L	M_T/Z_T	Load	
	(T)	(T)	(T)	(T)	(T)	(T)	
1	139.35	-9.74	-3.98	0.00	0.00	125.63	O.K
2	139.35	-9.74	0.00	0.00	0.00	129.61	O.K
3	139.35	-9.74	3.98	0.00	0.00	133.58	O.K
4	139.35	0.00	-3.98	0.00	0.00	135.37	O.K
5	139.35	0.00	0.00	0.00	0.00	139.35	O.K
6	139.35	0.00	3.98	0.00	0.00	143.32	O.K
7	139.35	9.74	-3.98	0.00	0.00	145.11	O.K
8	139.35	9.74	0.00	0.00	0.00	149.08	O.K
9	139.35	9.74	3.98	0.00	0.00	153.06	O.K

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Case 6

Pile No.	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total Load	Inference
				M_L/Z_L	M_T/Z_T		
	(T)	(T)	(T)	(T)	(T)	(T)	
1	134.45	-9.32	-17.55	0.00	0.00	107.57	O.K
2	134.45	-9.32	0.00	0.00	0.00	125.12	O.K
3	134.45	-9.32	17.55	0.00	0.00	142.68	O.K
4	134.45	0.00	-17.55	0.00	0.00	116.89	O.K
5	134.45	0.00	0.00	0.00	0.00	134.45	O.K
6	134.45	0.00	17.55	0.00	0.00	152.00	O.K
7	134.45	9.32	-17.55	0.00	0.00	126.22	O.K
8	134.45	9.32	0.00	0.00	0.00	143.77	O.K
9	134.45	9.32	17.55	0.00	0.00	161.33	O.K

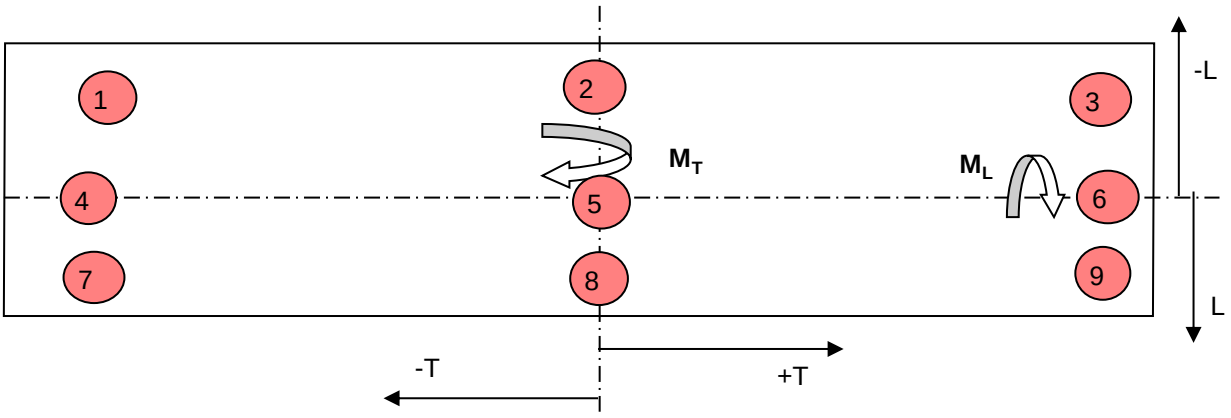
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Load calculation for design of pile (ULS)

		V	ML	MT
		t	t-m	t-m
Max. vertical load Combination 1	Case 1	1661	153	198
Max. longitudinal moment Combination 1	Case 2	1661	153	198
Max. transverse moment Combination 1	Case 3	1595	353	208
Max. vertical load Combination 2	Case 4	1242	54	173
Max. longitudinal moment Combination 2	Case 5	1242	54	173
Max. transverse moment Combination 2	Case 6	1184	227	182
Max. vertical load seismic/accidental combination	Case 7	1547	115	65
Max. longitudinal moment seismic/accidental combination	Case 8	1547	115	65
Max. transverse moment seismic/accidental combination	Case 9	1514	403	500

Diameter of Pile = 1.200 m
Shift = 0.000 m

SKETCH OF PILE CAP



No. of piles = 9

Calculations

Pile No.	Dist.(L)	Dist. (T)	L^2	T^2	Z_L	Z_T
1	-3.600	-3.600	12.960	12.960	-21.600	-21.600
2	-3.600	0.000	12.960	0.000	-21.600	0.000
3	-3.600	3.600	12.960	12.960	-21.600	21.600
4	0.000	-3.600	0.000	12.960	0.000	-21.600
5	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	3.600	0.000	12.960	0.000	21.600
7	3.600	-3.600	12.960	12.960	21.600	-21.600
8	3.600	0.000	12.960	0.000	21.600	0.000
9	3.600	3.600	12.960	12.960	21.600	21.600

I_{LL} = **77.76** sqm
I_{TT} = **77.760** sqm

Case 1

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	184.6	-7.1	-9.2	0.0	0.0	168.4
2	184.6	-7.1	0.0	0.0	0.0	177.5
3	184.6	-7.1	9.2	0.0	0.0	186.7
4	184.6	0.0	-9.2	0.0	0.0	175.4
5	184.6	0.0	0.0	0.0	0.0	184.6
6	184.6	0.0	9.2	0.0	0.0	193.7
7	184.6	7.1	-9.2	0.0	0.0	182.5
8	184.6	7.1	0.0	0.0	0.0	191.7
9	184.6	7.1	9.2	0.0	0.0	200.8

Case 2

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	184.6	-7.1	-9.2	0.0	0.0	168.4
2	184.6	-7.1	0.0	0.0	0.0	177.5
3	184.6	-7.1	9.2	0.0	0.0	186.7
4	184.6	0.0	-9.2	0.0	0.0	175.4
5	184.6	0.0	0.0	0.0	0.0	184.6
6	184.6	0.0	9.2	0.0	0.0	193.7
7	184.6	7.1	-9.2	0.0	0.0	182.5
8	184.6	7.1	0.0	0.0	0.0	191.7
9	184.6	7.1	9.2	0.0	0.0	200.8

Case 3

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	177.2	-16.3	-9.6	0.0	0.0	151.3
2	177.2	-16.3	0.0	0.0	0.0	160.9
3	177.2	-16.3	9.6	0.0	0.0	170.5
4	177.2	0.0	-9.6	0.0	0.0	167.6
5	177.2	0.0	0.0	0.0	0.0	177.2
6	177.2	0.0	9.6	0.0	0.0	186.9
7	177.2	16.3	-9.6	0.0	0.0	184.0
8	177.2	16.3	0.0	0.0	0.0	193.6
9	177.2	16.3	9.6	0.0	0.0	203.2

Case 4

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	138.0	-2.5	-8.0	0.0	0.0	127.5
2	138.0	-2.5	0.0	0.0	0.0	135.5
3	138.0	-2.5	8.0	0.0	0.0	143.5
4	138.0	0.0	-8.0	0.0	0.0	130.0
5	138.0	0.0	0.0	0.0	0.0	138.0
6	138.0	0.0	8.0	0.0	0.0	146.0
7	138.0	2.5	-8.0	0.0	0.0	132.5
8	138.0	2.5	0.0	0.0	0.0	140.5
9	138.0	2.5	8.0	0.0	0.0	148.5

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Case 5

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	138.0	-2.5	-8.0	0.0	0.0	127.5
2	138.0	-2.5	0.0	0.0	0.0	135.5
3	138.0	-2.5	8.0	0.0	0.0	143.5
4	138.0	0.0	-8.0	0.0	0.0	130.0
5	138.0	0.0	0.0	0.0	0.0	138.0
6	138.0	0.0	8.0	0.0	0.0	146.0
7	138.0	2.5	-8.0	0.0	0.0	132.5
8	138.0	2.5	0.0	0.0	0.0	140.5
9	138.0	2.5	8.0	0.0	0.0	148.5

Case 6

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	131.6	-10.5	-8.4	0.0	0.0	112.7
2	131.6	-10.5	0.0	0.0	0.0	121.1
3	131.6	-10.5	8.4	0.0	0.0	129.5
4	131.6	0.0	-8.4	0.0	0.0	123.2
5	131.6	0.0	0.0	0.0	0.0	131.6
6	131.6	0.0	8.4	0.0	0.0	140.0
7	131.6	10.5	-8.4	0.0	0.0	133.7
8	131.6	10.5	0.0	0.0	0.0	142.1
9	131.6	10.5	8.4	0.0	0.0	150.5

Case 7

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	171.9	-5.3	-3.0	0.0	0.0	163.6
2	171.9	-5.3	0.0	0.0	0.0	166.6
3	171.9	-5.3	3.0	0.0	0.0	169.6
4	171.9	0.0	-3.0	0.0	0.0	168.9
5	171.9	0.0	0.0	0.0	0.0	171.9
6	171.9	0.0	3.0	0.0	0.0	174.9
7	171.9	5.3	-3.0	0.0	0.0	174.3
8	171.9	5.3	0.0	0.0	0.0	177.3
9	171.9	5.3	3.0	0.0	0.0	180.3

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Case 8

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	171.9	-5.3	-3.0	0.0	0.0	163.6
2	171.9	-5.3	0.0	0.0	0.0	166.6
3	171.9	-5.3	3.0	0.0	0.0	169.6
4	171.9	0.0	-3.0	0.0	0.0	168.9
5	171.9	0.0	0.0	0.0	0.0	171.9
6	171.9	0.0	3.0	0.0	0.0	174.9
7	171.9	5.3	-3.0	0.0	0.0	174.3
8	171.9	5.3	0.0	0.0	0.0	177.3
9	171.9	5.3	3.0	0.0	0.0	180.3

Case 9

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	168.3	-18.6	-23.1	0.0	0.0	126.5
2	168.3	-18.6	0.0	0.0	0.0	149.6
3	168.3	-18.6	23.1	0.0	0.0	172.8
4	168.3	0.0	-23.1	0.0	0.0	145.1
5	168.3	0.0	0.0	0.0	0.0	168.3
6	168.3	0.0	23.1	0.0	0.0	191.4
7	168.3	18.6	-23.1	0.0	0.0	163.8
8	168.3	18.6	0.0	0.0	0.0	186.9
9	168.3	18.6	23.1	0.0	0.0	210.0

Moment on pile = 8.794 Q Ref. Appendix 2

	Combination 1			Combination 2			Seismic/Accidental Combination		
	P (t)	Q (t)	M (t.m)	P (t)	Q (t)	M (t.m)	P (t)	Q (t)	M (t.m)
MAX	**203.21**	17	147	**150.53**	14	125	**210.05**	23	198
MIN	**151.27**	17	147	**112.68**	14	125	**126.49**	23	198

Moments & shear force for pile cap

Width of abut. s	0.90	0.45	m			
Dia. Of pile =	1.20					
Length of Pile cap		8.70	m			
Width of Pile cap		8.70	m			
Thickness of pile cap		1.80	m			
Unit wt. of concrete		2.50	t/m3			
Unit wt. of Soil		2.00	t/m3			
Thickness of soil above pile cap		0.30	m	2.30	3.050	
Downward force & moment due to pile cap				2.30	0.00	

Longitudinal direction									
Self wt. of pile cap				3.05	8.70	1.80	2.50	119.41	182.10
Wt. due to soil				3.05	8.70	0.30	2.00	15.92	24.28
	Combination 1			Combination 2			Seismic/Accidental Combination		
	Max. Reaction from Pile	Shear (t)	Moment (t.m)	Max. Reaction from Pile	Shear (t)	Moment (t.m)	Max. Reaction from Pile	Shear (t)	Moment (t.m)
LL	581	51 /m	130 /m	426	33 /m	89 /m	561	49 /m	125 /m

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Verification of structural strength for pile cap
Calculation of Reinforcement

Width of section b = 8700 mm
Depth of section D = 1800 mm
Clear cover = 75

Moment on the section	**Moment at face of support**		
	Combination (1)	Seismic Combi.	Combination (2)
Actual moment (KNm)	11079	10627	7595
Grade of Concrete	30	30	30
Grade of steel fy	500	500	500
b	8700	8700	8700
D	1800	1800	1800
c	75	75	75
d	1687.5	1687.5	1687.5
f_{cd}	13.40	13.40	13.40
f_{Yd}	435	435	435
$xu_{,lim}/d$	0.62	0.62	0.62
$R_{sls} = M_{u,sls}/bd^2$	4.97	4.97	4.97
$M_{u,Lim}$ (KNm)	123233	123233	123233
	OK	**OK**	**OK**
Dia of bar (main tension) (mm)	25	25	25
Spacing (mm)	125	125	125
+ dia of bar (main tension) (mm)	0	0	0
Spacing (mm)	125	125	125
Ast provided (sq mm)	34165	34165	34165
Dia of bar(main compresion at top mm)	16	16	16
Spacing (mm)	125	125	125
Area of main compresion (mm^2)	13994	13994	13994
f_{ctm}	2.8	2.8	2.8
f_{yk}	435	435	435
cl. 16.6.1 (2) of IRC :112-2011			
$A_{S.min} = 0.26\ f_{ctm}\ b_t\ d\ /\ f_{yk} >= 0.0013\ b_t\ d$	24582	24582	24582
	OK	**OK**	**OK**
As.max = 0.025 Ac (main tension)	391500	391500	391500
cl. 16.5.1.1 (2) of IRC :112-2011	**OK**	**OK**	**OK**
As.max = 0.04 Ac (tension + compresion)	626400	626400	626400
Dia. of bar for (distribution)	25	25	25
S (mm)	125	125	125
A_{SW1}	982	982	982
	OK	**OK**	**OK**
No. of link for shear reinf.	16	16	16
Dia. of bar for shear reinf.	12	12	12
$S = Asw \times 0.9 \times d \times \cot\Theta \times fy\ /\ V_E$	200	200	200
A_{SW2}	1810	1810	1810
A_{SW}	2791	2791	2791
cl. 16.5.2(7) Eq. 16.6 of IRC :112-2011			
$S_{l.max} = 0.75\ d$	600	600	600
	OK	**OK**	**OK**
cl. 16.5.2(9) Eq. 16.8 of IRC :112-2011			
$S_{t.max} = 0.75\ d <= 600mm$	600	600	600
$S_{t.}$ mm	483	483	483
	OK	**OK**	**OK**
z (mm)	1622	1622	1622
f_{ywd}	348	348	348
cl. 10.3.3.2 Eq. 10.7 of IRC :112-2011			
$V_{Rd.s} = A_{SW}\ z\ f_{ywd}\ \cot\Theta\ /\ S$ (KN)	19686	19686	19686
	OK	**OK**	**OK**

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Combination for design of foundation (SLS Check)

Forces and moments are calculated at Pile cap bottom level

Following load combinations are con As per Annex B of IRC:112-2011

1 Rare Combination

2 Quasi-Perma. Combination

Skew Angle 0 cosθ 1.000 sinθ 0.000

		P (t)	ML (tm)	MT (tm)	HL(t)	HT(t)
1	DL Dead Load (due to superstructure+substructure+foundation)	1085	-597	0	0	0
2	DL_SUP	215	249	0	0	0
3	ABUT	870	-846	0	0	0
4	ABUT + RETURN WALL	0	0	0	0	0
5	FDN	0	0	0	0	0
6	SOIL ABOVE FDN	0	0	0	0	0
7	SIDL1 Super-imposed Dead Load due to wearing course	20	23	0	0	0
8	SIDL2 Super-imposed Dead Load other than wearing course	20	24	0	0	0
9	LL1 Live Load Maximum Reaction Case	81	94	85	0	0
10	LL2 Live Load Maximum Longitudinal Moment Case	81	94	85	0	0
11	LL3 Live Load Maximum Transverse Moment Case	39	45	92	0	0
12	LL1B Live Load Moment due to braking	2	85	0	21	0
13	LL2B Live Load Moment due to braking	2	85	0	21	0
14	LL3B Live Load Moment due to braking	0	0	0	0	0
15	W1 Wind vertically down without LL	10.40	7.25	34.47	1.18	7.14
16	W2 Wind vertically up without LL	-10.40	7.25	34.47	1.18	7.14
17	W3 Wind vertically down with LL	10.40	13.01	77.64	2.18	13.08
18	W4 Wind vertically up with LL	-10.40	13.01	77.64	2.18	13.08
19	HFL	-565	-550	0	0	0
20	Active earth pressure LWL	0	189	0	78	0
21	LL surcharge LWL	0	93	0	32	0
22	Active earth pressure HFL	0	94	0	39	0
23	LL surcharge HFL	0	47	0	16	0

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Note:- In combination 1 & 2 wind is taken as accompanying load and LL is taken as the leading load.

	Rare Combination				Quasi-Perma. Combination			
	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)
DL	1.00	1085	-597	0	1.00	1085	-597	0
SIDL1	1.00	20	23	0	1.00	20	23	0
SIDL2	1.00	20	24	0	1.00	20	24	0
LL1	1.00	81	94	85	0.00	0	0	0
LL2	1.00	81	94	85	0.00	0	0	0
LL3	1.00	39	45	92	0.00	0	0	0
LL1B	1.00	2	85	0	0.00	0	0	0
LL2B	1.00	2	85	0	0.00	0	0	0
LL3B	1.00	0	0	0	0.00	0	0	0
W1	0.60	6	4	21	0.00	0	0	0
W2	0.60	-6	4	21	0.00	0	0	0
W3	0.60	6	8	47	0.00	0	0	0
W4	0.60	-6	8	47	0.00	0	0	0
HFL	1.00	-565	-550	0	1.00	-565	-550	0
Active ear	1.00	0	189	0	1.00	0	189	0
LL surcha	0.80	0	75	0	0.00	0	0	0
Active ear	1.00	0	94	0	1.00	0	94	0
LL surcha	0.80	0	37	0	0.00	0	0	0

1 Rare Combination

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2	77.9	0.0	1125	-362	0
Case 2	DL+SIDL1+SIDL2+LL1	125.2	0.0	1208	-108	85
Case 3	DL+SIDL1+SIDL2+LL2	131.7	0.0	1208	-108	85
Case 4	DL+SIDL1+SIDL2+LL3	103.8	0.0	1164	-242	92
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	78.6	4.3	1131	-357	21
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	126.5	7.8	1215	-100	132
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	133.0	7.8	1215	-100	132
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	105.1	7.8	1171	-234	139
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	78.6	4.3	1119	-357	21
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	126.5	7.8	1202	-100	132
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	133.0	7.8	1202	-100	132
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	105.1	7.8	1158	-234	139
HFL case						
Case 13	DL+SIDL1+SIDL2 + HFL	38.9	0.0	560	-1006	0
Case 14	DL+SIDL1+SIDL2+LL1+ HFL	73.3	0.0	643	-789	85
Case 15	DL+SIDL1+SIDL2+LL2+ HFL	73.3	0.0	643	-789	85
Case 16	DL+SIDL1+SIDL2+LL3+ HFL	51.9	0.0	599	-923	92
Downward Wind Condition						
Case 17	DL+SIDL1+SIDL2 + Wind + HFL	39.7	4.3	566	-1001	21
Case 18	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	74.6	7.8	650	-782	132
Case 19	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	74.6	7.8	650	-782	132
Case 20	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	53.2	7.8	606	-915	139
Upward Wind Condition						
Case 21	DL+SIDL1+SIDL2 + Wind + HFL	39.7	4.3	554	-1001	21
Case 22	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	74.6	7.8	637	-782	132
Case 23	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	74.6	7.8	637	-782	132
Case 24	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	53.2	7.8	593	-915	139

133	8	1215	-100	132
126	8	1215	-100	132
		1171	-234	139

2 Quasi-Perma. Combination

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2	77.9	0.0	1125	-362	0
Case 2	DL+SIDL1+SIDL2+LL1	77.9	0.0	1125	-362	0
Case 3	DL+SIDL1+SIDL2+LL2	77.9	0.0	1125	-362	0
Case 4	DL+SIDL1+SIDL2+LL3	77.9	0.0	1125	-362	0
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	77.9	0.0	1125	-362	0
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	77.9	0.0	1125	-362	0
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	77.9	0.0	1125	-362	0
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	77.9	0.0	1125	-362	0
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	77.9	0.0	1125	-362	0
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	77.9	0.0	1125	-362	0
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	77.9	0.0	1125	-362	0
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	77.9	0.0	1125	-362	0
HFL case						
Case 13	DL+SIDL1+SIDL2 + HFL	38.9	0.0	560	-1006	0
Case 14	DL+SIDL1+SIDL2+LL1+ HFL	38.9	0.0	560	-1006	0
Case 15	DL+SIDL1+SIDL2+LL2+ HFL	38.9	0.0	560	-1006	0
Case 16	DL+SIDL1+SIDL2+LL3+ HFL	38.9	0.0	560	-1006	0
Downward Wind Condition						
Case 17	DL+SIDL1+SIDL2 + Wind + HFL	38.9	0.0	560	-911	0
Case 18	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	38.9	0.0	560	-1006	0
Case 19	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	38.9	0.0	560	-1006	0
Case 20	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	38.9	0.0	560	-1006	0
Upward Wind Condition						
Case 21	DL+SIDL1+SIDL2 + Wind + HFL	38.9	0.0	560	-1006	0
Case 22	DL+SIDL1+SIDL2+LL1+ Wind+ HFL	38.9	0.0	560	-1006	0
Case 23	DL+SIDL1+SIDL2+LL2+ Wind+ HFL	38.9	0.0	560	-1006	0
Case 24	DL+SIDL1+SIDL2+LL3+ Wind+ HFL	38.9	0.0	560	-1006	0

78	0	1125	-362	0
78	0	1125	-362	0
		1125	-362	0

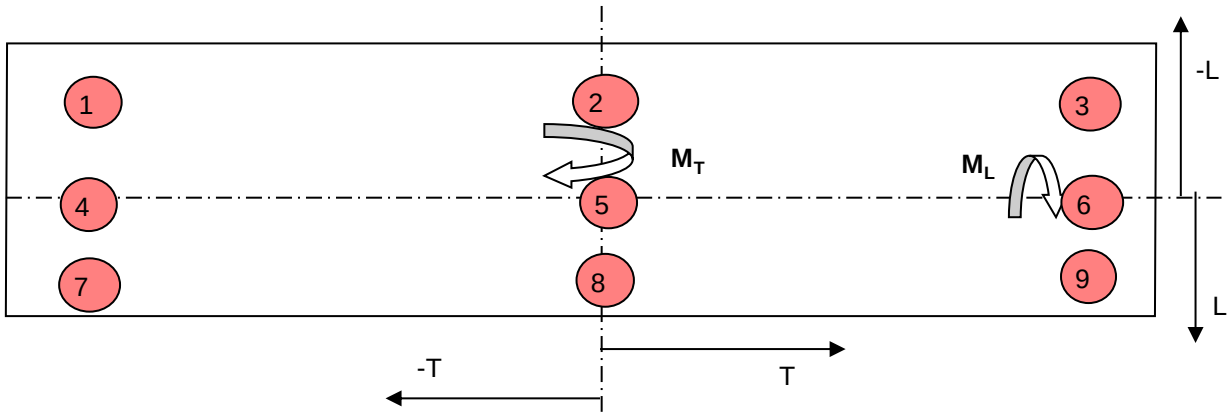
	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
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Load on pile for SLS

	V	ML	MT
	t	t-m	t-m
Case 1	1215	-100	132
Case 2	1215	-100	132
Case 3	1171	-234	139
Case 4	1125	-362	0
Case 5	1125	-362	0
Case 6	1125	-362	0

Diameter of Pile = 1.200 m
Shift = 0.000 m

SKETCH OF PILE CAP



No. of piles = 9

Calculations

Pile No.	Dist.(L)	Dist. (T)	L^2	T^2	Z_L	Z_T
1	-3.600	-3.600	12.960	12.960	-21.600	-21.600
2	-3.600	0.000	12.960	0.000	-21.600	0.000
3	-3.600	3.600	12.960	12.960	-21.600	21.600
4	0.000	-3.600	0.000	12.960	0.000	-21.600
5	0.000	0.000	0.000	0.000	0.000	0.000
6	0.000	3.600	0.000	12.960	0.000	21.600
7	3.600	-3.600	12.960	12.960	21.600	-21.600
8	3.600	0.000	12.960	0.000	21.600	0.000
9	3.600	3.600	12.960	12.960	21.600	21.600

I_{LL} = **77.76** sqm
I_{TT} = **77.760** sqm

Case 1

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	135.0	4.6	-6.1	0.0	0.0	133.5
2	135.0	4.6	0.0	0.0	0.0	139.6
3	135.0	4.6	6.1	0.0	0.0	145.7
4	135.0	0.0	-6.1	0.0	0.0	128.9
5	135.0	0.0	0.0	0.0	0.0	135.0
6	135.0	0.0	6.1	0.0	0.0	141.1
7	135.0	-4.6	-6.1	0.0	0.0	124.2
8	135.0	-4.6	0.0	0.0	0.0	130.3
9	135.0	-4.6	6.1	0.0	0.0	136.4

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Case 2

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	135.0	4.6	-6.1	0.0	0.0	133.5
2	135.0	4.6	0.0	0.0	0.0	139.6
3	135.0	4.6	6.1	0.0	0.0	145.7
4	135.0	0.0	-6.1	0.0	0.0	128.9
5	135.0	0.0	0.0	0.0	0.0	135.0
6	135.0	0.0	6.1	0.0	0.0	141.1
7	135.0	-4.6	-6.1	0.0	0.0	124.2
8	135.0	-4.6	0.0	0.0	0.0	130.3
9	135.0	-4.6	6.1	0.0	0.0	136.4

Case 3

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	130.1	10.8	-6.4	0.0	0.0	134.5
2	130.1	10.8	0.0	0.0	0.0	140.9
3	130.1	10.8	6.4	0.0	0.0	147.3
4	130.1	0.0	-6.4	0.0	0.0	123.7
5	130.1	0.0	0.0	0.0	0.0	130.1
6	130.1	0.0	6.4	0.0	0.0	136.5
7	130.1	-10.8	-6.4	0.0	0.0	112.8
8	130.1	-10.8	0.0	0.0	0.0	119.2
9	130.1	-10.8	6.4	0.0	0.0	125.7

Case 4

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	125.0	16.7	0.0	0.0	0.0	141.8
2	125.0	16.7	0.0	0.0	0.0	141.8
3	125.0	16.7	0.0	0.0	0.0	141.8
4	125.0	0.0	0.0	0.0	0.0	125.0
5	125.0	0.0	0.0	0.0	0.0	125.0
6	125.0	0.0	0.0	0.0	0.0	125.0
7	125.0	-16.7	0.0	0.0	0.0	108.3
8	125.0	-16.7	0.0	0.0	0.0	108.3
9	125.0	-16.7	0.0	0.0	0.0	108.3

Case 5

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	125.0	16.7	0.0	0.0	0.0	141.8
2	125.0	16.7	0.0	0.0	0.0	141.8
3	125.0	16.7	0.0	0.0	0.0	141.8
4	125.0	0.0	0.0	0.0	0.0	125.0
5	125.0	0.0	0.0	0.0	0.0	125.0
6	125.0	0.0	0.0	0.0	0.0	125.0

7	125.0	-16.7	0.0	0.0	0.0	108.3
8	125.0	-16.7	0.0	0.0	0.0	108.3
9	125.0	-16.7	0.0	0.0	0.0	108.3

Case 6

Pile	P_T / N	M_L / Z_L	M_T / Z_T	Due to Shift		Total
No.				M_L/Z_L	M_T/Z_T	Load
	(T)	(T)	(T)	(T)	(T)	(T)
1	125.0	16.7	0.0	0.0	0.0	141.8
2	125.0	16.7	0.0	0.0	0.0	141.8
3	125.0	16.7	0.0	0.0	0.0	141.8
4	125.0	0.0	0.0	0.0	0.0	125.0
5	125.0	0.0	0.0	0.0	0.0	125.0
6	125.0	0.0	0.0	0.0	0.0	125.0
7	125.0	-16.7	0.0	0.0	0.0	108.3
8	125.0	-16.7	0.0	0.0	0.0	108.3
9	125.0	-16.7	0.0	0.0	0.0	108.3

Moment on pile = 8.794 Q Ref. Appendix 2

	Rare Combination			Quasi-Perma. Combination		
	P (t)	Q (t)	M (t.m)	P (t)	Q (t)	M (t.m)
MAX	**147.31**	15	130	**141.77**	9	76
MIN	**112.83**	15	130	**108.28**	9	76

Moments & shear force for pile cap

Width of abut. s	0.90		0.45	m		
Dia. Of pile =	1.20					
Length of Pile cap			8.70	m		
Width of Pile cap			8.70	m		
Thickness of pile cap			1.80	m		
Unit wt. of concrete			2.50	t/m3		
Unit wt. of Soil			2.00	t/m3		
Thickness of soil above pile cap			0.30	m		
Downward force & moment due to pile cap			2.30		3.05	

Longitudinal direction						
Self wt. of pile cap	3.05	8.70	1.80	2.50	119.41	182.10
Wt. due to soil	3.05	8.70	0.30	2.00	15.92	24.28

	Rare Combination			Quasi-Perma. Combination		
	Max. Reaction from Pile	Shear (t)	Moment (t.m)	Max. Reaction from Pile	Shear (t)	Moment (t.m)
LL	423	33 /m	88 /m	425	33 /m	89 /m
Max		287	766		290	772

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Verification for serviceability limit state for pile cap
Calculation of Reinforcement

Width of section b = 8700 mm
Depth of section d = 1800 mm
Clear cover = 75

Moment on the section	**Pile cap bottom**		
	For Rare Combination		For Quasi-Perma. Combination
Actual moment (KNm)	7512.2		7572.0
b	8700		8700
D	1800		1800
c	75		75
d	1687.5		1687.5
f_{cd}	16.80		12.60
f_{Yd}	300		300
xu,sls/d	0.70		0.70
$R_{sls} = M_{u,sls}/bd^2$	4.52		3.39
$M_{u,sls}$ (KNm)	112024		84018
	OK		**OK**
Ast Req.	15020		15204
Dia of bar (main tension) (mm)	25		25
Spacing (mm)	125		125
+ dia of bar (main tension) (mm)	0		0
Spacing (mm)	125		125
Ast provided (sq mm)	34165		34165
Dia of bar (main compresion) (mm)	16		16
Spacing (mm)	125		125
Area of main compresion (mm^2)	13994		13994
f_{ctm}	2.5		2.5
x (mm)	140.2		187.0
x/d	0.083		0.111
	OK		**OK**
z (mm)	1641		1626
MR_{sls} (KNm)	16822		16663
	OK		**OK**
$\sigma_{sc} = M/(A_s z)$	134		136
	OK		**OK**
$\sigma_{ca} = M/(0.8095 z b x_u)$	7.50		5.73
	OK		**OK**

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Calculation of crack width	**Pile cap bottom**		
n_1			8
n_2			8
$\phi_{eq} = (n_1\phi_1^2 + n_2 \phi_2^2) / (n_1\phi_1+n_2\phi_2)$			25
cl. 12.3.4 (3) of IRC :112-2011			
c			75
k1			0.8
k2			0.50
For skew slab refer eq. 12.10 of IRC :112-2011			
$\rho_{p.eff} = A_s / A_{c,eff}$			0.018
$S_{r,max} = \{ 3.4 c + (0.425 k_1 k_2 \phi) / \rho_{p.eff} \}$			492
cl. 12.3.4 (3) of IRC :112-2011			
k_t			0.5
$f_{ct,eff}$			2.90
E_s			200000
E_{cm}			31000
$\alpha_e = E_s / E_{cm}$			6.45
$(\varepsilon_{sm}-\varepsilon_{cm})=(\sigma_{sc}-k_t f_{ct,eff}(1+\alpha_e \rho_{p,eff})/\rho_{p,eff})/E_s >=0.6\sigma_{sc}/E_s$			0.0004
cl. 12.3.4 (2) of IRC :112-2011			
$W_k = S_{r,max} (\varepsilon_{sm} - \varepsilon_{cm})$			0.20
cl. 12.3.4 (1) of IRC :112-2011			
			OK

1.15 Check for safety of section for pile

LOADS & MOMENTS			
P	KN	1418	1083
M	KNm	761	761
Dia. Of section	mm	1200	1200
Radius of section	mm	600	600
Effective cover c		100	100
R - c		500	500
Modular ratio		10	10
No. of bar		30	30
Dia. Of bar	mm	25	25
Area of steel	mm^2	14726.22	14726.22
% of steel		1.30	1.30
Area of concrete section	mm^2	1130973	1130973
Net area of concrete	mm^2	1116247	1116247
Transformed area = Ac + m. As =	mm^2	1263509	1263509
Transformed M. I. = Ig + (m - 1) * As * (R - c) ^2	mm^4	134921586995	134921586995
Section modulus	mm^3	224869312	224869312
Actual stress	Mpa		
σ co cal (P / A)		1.12	0.86
σ cbc,cal (M / Z)		3.38	3.38
Permissible stress		35	35
σ co	Mpa	8.75	8.75
σ cbc	Mpa	11.67	11.67
Check for safety of section			
σ co cal + σ cbc,cal		0.42	0.39
σ co σ cbc		OK	OK
CHECK FOR CRACKED / UNCRACKED SECTION			
σ co cal - σ cbc,cal		-2.26	-2.53
Permissible tensile stress in concrete		-3.83	-3.83
Section is to be design as a		UNCRACKED	UNCRACKED

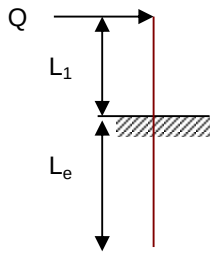
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	Client	MP PWD		Checked by:	-
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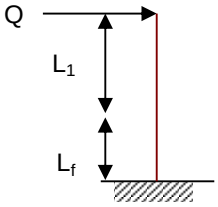
Appendex 1

	Ah	=	(Z/2)*(Sa/g)*(I/R)					
For this region								
Seismic Zone	II							
	Z	=	0.1					
	I	=	1.2					
(RCC Column with out Ductile Detailing)	R	=	2.5					
For founding on hard soil	I	Type of soil						
	Sa/g	=	2.50			2.50		
	Ah	**=**	**0.060**			**0.060**		

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Appendex 2





Calculating Depth of Fixity :

$T = 5\sqrt{(EI/K_1)}$, and $R = 4\sqrt{(EI/K_2)}$

E	=	31000	MN/m^2
I	=	0.1018	m^4
K_1	=	3.2	MN/m^3
K_2	=	9	MN/m^3

T	=	3.970	**OK**
R	=	4.134	

Pile length above sand layer , L_1 = 12.01 m

∴ Case(I) - L_1/R = 2.905

∴ Case(II) - L_1/T = 3.025

Case(I) - Corresponding value of L_f/R = 2.1

Case(II) - Corresponding value of L_f/T = 2.1

∴ Case(I) - Depth of fixity, L_f = 8.337 m = **8.337** m

∴ Case(II) - Depth of fixity, L_f = 8.682 m = **8.682** m

$L_1 + L_f$ = 20.692 m

Reduction factor m = 0.85

M_T = $Q (L_1 + L_f)/2*m$ (for fixed head pile)

= 8.794 Q

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

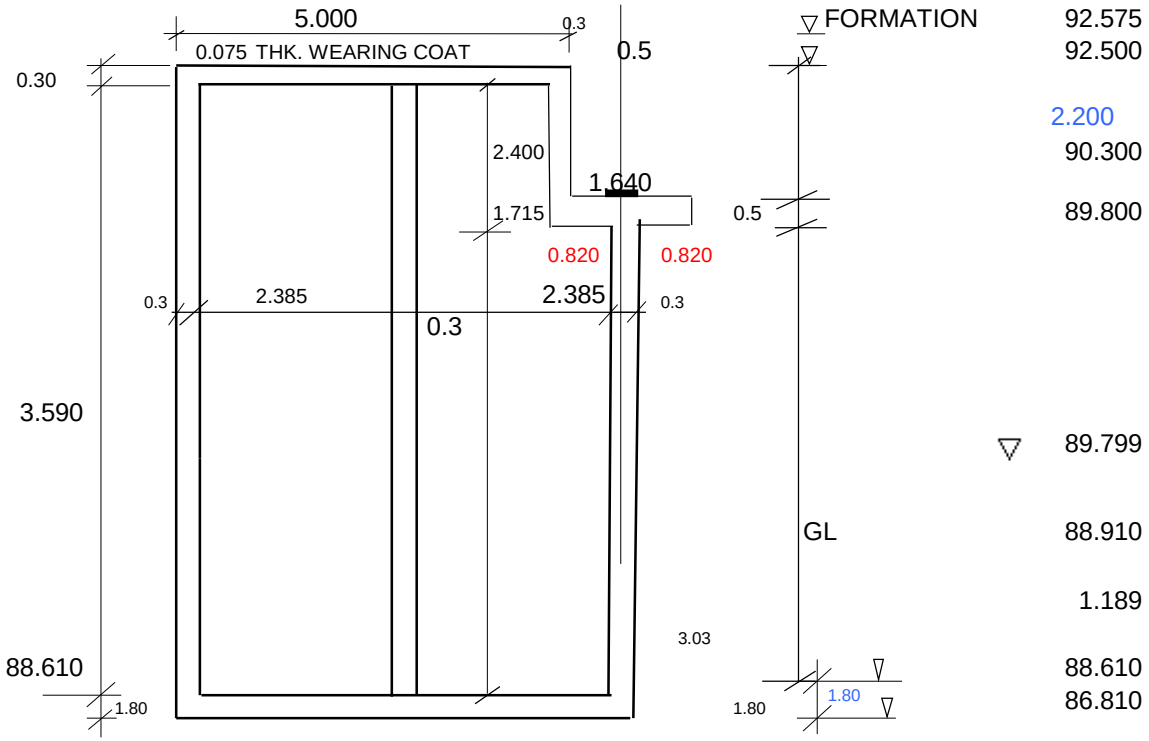
DESIGN OF ABUTMENT A1 & A2

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

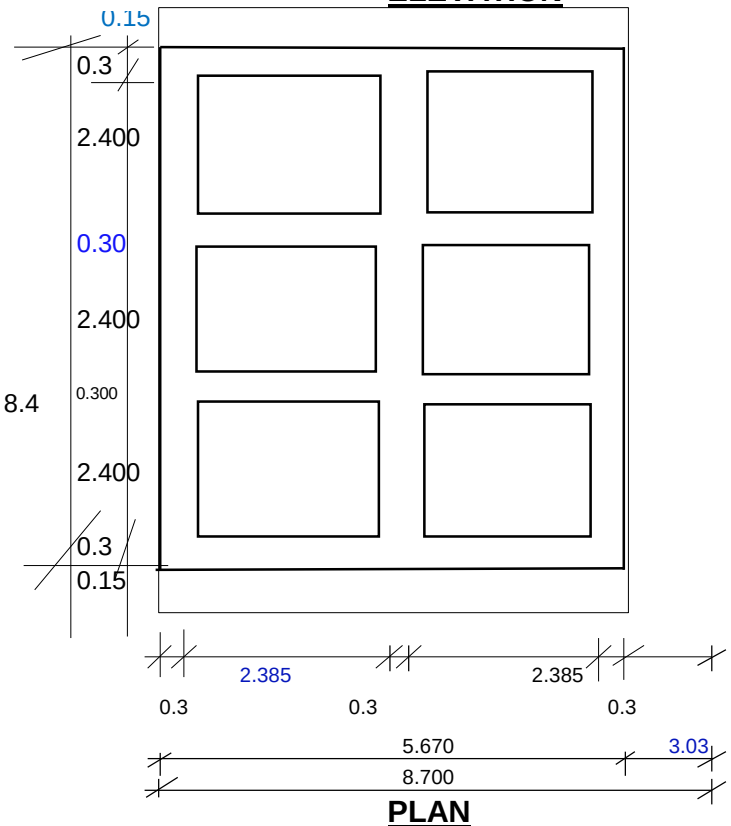
Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

DESIGN OF ABUTMENT for LWL CASE A1



ELEVATION



PLAN

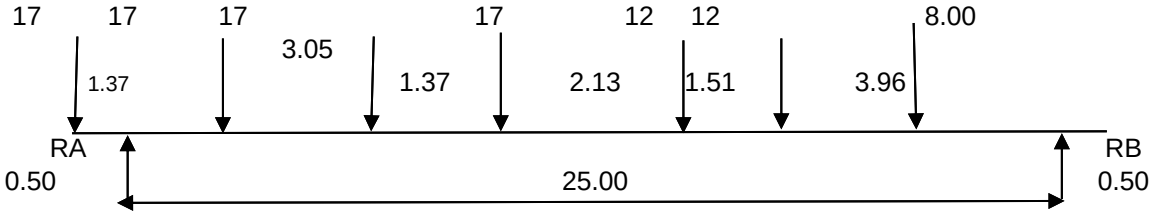
	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF ABUTMENT A1 & A2	Date & Rev.	29-06-2023

(As per STAAD Output)

26 m span Reactions (KN)			
Bearing No.	**DL**	**SIDL1 (WC)**	**SIDL2(CB)**
	2106	**197**	**200**

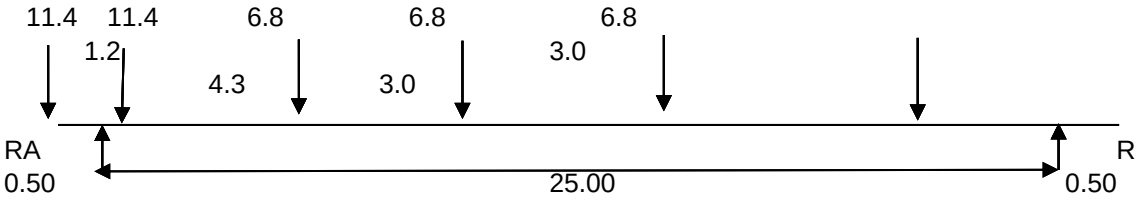
Live load

a) 70 R 1lane loading



Ra		*		25.00				
		4.00	*	17.00	*	22.61	=	1537.1
		2.00	*	12.00	*	16.83	=	403.80
		1.00	*	8.00	*	12.11	=	96.88
							Total :	2037.8
Ra		=	2037.82	/	25.00		=	**81.5** **t**
			Impact	1.145				**93.3** **t**
								18.49 **t**

b) Class A 1 lane loading



Ra		*	25.0 =					
		3	*	6.8	*	17.00	=	347
		2	*	11.4	*	24.9	=	568
							Total :	915
	Ra =						=	**36.6** **t**
				Impact	1.145			**41.9** **t**
								13.4 **t**

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.
Client:- MP PWD
DESIGN OF ABUTMENT A1 & A2

TAKE MOMENTS ABOUT TOE (PT. A)

SL. NO.	ITEM	UNIT WT. T / m^3	WEIGHT T.	L. A. m	MOMENT T.m
$Z_{(LONG.)}$					
1	FIXED ITEMS (RAILING ETC.)				
	2.0 T/M X 5.000	-	10	1.8500	18.50
2	APPROACH SLAB I/C WEARING COAT				
	5.000 X 8.4 X 0.375	2.5	39.38	1.850	72.84
3	BEARING BEAM				
	8.4 X 1.640 X 0.5	2.5	17.22	-1.170	-20.15
4	SIDE WALLS				
	2 X 5.67 X 1.189 X 0.3	2.5	10.11	1.52	15.32
	2 X 5.67 X 0.001 X 0.3	2.5	0.01	1.52	0.01
	2 X 4.700 X 2.400 X 0.3	2.5	16.92	2.00	33.84
5	INTERMEDIATE WALLS				
	1 X 7.80 X 1.189 X 0.30	2.5	6.96	1.515	10.54
	1 X 7.80 X 2.401 X 0.3	2.5	14.05	1.515	21.28
	1 X 4.77 X 1.189 X 0.30	2.5	4.25	1.5150	6.44
	1 X 4.77 X 0.001 X 0.30	2.5	0.00	1.5150	0.01
	1 X 2.385 X 2.400 X 0.30	2.5	4.29	2.858	12.27
	1 X 1.72 X 2.400 X 0.30	2.5	3.09	0.508	1.57
6	FRONT WALL				
	1 X 7.80 X 1.189 X 0.3	2.5	6.96	-1.170	-8.14
	1 X 7.80 X 0.001 X 0.3	2.5	0.01	-1.170	-0.01
	1 X 7.80 X 1.900 X 0.3	2.5	11.12	-0.500	-5.56
7	REAR WALL				
	1 X 7.80 X 1.189 X 0.3	2.5	6.96	4.200	29.21
	1 X 7.80 X 0.001 X 0.3	2.5	0.01	4.200	0.02
	1 X 7.80 X 2.400 X 0.3	2.5	14.04	4.200	58.97
		Σ	165.35	Σ	246.97

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

TAKE MOMENTS ABOUT TOE (PT. A)

SL. NO.	ITEM	UNIT WT. T / m ^{3}	WEIGHT T	L.A. m	MOMENT T.m
			165.35		246.97
10	BASE SLAB				
	1 X 8.70 X 1.800 X 8.70	2.5	340.61	0.000	0.00
11	EARTH FILLING upto RL 92.050				
	6 X 2.385 X 2.40 X 1.189	2	81.67	1.515	123.73
	6 X 2.385 X 2.40 X 0.001	2	0.07	1.515	0.10
	3 X 2.385 X 2.40 X 2.400	2	82.43	2.858	235.53
	3 X 2.385 X 1.72 X 2.400	2	58.90	0.508	29.89
	2 X 8.70 X 0.15 X 3.89	2	20.31	0.00	0.00
		TOTAL	870.10		846.31

C.G. = 846.31 / 870.10 = 0.973 M

= 0.973 M < b/6 = 1.450 **OK** 846.31

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

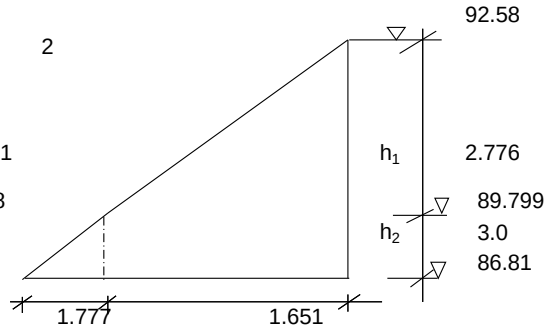
Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

5. EARTH PRESSURE :-

USE θ = 30 ^{o} δ = 20.000 ^{o} γ = 2

Kah= 0.297 Kav = 0.102



i) ACTIVE EARTH PRESSURE :

h_1 = 0.297 X 2 X 2.776 = 1.651

h_2 = 0.297 X 2 X 2.99 = 1.78

a) HORIZONTAL FORCE : -

$$F_h = \left[\frac{1.651 \times 2.78}{2} + 3.0 \times 1.651 + \frac{1.777 \times 3.0}{2} \right] \times 8.4$$

$$= [\ 2.3 + 4.93 + 2.66\ ] \times 8.4$$

$$= 83.00 \text{ T}$$

$$Fv = 83.0 \times \frac{0.102}{0.297} = 28.39 \text{ T}$$

b) OVERTURNING MOMENT DUE TO HORIZONTAL FORCES : -

$$M_{OT} = \left[2.3 \times (0.42 \times 2.78 + 3.0) + 4.93 \times \frac{3.0}{2} + 2.66 \times 3.0 \times 0.42 \right] \times 8.4$$

$$= [\ 9.5 + 7.37 + 3.33\ ] \times 8.40$$

$$= 169.91 \text{ T.m}$$

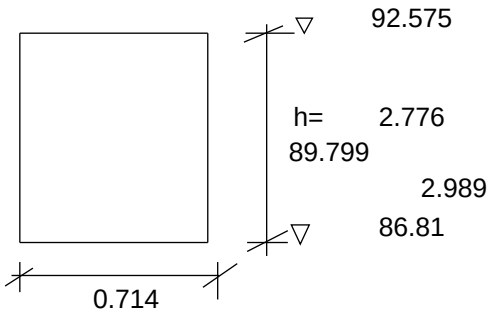
c) STABILIZING MOMENT DUE TO VERTICAL COMPONENT OF EARTH PRESSURE : -

M_{ST}= 28.39 x 8.7 = 247.0 T.m

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2



6. DUE TO L.L. SURCHARGE
(Ref. IRC 78-2000 cl.710.4.4)

h_3 = 0.29731 X 2 X 1.2

= 0.7136 T/m^2

0.29731 x 2 x 1.2

= 0.71355 T/m^2

a) HORIZONTAL FORCE

F_H = 0.7136 X 2.776 X 8.4 + 0.71 x 3 x 8.4

= 34.55 T

Fv = 34.6 x $\frac{0.102}{0.297}$ = 11.82 T

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

USE JENSEN'S FORMULA FOR INTERNAL EARTH PRESSURE

$$P_h = \frac{\gamma \times r}{\mu'}\left[1 - \frac{e^{-K\mu' h}}{r}\right]$$

γ = DENSITY = 2 T/m^3

r = HYDRAULIC MEAN DEPTH (A/P)

$$= \frac{2.39 \times 2.4}{2\ (\ 2.39 + 2.4\ )} = 0.60$$

$$K = \frac{1 - \sin\phi}{1 + \sin\phi}, \quad \phi = 35 \quad K = 0.33$$

μ' = $\tan\delta$, δ = 23.33 (ANGLE OF FRICTION) = 0.224

$$Pi = \frac{1.8 \times 0.598}{1\ 0.224}\left[1 - e^{-\left(\frac{0.33 \times 0.224 \times h}{0.598}\right)}\right]$$

$$= 4.806\left[1 - e^{-0.1236\ h}\right]$$

VALUES ARE TABULATED :

h(m)	Pi	LEVEL
3.965	1.86	88.610
3.816	1.81	88.759
3.668	1.75	88.908
3.519	1.69	89.056
3.370	1.64	89.205
3.221	1.58	89.354
3.072	1.52	89.503
2.924	1.46	89.651
2.775	1.40	89.800
1.388	0.76	91.188
Top RL		92.6

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD
DESIGN OF ABUTMENT A1 & A2

SIDE WALL :-

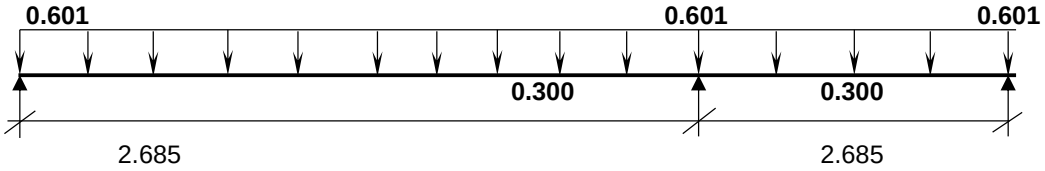
WALL IS SUBJECTED TO INTERNAL PRESSURE (I.E.P.) OF Pi AND EXTERNAL EARTH PRESSURE (E.E.P.) INCLUDING SURCHARGE LOAD OF Pe.

LOADS AND MOMENTS ARE CALCULATED AT APPROX. 2.5 TO 3M HEIGHT.

E.E.P. (FROM PRESSURE DIAG.)

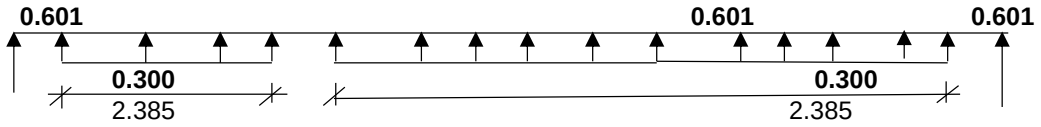
FROM ELEV. 92.575 TO 88.610 = 0.448 h_1 + 0.71355

MOMENTS DUE TO THE UNIT LOAD (REFER APPENDIX 1)



MOMENTS DUE TO THE UNIT LOAD FOR E.E.P.

TAKEN SAME AS E.E.P.



MOMENT DUE TO THE UNIT LOAD FOR I.E.P.

MOMENT M = UNIT LOAD MOMENT. x Pi OR (Pi - Pe)

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

TABLE FOR SIDE WALL MOMENTS

(SIDE WALL CONTD.)

LEVEL	P_i max	P_e max	$P_e - P_i$		ONLY INT. EARTH PRESS.					INT. + EXT EARTH PRESS.			
					AT END SUPT.	AT MID SUPT.		END SPAN	MID SPAN	AT END SUPT.	AT MID SUPT.	END SPAN	MID SPAN
					FOR UNIT LOAD OF 0.60	FOR UNIT LOAD OF 0.60		FOR UNIT LOAD OF 0.300	FOR UNIT LOAD OF 0.300	FOR UNIT LOAD OF 0.60	FOR UNIT LOAD OF 0.60	FOR UNIT LOAD OF 0.300384	FOR UNIT LOAD OF 0.300384
					INNER FACE	INNER FACE		OUTER FACE	OUTER FACE	OUTER FACE	OUTER FACE	INNER FACE	INNER FACE
88.610 TO 88.908 (3.97 - 3.67)	1.86	2.49	0.63	MOMENT	1.12	1.12		0.56	0.56	0.38	0.38	0.19	0.19
88.908 TO 89.205 (3.67 - 3.37)	1.75	2.36	0.60	MOMENT	1.05	1.05		0.53	0.53	0.36	0.36	0.18	0.18
89.205 TO 89.503 (3.37 - 3.07)	1.64	2.22	0.59	MOMENT	0.98	0.98		0.49	0.49	0.35	0.35	0.18	0.18
89.503 TO 89.800 (3.07 - 2.78)	1.52	2.09	0.57	MOMENT	0.91	0.91		0.46	0.46	0.34	0.34	0.17	0.17
89.800 TO 92.575 (2.78 - 0)	1.40	1.96	0.56	MOMENT	0.84	0.84		0.42	0.42	0.34	0.34	0.17	0.17

AT Depth

(3.97 - 3.67) 300 x 300 MM Haunch

Grade of concrete M 30 $\beta 1$ = 0.810 $\beta 2$ = 0.41597 fcd 13.40 M 16 KNm

Grade of steel Fe 500 cover 50 mm

depth 300 d 244 x = 6.3 mm < x lim = 150 mm Z= 241

Width 1000 mm Ast. Req. = 157 mm^2 **OK**

$\rho 1$ = 0.00232 < 0.02 **OK** 565 Vertical reinforcement 0.2

K= 1.90536 **12** @ **200** **565** mm^2/m **OK** F 16 KN **12** @ **200** **OK**

Axial Load = 0 σ_{cp} = 0 V_{Rdc} 98 KN

$V_{Rdc,min}$ 109 KN **OK**

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD
DESIGN OF ABUTMENT A1 & A2

AT Depth
(3.67 - 3.37) 300 x 300 MM Haunch

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		15	KNm			
Grade of steel	Fe	500			cover	50	mm											
depth			300		d		244		x =		5.9	mm	<	x lim =		150	mm	Z=	242
Width			1000	mm				Ast. Req. =		147	mm^2			**OK**				
$\rho 1$	=		0.00232	<		0.02	**OK**			565							Vertical reinforcement		0.2
K=			1.90536			**12**	@	**200**		**565**	**mm^2/m**	**OK**		F		15	KN	**12**	@ **200** **OK**
Axial Load =		0		σ_{cp} =	0		V_{Rdc}		98	KN									
							$V_{Rdc,min}$		109	KN	**OK**								

AT Depth
(3.37 - 3.07) 300 x 300 MM Haunch

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		14	KNm			
Grade of steel	Fe	500			cover	50	mm											
depth			300		d		244		x =		5.5	mm	<	x lim =		150	mm	Z=	242
Width			1000	mm				Ast. Req. =		138	mm^2			**OK**				
$\rho 1$	=		0.00232	<		0.02	**OK**			565							Vertical reinforcement		0.2
K=			1.90536			**12**	@	**200**		**565**	**mm^2/m**	**OK**		F		14	KN	**12**	@ **200** **OK**
Axial Load =		0		σ_{cp} =	0		V_{Rdc}		98	KN									
							$V_{Rdc,min}$		109	KN	**OK**								

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

AT Depth **(3.07 - 2.78)** 300 x 300 MM Haunch

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		13	KNm			
Grade of steel	Fe	500			cover	50	mm											
depth			300		d		244		x =		5.1	mm	<	x lim =		150	mm	Z=	242
Width			1000	mm					Ast. Req. =		128	mm^2				**OK**			
$\rho 1$	=		0.00232	<		0.02	**OK**				565						Vertical reinforcement		0.2
K=			1.90536			**12**	@	**200**			**565**	**mm^2/m**	**OK**		F	13	KN	**12**	@ **200** **OK**
Axial Load =		0		σ_{cp} =	0		V_{Rdc}		98	KN									
							$V_{Rdc,min}$		109	KN		**OK**							

AT Depth **(2.78 - 0)** 300 x 300 MM Haunch

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		12	KNm			
Grade of steel	Fe	500			cover	50	mm											
depth			300		d		244		x =		4.7	mm	<	x lim =		150	mm	Z=	242
Width			1000	mm					Ast. Req. =		117	mm^2				**OK**			
$\rho 1$	=		0.00232	<		0.02	**OK**				565						Vertical reinforcement		0.2
K=			1.90536			**12**	@	**200**			**565**	**mm^2/m**	**OK**		F	12	KN	**12**	@ **200** **OK**
Axial Load =		0		σ_{cp} =	0		V_{Rdc}		98	KN									
							$V_{Rdc,min}$		109	KN		**OK**							

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD
DESIGN OF ABUTMENT A1 & A2

FRONT WALL :-

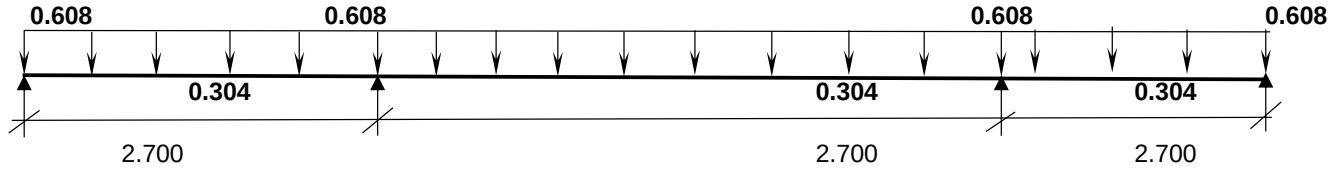
WALL IS SUBJECTED TO INTERNAL PRESSURE (I.E.P.) OF Pi AND EXTERNAL EARTH PRESSURE (E.E.P.) INCLUDING SURCHARGE LOAD OF Pe.

LOADS AND MOMENTS ARE CALCULATED AT APPROX. 2.5 TO 3M HEIGHT.

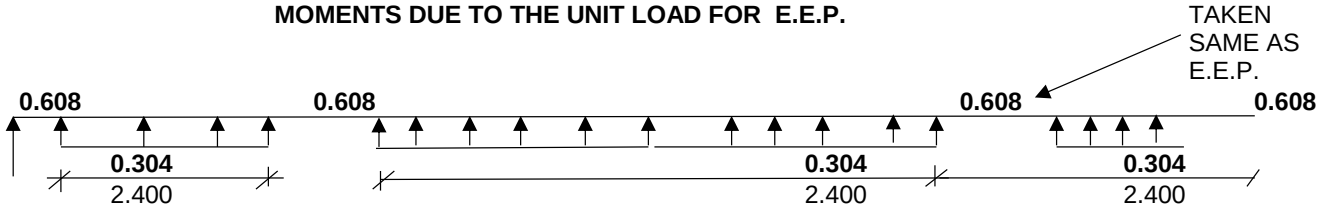
E.E.P. (FROM PRESSURE DIAG.)

FROM ELEV. 92.575 TO 88.610 = 0.448 h_1 + 0.71355

MOMENTS DUE TO THE UNIT LOAD (REFER APPENDIX 1)



MOMENTS DUE TO THE UNIT LOAD FOR E.E.P.



MOMENT DUE TO THE UNIT LOAD FOR I.E.P.

MOMENT M = UNIT LOAD MOMENT. x Pi OR (Pi - Pe)

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

TABLE FOR FRONT WALL MOMENTS
(FRONT WALL CONTD.)

LEVEL			P_i max	P_e max	$P_e - P_i$		ONLY INT. EARTH PRESS.					INT. + EXT EARTH PRESS.			
							AT END SUPT.	AT MID SUPT.		END SPAN	MID SPAN	AT END SUPT.	AT MID SUPT.	END SPAN	MID SPAN
							FOR UNIT LOAD OF 0.61	FOR UNIT LOAD OF 0.61		FOR UNIT LOAD OF 0.304	FOR UNIT LOAD OF 0.304	FOR UNIT LOAD OF 0.61	FOR UNIT LOAD OF 0.61	FOR UNIT LOAD OF 0.30375	FOR UNIT LOAD OF 0.30375
							INNER FACE	INNER FACE		OUTER FACE	OUTER FACE	OUTER FACE	OUTER FACE	INNER FACE	INNER FACE
88.610 (3.97 - 3.67)	TO	88.908	1.86	2.49	0.63	MOMENT	1.13	1.13		0.57	0.57	0.38	0.38	0.19	0.19
88.908 (3.67 - 3.37)	TO	89.205	1.75	2.36	0.60	MOMENT	1.06	1.06		0.53	0.53	0.37	0.37	0.18	0.18
89.205 (3.37 - 3.07)	TO	89.503	1.64	2.22	0.59	MOMENT	0.99	0.99		0.50	0.50	0.36	0.36	0.18	0.18
89.503 (3.07 - 2.78)	TO	89.800	1.52	2.09	0.57	MOMENT	0.92	0.92		0.46	0.46	0.35	0.35	0.17	0.17
89.800 (2.78 - 0)	TO	92.575	1.40	1.96	0.56	MOMENT	0.85	0.85		0.42	0.42	0.34	0.34	0.17	0.17

AT Depth

(3.97 - 3.67) 300 x 300 MM Haunch

Grade of concrete M 30 $\beta 1$ = 0.810 $\beta 2$ = 0.41597 fcd 13.40 M 17 KNm

Grade of steel Fe 500 cover 50 mm

depth 300 d 244 x = 6.4 mm < x lim = 150 mm Z= 241

Width 1000 mm Ast. Req. = 159 mm^2 **OK**

$\rho 1$ = 0.00232 < 0.02 **OK** 565 Vertical reinforcement 0.2

K= 1.90536 **12** @ **200** **565** **mm^2/m** **OK** F 17 KN **12** @ **200** **OK**

Axial Load = 0 σ_{cp} = 0 V_{Rdc} 98 KN

$V_{Rdc,min}$ 109 KN **OK**

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

AT Depth

(3.67 - 3.37)			300	x		300	MM Haunch														
Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		16	KNm						
Grade of steel	Fe	500			cover	50	mm														
depth			300		d	244		x =			6.0	mm	<	x lim =		150	mm		Z=		242
Width			1000	mm				Ast. Req. =		149	mm^2				**OK**						
$\rho 1$	=	0.00232	<	0.02	**OK**					565									Vertical reinforcement		0.2
K=		1.90536		**12**	@	**200**				**565**	**mm^2/m**	**OK**		F		16	KN		**12**	@ **200**	**OK**
Axial Load =		0	σ_{cp} =	0		V_{Rdc}	98	KN													
						$V_{Rdc,min}$	109	KN		**OK**											

AT Depth

(3.37 - 3.07)			300	x		300	MM Haunch														
Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		15	KNm						
Grade of steel	Fe	500			cover	50	mm														
depth			300		d	244		x =			5.6	mm	<	x lim =		150	mm		Z=		242
Width			1000	mm				Ast. Req. =		139	mm^2				**OK**						
$\rho 1$	=	0.00232	<	0.02	**OK**					565									Vertical reinforcement		0.2
K=		1.90536		**12**	@	**200**				**565**	**mm^2/m**	**OK**		F		15	KN		**12**	@ **200**	**OK**
Axial Load =		0	σ_{cp} =	0		V_{Rdc}	98	KN													
						$V_{Rdc,min}$	109	KN		**OK**											

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

AT Depth

(3.07 - 2.78) 300 x 300 MM Haunch

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		14	KNm	
Grade of steel	Fe	500			cover	50	mm									
depth			300		d		244		x =		5.2	mm	<	x lim =		150 mm
Width			1000	mm					Ast. Req. =		129	mm^2				
$\rho 1$	=		0.00232	<		0.02	**OK**				565					
K=			1.90536			**12**	@	**200**			**565**	**mm^2/m**	**OK**			
Axial Load =			0	σ_{cp} =		0			V_{Rdc}		98	KN				
									$V_{Rdc,min}$		109	KN		**OK**		

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		14	KNm	
Grade of steel	Fe	500			cover	50	mm									
depth			300		d		244		x =		5.2	mm	<	x lim =		150 mm
Width			1000	mm					Ast. Req. =		129	mm^2				
$\rho 1$	=		0.00232	<		0.02	**OK**				565					
K=			1.90536			**12**	@	**200**			**565**	**mm^2/m**	**OK**			
Axial Load =			0	σ_{cp} =		0			V_{Rdc}		98	KN				
									$V_{Rdc,min}$		109	KN		**OK**		

AT Depth

(2.78 - 0) 300 x 300 MM Haunch

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		12	KNm	
Grade of steel	Fe	500			cover	50	mm									
depth			300		d		244		x =		4.8	mm	<	x lim =		150 mm
Width			1000	mm					Ast. Req. =		118	mm^2				
$\rho 1$	=		0.00232	<		0.02	**OK**				565					
K=			1.90536			**12**	@	**200**			**565**	**mm^2/m**	**OK**			
Axial Load =			0	σ_{cp} =		0			V_{Rdc}		98	KN				
									$V_{Rdc,min}$		109	KN		**OK**		

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =	0.41597		fcd	13.40		M		12	KNm	
Grade of steel	Fe	500			cover	50	mm									
depth			300		d		244		x =		4.8	mm	<	x lim =		150 mm
Width			1000	mm					Ast. Req. =		118	mm^2				
$\rho 1$	=		0.00232	<		0.02	**OK**				565					
K=			1.90536			**12**	@	**200**			**565**	**mm^2/m**	**OK**			
Axial Load =			0	σ_{cp} =		0			V_{Rdc}		98	KN				
									$V_{Rdc,min}$		109	KN		**OK**		

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.
Client:- MP PWD
DESIGN OF ABUTMENT A1 & A2

INTERIOR WALLS NEXT TO REAR WALL & SIDE WALL

DESIGN WALLS FOR A DIFFERENCE OF 1.5 M EARTH FILLING

-M= 0.1 x 0.2973139 x 2 x 1.5 x 2.39 x 2.385 = 0.213 Tm

+M= 0.1 x 0.2973139 x 2 x 1.5 x 2.39 x 2.385 = 0.17 Tm

Grade of concrete	M	30	$\beta 1$ =	0.810	$\beta 2$ =	0.41597	fcd	13.40	M	2	KNm		
Grade of steel	Fe	500	cover	50	mm								
depth		300	d	244			x =	8.0	mm <	x lim =	150	mm	Z= 241
Width		1000	mm			Ast. Req. =	199	mm^2		**OK**			
$\rho 1$	=	0.002	<	0.02	**OK**		565						
K=		1.905		**12**	**@**	**200**	**565**	**mm^2/m**	**OK**	F	2	KN	
Axial Load =		0	σ_{cp} =	0	V_{Rdc}	98	KN		Shear Stress =	0.009	N/mm^2		
					$V_{Rdc,min}$	109	KN	**OK**					

USE - 12 mmϕ @ 200 mm C/C REINF. BOTH FACES BOTHWAYS FOR FULL HEIGHT

Calculation for vertical reinforcement in wall per meter length

Thickness of wall **=** 0.3 m **Fe 500**

Pu = 0.4.fck.Ac + 0.67.fy.Asc

Total area Ac= 5.67 x 0.3 x 2 + 8.4 x 0.3 x 2 + 2.4 x 0.3 x 4 + 2 x 0 x 3

= 11.322 sqm

Total wt. = 165 + 93.3 = 258.70 ton DL + LL

atctual stress = 259 / 11.32 = 22.85 t/sqm

12 @ 200 mm c/c in as a vertical reinforcement = 565 mm2 **Same reinforcement provided in dirt wall also.**

fck = 30 Ac = 300000 - 1131 = 298869 Asc = 1131 both face

Pu = 0.4 x 30 x 298869.03 + 0.67 x 434.78 x 1131

399 ton

Safety factor **=** 1.5

permissible load **=** 266 ton

permissible stress **=** 887 t/sqm **>** **22.85** **OK**

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

APPROACH SLAB

DESIGNED AS TWO WAY SLABS - REF. IS 456 PANEL SIZE - 2.400 X 2.385

$\frac{ly}{lx} = \frac{2.400}{2.385} \approx 1.01$

SLAB THICKNESS D = 300.00 mm

DL +Wearing coat = 0.375 x 2.4 = 0.90 t/m2

LL 20 t boggie load 20 / (1.97 x 3.54) = 2.87 t/m2

BY INTERPOLATION		ly/lx =	ly/lx =	ly/lx =	
SHORT SPAN COEFF.		**1.00**	**1.10**	**1.01**	LONG SPAN COEFF.
	-VE	0.047	0.053	0.047	**0.047**
	+VE	0.035	0.04	0.035	**0.035**

APPROACH SLAB

W = 3.77 T/ m^2

Case No.of slab = **4** (OUTER PANEL)
(as per IS -456 Table 22)

SHORT SPAN (- M)	=	0.047	X	3.77	(2.40	) ^{2}	= 1.03
SHORT SPAN (+M)	=	0.035	X	3.77	(2.40	) ^{2}	= 0.77
LONG SPAN (-M)	=	0.047	X	3.77	(2.40	) ^{2}	= 1.02
LONG SPAN (+M)	=	0.035	X	3.77	(2.40	) ^{2}	= 0.76

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD

DESIGN OF ABUTMENT A1 & A2

PROVIDE REINF. :-

PROV. 12 mm dia @ 150 mm c/c AT BOTT. SPAN SHORT SPAN DIR.

PROV. 12 mm dia @ 150 mm c/c AT TOP SPAN SHORT SPAN DIR.

PROV. 12 mm dia @ 150 mm c/c AT BOTT. SPAN LONG SPAN DIR.

PROV. 12 mm dia @ 150 mm c/c AT TOP SPAN LONG SPAN DIR.

SHORT SPAN (- M)

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =		0.41597		fcd	13.40		M			15	KNm		
Grade of steel	Fe	500				cover		50	mm										
depth			300	d			244			x =		5.8	mm	<	x lim =		150	mm	Z= 242
Width			1000	mm					Ast. Req. =		144	mm^2				**OK**			
$\rho 1$	=	0.003090091	<		0.02	**OK**					754								
K=		1.90535746			**12**	@		**150**			**754**	**mm^2/m**	**OK**		F	56	KN		
Axial Load =		0	σ_{cp} =		0	V_{Rdc}			108	KN									
						$V_{Rdc,min}$			109	KN		**OK**							

LONG SPAN (-M)

Grade of concrete	M	30		$\beta 1$ =	0.810	$\beta 2$ =		0.41597		fcd	13.40		M			15.0	KNm		
Grade of steel	Fe	500				cover		50	mm										
depth			300	d			244			x =		5.7	mm	<	x lim =		150	mm	Z= 242
Width			1000	mm					Ast. Req. =		143	mm^2				**OK**			
$\rho 1$	=	0.003090091	<		0.02	**OK**					754								
K=		1.90535746			**12**	@		**150**			**754**	**mm^2/m**	**OK**		F	56	KN		
Axial Load =		0	σ_{cp} =		0	V_{Rdc}			108	KN									
						$V_{Rdc,min}$			109	KN		**OK**							

	Project	CONTRUCTION OF HIGH LEVEL BRIDGE ACROSS BANKURI RIVER ON BANDA (BAILAI) - BANDARI ROAD AT	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & OPEN FOUNDATION OF PIER AT KM 10/6	Date & Rev.	12/12/2018

Abutment cap is fully resting on front wall, so provide min. reinforcement. size 1.640m x 0.5m

cl. 16.5.1.1 (1) of IRC :112-2011 **Grade of concrete** **M 30**

$A_{S.min} = 0.26\ f_{ctm}\ b_t\ d / f_{yk} >= 0.0013\ b_t\ d$

f_{ctm}	2.5	
b_t	1640	
d	500	
f_{yk}	500	
$0.26\ f_{ctm}\ b_t\ d / f_{yk}$	1066	mm^2
$0.0013\ b_t\ d$	1066	mm^2

Provide

12 ϕ @ 150 mm c/c 754 mm^2 at top and bottom

OK

4 Legged 12 ϕ @ 150 mm c/c 3016 mm^2 as distribution in the form of stirrup

OK

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

Client:- MP PWD
DESIGN OF ABUTMENT A1 & A2

Calculation for vertical reinforcement in dirt wall per meter length

Thickness of wall	=	0.3	m			**Fe**		**500**								
Pu = 0.4.fck.Ac + 0.67.fy.Asc																
Total area Ac=		1	x	0.3												
	=	DL		LL				0.3	sqm							
Total wt.	=	1.706	+	7.17	=			8.87	ton	DL + LL						
actual stress	=	9	/	0.30	=			29.58	t/sqm							
	12	**@**	**150**	**mm c/c in as a vertical reinforcement =**							754	mm2				
fck	=	30			Ac =	300000	-	1508	=	298492		Asc =	1508	both face		
Pu =		0.4	x	30	x	298492.04	+	0.67	x	434.78	x	1508				
						410	ton									
Safety factor	**=**	1.5														
permissible load	**=**	273	ton													
permissible stress	**=**	911	t/sqm		**>**	**29.58**		**OK**								

DESIGN CURVE OF PILE SHAFT FOR MAXIMUM BENDING MOMENTS

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by	-
	Client	MP PWD	Checked by	-
	Job Name	DESIGN OF ABUTMENT A1 & A2	Date & Rev	09-01-2024

Design of Pile Shaft Subjected to Axial Load & Maximum Bending Moments-

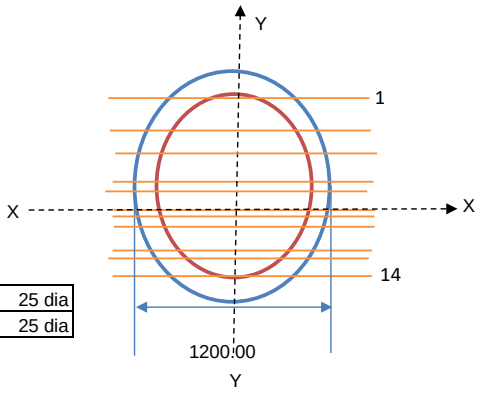
Given Data :

Pile Size:

Diameter (D)	=	1200.00	mm
Area of Pile (Ag)	=	1.13E+06	mm^2

Section Reinforcement:

			Area (mm^2)
Outer Peripheri			14726.22
			0.00



PROVIDE	30 x	25 dia +	0 x	25 dia
Astprovided =	14726.22 mm^2		0 x	25 dia

Total Ast	=	14726.216	mm^2	
% of Reinforcement (Pt)	=	1.30	%	
Effective Spacing Bars	=	120.73	mm	
Clear Cover	=	75	mm	
Dia of Tie Bar	=	12	mm	
Effective Cover (d')	=	100	mm	
d'/D ratio	=	0.084		

Vertical Bar Curtailment :

Reinforcement : 30 x 25 dia

0 x 25 dia

Transverse Reinforcement

As per Clause 16.2.3 of IRC 112:2011

Minimum Diameter of Tie bars	=	0.250	x	25	=	6.25	mm
	or	8.000	mm		whichever is greater		
Adopt	10 mm dia Helical Reinforcement						
Pitch of the Lateral Ties shall not be excee	=	12.000	x	25	=	300	mm
	or =	1200.000	mm				
	or =	200.000	mm	whichever is less			
	OR =	150.000	mm	as per IRC:78-2000 Clause 709.4.4			

provide 10 mm dia bar @ 150 mm c/c as Spiral Helical Tie Bar beyond point of fixity

Material Properties:

Concrete Grade	=	M35		Factor of Safety For Concrete	=	1.5	Clause 6.4.2.8
Steel Grade	=	Fe500	D	Factor of Safety For Steel	=	1.15	Clause 6.2.2
Characteristic Compressive Strength of Concrete at 28 days,fck	=	35.00	N/mm^2	Maximum Strain in Concrete,ε_{cu}	=	0.0035	As per Tabel 6.5
Design Compressive strength of Concrete, fcd	=	15.63	N/mm^2	Yield Strain in Concrete, ε_{cy}	=	0.002	As per Tabel 6.5
Yield Strength of Reinforcement Steel ,fy	=	500.00	N/mm^2	Yield Strain in Steel,ε_{st}	=	0.002174	(fyd/Es)
Design Yield Strength of Reinforcement Steel ,fyd	=	434.78	N/mm^2	Max. Strain in steel, ε_{stu}	=	0.004174	(ε_{cy} + ε_{st})
Modulus of Elasticity of Concrete (E_c)	=	29580.40	N/mm^2				
Modulus of Elasticity of Steel (Es)	=	200000	N/mm^2				

Summary of Design Forces at Pile Fixity

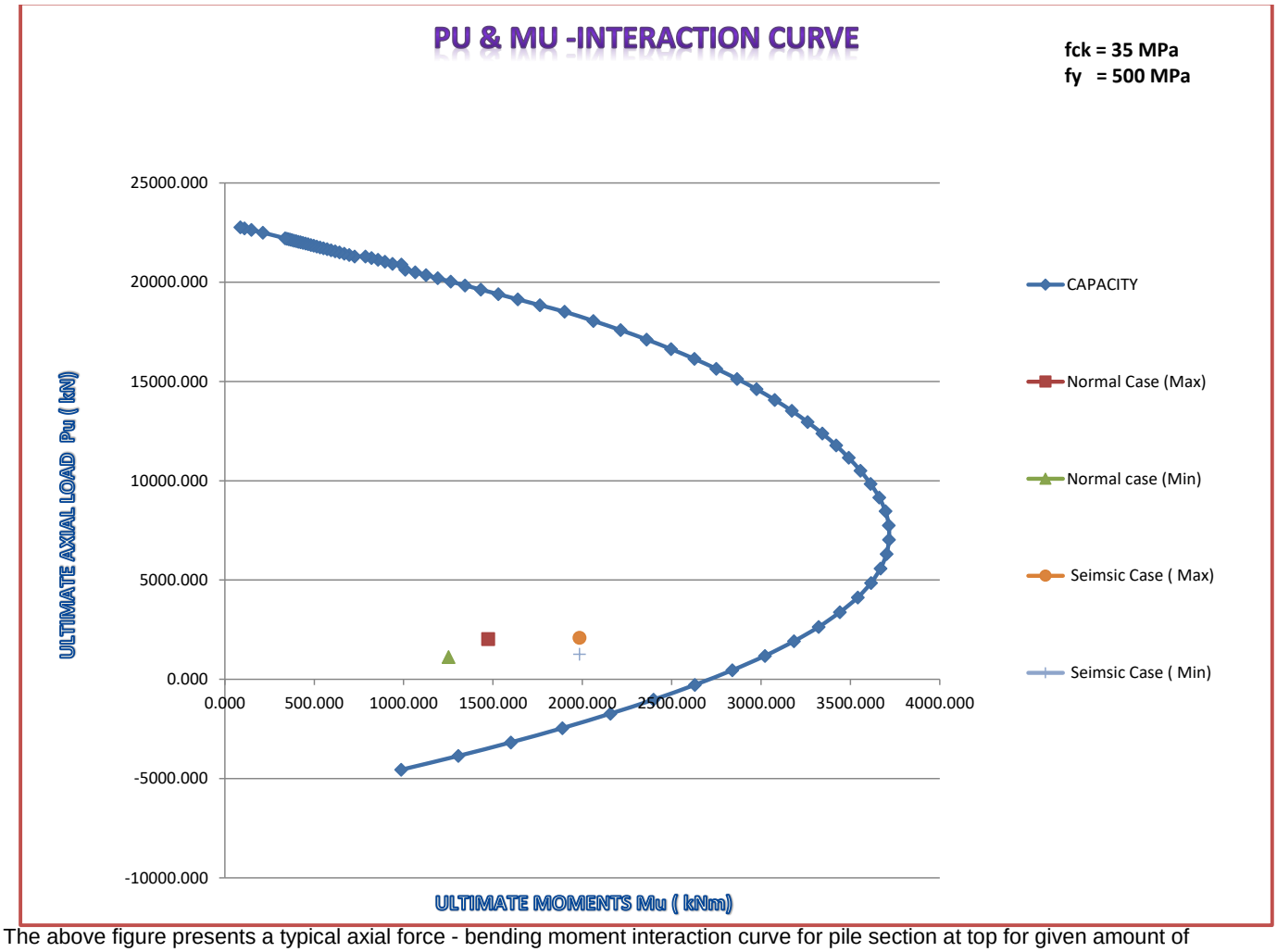
Cases		Axial Load (kN)	Initial Resultant Moment (kNm)	Final Design Moment (kNm)
		Pu	M	Mu
Normal Dry Case	Max	**2032.090**	**1473.246**	1473.246
	Min	**1126.767**	**1251.880**	1251.880
Seismic Case	Max	**2100.498**	**1984.597**	1984.597
	Min	**1264.870**	**1984.597**	1984.597

Summary of Results:

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by	-
	Client	MP PWD	Checked by	-
	Job Name	DESIGN OF ABUTMENT A1 & A2	Date & Rev	09-01-2024

Load Cases	Axial Load (kN)	Final Design Moment (kNm)	Permissible Moment (kNm)	Max. Tensile Stress in Steel (N/mm^2)	**Section Status**	
	Pu	Mu	Mu1	Fs		
Normal Dry Case	2032.090	1473.246	3205.720	434.800	**SAFE**	0.46
	1126.767	1251.880	3007.610	434.800	**SAFE**	0.42
Seismic Case	2100.498	1984.597	3218.870	434.800	**SAFE**	0.62
	1264.870	1984.597	3039.870	434.800	**SAFE**	0.65

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by	-
	Client	MP PWD	Checked by	-
	Job Name	DESIGN OF ABUTMENT A1 & A2	Date & Rev	09-01-2024



The above figure presents a typical axial force - bending moment interaction curve for pile section at top for given amount of reinforcement. The actual demands due to various load cases as presented in table are shown by filled circles. This provides a graphical representation of capacity of the section and demands due to various load cases. Since demand points are within the interaction curve, the section is safe for the present load cases.

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

1.01 Introduction

This design note present the detailed calculations for pier having Elastomeric Bearing . Design note contains the typical design of pier cap, pier & pier foundation. In the analysis the forces have been worked out at the bottom of foundation for foundation design and at the top of foundation for pier shaft design.

This design is applicable for piers

Pier no.	Road Top Level (M)	GROUND LEVEL (M)	WELL CAP BOTTOM (M)	DIFF (M)
P1	92.575	86.500	84.500	8.075

Design Data

		Left		Right	
Span Arrangement C/C of exp. Joint	=	**25**	m	**25**	m
Structural system	=	simply supported			
construction type	=	RCC			
Effective Span	=	**24.00**	m	**24.00**	m
Expansion gap in abutment	=	**40**	mm		
Expansion gap in pier	=	**40**	mm		
No of Lane considered for design	=	**2**			
Bearing type	=	**Elastomeric Bearing**			
Skew Angle	=	**0**	Degree		

Salient Reduced Levels (RL's)

Deck Level (cL of carriageway)	=	**92.575**	
Soffit Level (cL of cariageway)	=	90.800	
Pier Cap Top Level	=	90.300	
Pier Cap Bottom Level	=	89.300	
Ground Level (LBL)	=	**86.500**	
Well cap Bottom level	=	**84.500**	

Details of superstructure and other parameters

Type of structure		**RCC**		
Depth of Girder	=	**1.200**	m	
Depth of deck slab	=	**0.500**	m	
Thickness of wearing coat	=	**0.075**	m	
Camber in both direction from CL of deck	=	**2.5%**		
Expansion joint Type	=	**Strip seal**		
Impact Factor	=	**1.150**		
Length of pier cap		7.100	m	
Width of pier cap		2.500	m	
Depth of pier cap at tip		1.000	m	
Depth of pier cap at root		1.000	m	
Length of the pier		6.000	m	
outer dia. well of foundation		7.50	m	
Max. scour level		86.00	m	
Foundation level of well foundation		54.50	m	
water velocity		2.90	m/s	
Net safe bearing capacity req.		56.50	t/m2	Ref: Soil report

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

Material Parameters

Concrete **Refer Table 6.5, IRC:112-2011**

Grade of concrete for sub structure and foundation			=	**M35**
Cube strength of concrete at 28 days		f_{ck}	=	**35** MPa
Design value of concrete compressive strength		f_{cd}	=	$\alpha f_{ck}/\gamma_m$
			α =	0.67
	For Basic Combination	f_{cd}	=	15.63 MPa
	For Accidental Combination	f_{cd}	=	19.54 MPa
	For Seismic Combination	f_{cd}	=	15.63 MPa
Modulus of Elasticity		E_c	=	32000 MPa
Mean value of axial tensile strength of concrete		f_{ctm}	=	2.8 MPa
Wet Density	**2.00** t/m^3	Dry Density	=	**2.50** t/m^3

Reinforcing Steel

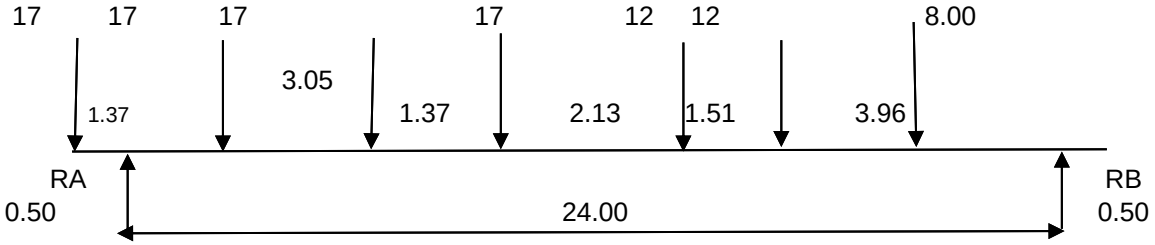
Grade			=	**Fe500**
Characteristics yield strength		f_{yk}	=	500 MPa
Design yield strength		f_{yd}	=	f_{yk}/γ_m
	For Basic Combination	f_{yd}	=	434.8 MPa
	For Accidental Combination	f_{yd}	=	500.0 MPa
	For Seismic Combination	f_{yd}	=	434.8 MPa
Modulus of Elasticity		E_s	=	**2.0E+05** MPa
Density			=	**7.85** t/m^3
Soil density			=	**2.00** t/m^3

(As per STAAD Output)

25 m span Reactions (KN)				25 m span Reactions (KN)				
Bearing No.	**DL**	**SIDL1 (WC)**	**SIDL2(CB)**	**Bearing No.**	**DL**	**SIDL1 (WC)**	**SIDL2(CB)**	FPLL
	1603	**176**	**200**		**1603**	**176**	**200**	**0**

Live load

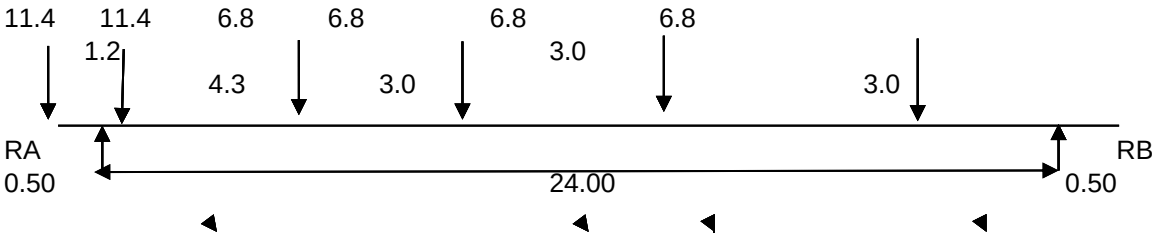
a) 70 R 1lane loading



Ra	*		24.00					
	4.00	*	17.00	*	21.61	=	1469.1	
	2.00	*	12.00	*	15.83	=	379.80	
	1.00	*	8.00	*	11.11	=	88.88	
						Total :	1937.8	
Ra	=	1937.82	/	24.00		=	**80.7**	**t**
		Impact	1.150				**92.9**	**t**
							19.26	**t**

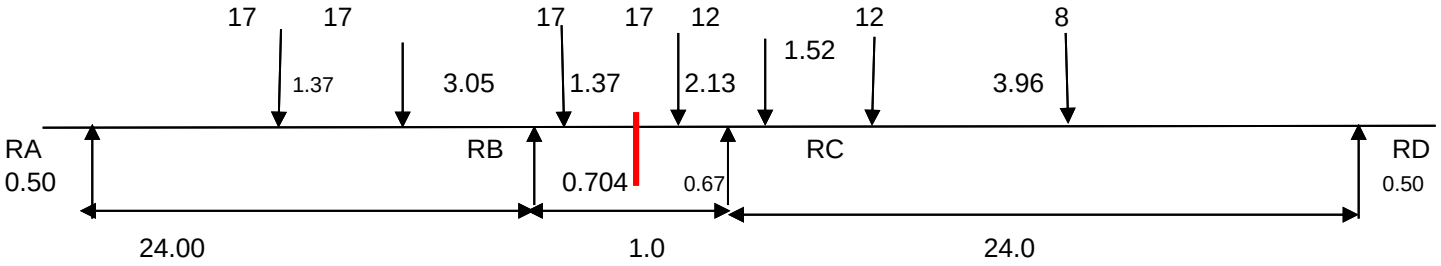
	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

b) Class A 1 lane loading



Ra		*	24.0	=				
	4	*	6.8	*	14.50	=	394	
	2	*	11.4	*	23.9	=	545	
						Total :	939	
Ra =						=	**39.1**	**t**
			Impact		1.150		**45.0**	**t**
							10.9	**t**

Live Load on both span (70 R)



Rb		*	24.0	RC				
	3	*	17	*	21.306	=	1087	
						Total :	1087	
Rb	=		1087	/	24.0	=	**45.3**	**t**
			Impact		1.150		**52.1**	**t**
Rc		*	24.0					
	1	*	17	*	23.834	=	405	
	2	*	12	*	20.944	=	503	
	1	*	8	*	16.22	=	130	
						Total :	1038	
Rc	=		1038	/	24.0	=	**43.2**	**t**
RB=					=	45.28	t	**49.7** **t**
RC=					=	43.23	t	
				Total RB+RC	=	**88.51**	t	

Pier cap is fully resting on Pier wall, so provide min. reinforcement.
cl. 16.5.1.1 (1) of IRC :112-2011

$A_{S.min} = 0.26\ f_{ctm}\ b_t\ d / f_{vk} >= 0.0013\ b_t\ d$

f_{ctm}	2.8	
b_t	2500	
d	1000	
f_{vk}	500	
$0.26\ f_{ctm}\ b_t\ d / f_{vk}$	3640	mm^2
$0.0013\ b_t\ d$	3250	mm^2

Provide

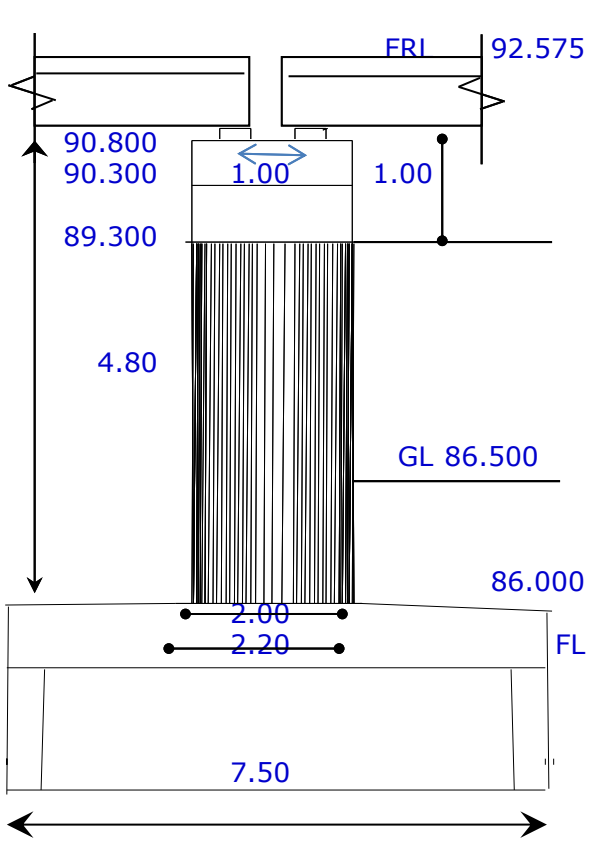
16 ϕ @	**125 mm c/c**	**1608 mm^2 OK**	**at top and bottom**	
12 ϕ @ 6 LEG	**150 mm c/c**	**4524 mm^2 OK**	**as distribution in the form of stirrup**	

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1.06 Load calculation for pier & pier foundation

1. Salient Reduced Levels (RL's)

Deck Level	=	92.575	m	
Bearing Top Level	=	90.800	m	
Pier Cap Top Level	=	90.300	m	
Pier Bottom Level	=	86.000	m	
GL/LBL	=	86.500	m	
Foundation bottom Level		84.500	m	
HFL	=	94.500	m	
Expansion Gap	=	0.040	m	
dia. of the pier	=	2.000	m	l= 31.50 K= 0.03
Length of the pier	=	6.000		Steining th. h=Kd(l)^0.5
outer dia. well of foundation	=	7.500	OD= 7.50	Req. 1.26 m
Width of foundation	=	7.500	ID= 4.70	**Take 1.40 m**
Thickness of Wearing coat (W/c)	=	0.075	As per cl. 708.2.3.2 of IRC78 For 10 % extra	
Height of Deck+Girder	=	1.700	m	
Thickness of bearingBearing	=	0.052	m	
Height of Pedestal	=	0.448	m	
Total Height from w/c to Cap top	=	2.275	m	
Span EJ to EJ	=	25.000	m	



Max. scour level 76.000

Min. 1/3rd grip Length = 21.50 **OK**

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Load Calculation
Taking Moments about founding level

	LHS	RHS	L (Long)	L (Trans)	ML	Mt	ML
	t	t	m	m	t-m	t-m	OSD
Dead Load	163.40	163.40	0.50	0.00	0.00	0.00	81.70
SIDL1 w/c	17.92	17.92	0.50	0.00	0.00	0.00	8.96
SIDL 2 exl w/c	20.39	20.39	0.50	0.00	0.00	0.00	10.19
FPLL	0.00	0.00	0.50	0.00	0.00	0.00	
LL1 (Max Reaction)							
Class A	22.88	21.86	0.50	2.400	0.51	107.39	
Class 2 A	45.77	43.73	0.50	0.650	1.02	58.17	
70 R	45.28	43.23	**0.50**	**1.105**	**1.02**	**97.80**	
LL2 (Max Moment)							
Class A	40.38	0.00	0.50	2.400	20.19	96.91	
Class 2 A	80.76	0.00	0.50	0.650	40.38	52.49	
70 R	82.98	0.00	0.50	1.105	41.49	91.69	

Horizontal Force Due to braking Elastomeric Bearing

			(as per cl.211.5 IRC 6:2014)	Fh/2+Vr ltc	@foundation		Pier Bottom	
	Fh	Rg	Rq	H	Lev	ML	Lev	ML
	t	t	t	t	m	t-m	m	t-m
Class 2 A	22.16	163.40	80.76	12.26	6.30	77.25	4.80	58.86
Class 70 R	20.00	163.40	82.98	11.18	6.30	70.45	4.80	53.67

DUE TO BEARING

TEMPERATURE CORRECTIONS L a T

L	=	SPAN	25.00	M
a	=	EXPANSION COEFFICIENT	1.20E-05	/0C As per cl. 215.4 of IRC 6-201
T	=	TEMPERATURE DIFFERENCE	50.00	^{0}C

TEMPERATURE CORRECTIONS L α T 0.015 M

ON EACH SIDE OF THE PIER 0.015/2 *1000 7.50 mm

DUT TO SHRINKAGE

SHRINKAGE 2.0E-04 As per cl. 217.3 of IRC 6-2014

TOTAL DEFORMATION 0.0002*25*1000/2+7.5 10.00 mm

NO OF BEARINGS one side 3.00 NOS

F = G*A* d /T

WHERE

F =	FORCE IN BEARING				
G =	SHEAR MODULUS			10.00	KG/CM^2
d =	DEFORMATION			1.00	CM
A =	AREA OF BEARING	450	350	1575	CM^2
T =	THICKNESS OF BEARING		52	40	CM

F = 10*1575*1/1000/40 = 0.394 T

TOTAL FORCE ON BEARINGS 0.394 * 3 = 1.18 T

WIND FORCES ON SUPER STRUCTURE REF: IRC 6-2014 CL NO:209.2

Basic wind speed	=	47	m/s	Plain terrain			1
Design wind speed	=	43.15	m/s	at height	20.00	m	30.30
Design wind pressure P_Z		1117.39	N/m^2				

$F_T = P_Z A_1 G C_D$	547	Kg/m^2	G =	**2.00**	**C_D =**	**2.40**	
F_L =	136.68	Kg/m^2					
$F_V = P_Z A_3 G C_L$	171	Kg/m^2	G =	**2.00**	**C_L =**	**0.75**	

Up ward or downward vertical wind load Fv

F_V =	25.00	x	**8.40**	x	170.85	=	**35.88**	**t**

WIND FORCES ON LIVE LOAD

Basic wind speed	=	47	m/s	Plain terrain			1
Design wind speed	=	43.15	m/s	at height	20.00	m	30.30
Design wind pressure P_Z		1117.39	N/m^2				

$F_T = P_Z A_1 G C_D$	273	Kg/m^2	G =	**2.00**	**C_D =**	**1.20**	
F_L =	68.34	Kg/m^2					
$F_V = P_Z A_3 G C_L$	171	Kg/m^2	G =	**2.00**	**C_L =**	**0.75**	

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WIND FORCES ON SUB STRUCTURE

Basic wind speed = 47 m/s Plain terrain 1

Design wind speed = 43.15 m/s at height 20.00 m 30.30

Design wind pressure P_Z 1117.39 N/m^2

REF: IRC 6-2014 CL NO:209.4

$F_T = P_Z A_1 G C_D$ 114 Kg/m^2 G = **2.00** **C_D = 0.50**

F_L = 28.48 Kg/m^2

$F_V = P_Z A_3 G C_L$ 171 Kg/m^2 G = **2.00** **C_L = 0.75**

TRANSVERSE DIRECTION		**Moments about FL**			**Pier Bottom**	
DESCRITPTION		**H**	**Lev**	**Mt**	**Lev**	**Mt**
		t	**M**	**t-m**	**M**	**t-m**
WIND ON SUPER STR ACT @	91.69	24.26	37.19	902.22	5.69	137.99
WIND ON MOVING LOAD @	94.08	20.50	39.58	811.39	8.08	165.56
WIND ON PIER CAP @	89.80	0.28	35.30	10.05	3.80	1.08
WIND ON PIER WALL @	87.90	0.64	33.40	21.30	1.90	1.21
	TOTAL	**45.69**		**1744.96**		**305.84**
		25.18		**933.58**		**140.28**

LONGITUDIANL DIRECTION		**Moments about FL**			**Pier Bottom**	
DESCRITPTION		**H**	**Lev**	**Mt**	**Lev**	**Mt**
		t	**M**	**t-m**	**M**	**t-m**
WIND ON SUPER STR ACT @	91.69	2.04	37.19	75.79	5.69	11.59
WIND ON MOVING LOAD @	94.08	1.72	39.58	68.16	8.08	13.91
WIND ON PIER CAP @	89.80	0.20	35.30	7.14	3.80	0.77
WIND ON PIER WALL @	87.90	0.48	33.40	15.98	1.90	0.91
	TOTAL	**4.44**		**167.06**		**27.17**
		2.72		**91.27**		**15.58**

Horizontal force due to water currents REF: IRC 6-2014 CL NO:210

Velocity of water in m/s 2.900 Scour Level = 76.000

V_L = 1.40 m/s V_T = 3.85 m/s

Water current force in longitudinal direction

		K_L	V_L	P (Kg/m2)	A_L	Force (t)	L. A. (m)	M_L (tm)
HFL	94.500	0.00	1.403	0.00		0.000		
super str top	92.875	1.50	1.279	127.69				
super str bott.	91.300	1.50	1.160	104.97	13.23	1.539	6.088	9.369
Pier cap top	90.300	1.50	1.084	91.70				
Pier cap middle	89.300	1.50	1.008	79.32	7.100	0.607	3.800	2.307
Pier cap bottom	89.300	1.50	1.008	79.32	0.000	0.000	5.900	0.000
Pier top	89.300	0.00	1.008	0.00				
Pier middle	87.650	0.66	0.883	26.78	9.900	0.133	2.475	0.328
Pier bottom	86.000	0.66	0.758	19.73	9.900	0.230	0.825	0.190
						2.509		**12.194**
Foundation top	86.000	0.66	0.758	19.73				
Max. scour level	76.000	0.66	0.000	0.00	75.000	0.740	5.000	3.699
						3.249		**19.657**

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Water current force in transverse direction

		K_T	V_T	P (Kg/m2)	A_T	Force (t)	L. A. (m)	M_T (tm)
HFL	94.500	0.00	3.854	0.00				
super str top	92.875	1.50	3.515	963.91				
super str bott.	91.300	1.50	3.187	792.38	39.38	34.577	6.088	210.487
Pier cap top	90.300	1.50	2.979	692.18				
Pier cap middle	89.300	1.50	2.771	598.76	2.500	1.614	3.800	6.132
Pier cap bottom	89.300	1.50	2.771	598.76	0.000	0.000	5.900	0.000
Pier top	89.300	0.00	2.771	0.00				
Pier middle	87.650	0.66	2.427	202.14	3.300	0.334	2.475	0.825
Pier bottom	86.000	0.66	2.083	148.94	3.300	0.579	0.825	0.478
						37.103		**217.92**
Foundation top	86.000	0.66	2.083	148.94				
Max. scour level	76.000	0.66	0.000	0.00	75.000	5.585	5.000	27.926
						42.688		**301.50**

Dead Weight of Pier Cap

		L	B	H	Qty	wt
		m	**m**	**m**	**m3**	**t**
	Part1	7.10	2.50	1.00	17.75	44.38
	Part 2	17.75	15.62	0.00	0.00	0.00
				1.00	17.75	

Dead wt of pier & Foundation

	Nos.	L	B	H	Qty	wt
		m	**m**	**m**	**m3**	**t**
Pier Stem	1	11.14		0.00	0.00	0.00
	1	11.14		2.80	31.20	77.99
	1	11.14		0.50	5.57	13.93
well cap		7.50	7.50	1.50	66.27	165.67
Top pluge		4.70	4.70	**0.50**	8.67	21.69
intermediate pluge		4.70	4.70	**0.00**	0.00	0.00
Bottom pluge		4.70	4.70	**3.00**	52.05	130.12
well staining		7.50	4.70	30.00	804.88	2012.19
						2421.59

Soil Above the Foundation **ht. of empty well** **0.00 m**

Soil in well		4.70	4.70	26.500	459.76	919.52
Deduction	Pier Stem				-5.57	-11.14
	Foundation				0.00	0.00
	Foundation				0.00	0.00
					454.19	**908.38**

HFL Condition

		L	B	H	Qty	wt
Pier Stem		**m**	**m**	**m**	**m3**	**t**
Pier Stem		11.14	1.00	0.00	0.00	0.00
Pier Stem		11.14	1.00	2.80	31.20	46.79
Pier Stem		11.14	1.00	0.50	5.57	8.36
well cap					66.27	99.40
Top pluge					8.67	13.01
intermediate pluge					0.00	0.00
Bottom pluge					52.05	78.07
well staining					804.88	1207.31
Soil in well					459.76	459.76
					1428.39	**2142.59**

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Earth pressure			Active earth pressure		Passive earth pressur	
			LBL - FL		MSL - FL	
Height	m		31.500		21.50	
Dry unit wt. of soil	t/m3		1.00		1.00	
wet unit wt. of soil	t/m3					
Coulomb's theory Ka (Normal case)						
ϕ Angle of friction			30.00			
α wall inclination to horizontal			90.00			
α wall inclination to vertical			0.00			
β soil surcharge inclination to horizontal			0.00			
δ			20.00			
Ka (Normal case)			0.279			
Kp (Normal case)			3.584			
Coefficient of active earth pressure			0.279		3.584	
earth pressure			8.79		77.06	
Force due to earth pressure	t/m		1038		6213	
Moment due to earth pressure	t.m/m		13735		56104	
					42369	

Tilt & shift

Tilt = 623 $\frac{\times 6.58}{80.00}$ + 3249.19 $\frac{\times 31.50}{150}$ = **734 tm**

shift = 622.67 x 0.15 = **93 tm**

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Seismic Force Calculations	**For Foundation**					
	Horizontal	Vertical (2/3 of Horizontal)				
Long Seismic Coefficient	0.061	0.041	*Refer Seismic Calculations Appendex 1*			
Trans Seismic Coefficient	0.061	0.041				
For one span Condition						
	Horizontal	**Vertical**				
Long Seismic Coefficient	0.061	0.041				
Trans Seismic Coefficient	0.061	0.041				

Seismic

						@ Foundation	
	V (t)	**H Comp**	**V Comp**	**Acting At**	**Lever arm (For H)**	**ML Due to H**	**Moment Due to V**
						ML	**MT**
DL	326.8	19.9	13.2	90.80	36.300	720.7	0.0	0.0
SIDL1	35.8	2.2	1.5	90.80	36.300	79.0	0.0	0.0
SIDL2	40.8	2.5	1.7	90.80	36.300	89.9	0.0	0.0
Pier Cap	44.4	2.7	1.8	89.80	35.300	95.2		
Pier Stem	91.9	5.6	3.7	87.65	33.150	185.1		
LL1	17.9	1.1	0.7	90.80	36.300	39.5	0.0	0.5
LL2	16.6	1.0	0.7	90.80	36.300	36.6	1.7	0.7
LL3	8.9	0.5	0.4	90.80	36.300	19.7	0.0	0.9

Longitudinal seismic

	V (t)	ML (t.m)	MT (t.m)
DL	18.76	1000.93	0.00
SIDL1	1.45	79.01	0.00
SIDL2	1.65	89.92	0.00
LL1	0.72	39.51	0.47
LL2	0.67	38.28	0.74
LL3	0.36	19.76	0.87

Tranverse seismic

	V (t)	ML (t.m)	MT (t.m)
DL	18.76	0.00	1000.93
SIDL1	1.45	0.00	79.01
SIDL2	1.65	0.00	89.92
LL1	0.72	0.04	39.94
LL2	0.67	1.68	37.34
LL3	0.36	0.02	20.61

For one Span Condition

	V (t)	ML (t.m)	MT (t.m)
DL	6.62	520.49	0.00
	6.62	0.00	520.49

Combination of Orthogonal Seismic Force

		V (t)	ML (t.m)	MT (t.m)	V (t)	ML (t.m)	MT (t.m)
DL		18.76	1301.21	0.00	**18.8**	**1301.2**	**0.0**
SIDL1		1.45	102.72	0.00	**1.5**	**102.7**	**0.0**
SIDL2		1.65	116.89	0.00	**1.7**	**116.9**	**0.0**
LL1		0.72	51.50	0.47	**0.7**	**51.5**	**0.5**
LL2		0.67	49.48	0.74	**0.7**	**49.5**	**0.7**
LL3		0.36	25.94	0.87	**0.4**	**25.9**	**0.9**
DL		18.76	0.00	1301.21	**18.8**	**0.0**	**1301.2**
SIDL1		1.45	0.00	102.72	**1.5**	**0.0**	**102.7**
SIDL2		1.65	0.00	116.89	**1.7**	**0.0**	**116.9**
LL1		0.72	0.04	51.80	**0.7**	**0.0**	**51.8**
LL2		0.67	1.68	48.82	**0.7**	**1.7**	**48.8**
LL3		0.36	0.02	26.53	**0.4**	**0.0**	**26.5**

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	V (t)	ML (t.m)	MT (t.m)
DL+SIDL1+SIDL2+LL1	22.58	1572.31	0.47
DL+SIDL1+SIDL2+LL2	22.53	1570.30	0.74
DL+SIDL1+SIDL2+LL3	22.22	1546.76	0.87
DL+SIDL1+SIDL2+LL1	22.58	0.04	1572.61
DL+SIDL1+SIDL2+LL2	22.53	1.68	1569.64
DL+SIDL1+SIDL2+LL3	22.22	0.02	1547.35
DL+SIDL1+SIDL2+LL1	**23**	**1572**	**0**
DL+SIDL1+SIDL2+LL2	**23**	**1572**	**0**
DL+SIDL1+SIDL2+LL3	**23**	**0**	**1573**

Seismic Force Calculations	**For Pier**		
	Horizontal	**Vertical (2/3 of Horizontal)**	
Long Seismic Coefficient	0.061	0.041	***Refer Seismic Calculations Appendex 1***
Trans Seismic Coefficient	0.061	0.041	
For one span Condition			
	Horizontal	**Vertical**	
Long Seismic Coefficient	0.061	0.041	
Trans Seismic Coefficient	0.061	0.041	

Seismic

						@ Stem Bottom		
	V (t)	**H Comp**	**V Comp**	**Acting At**	**Ecc (For H)**	**ML Due to H**	**Moment Due to V**	
							ML	**MT**
DL	326.8	19.9	13.2	**90.8**	**4.8**	95.3	0.0	0.0
SIDL1	35.8	2.2	1.5	**90.8**	**4.8**	10.4	0.0	0.0
SIDL2	40.8	2.5	1.7	**90.8**	**4.8**	11.9	0.0	0.0
Pier Cap	44.4	2.7	1.8	**89.8**	**3.8**	10.2		
Pier Stem	91.9	5.6	3.7	**87.7**	**1.7**	9.2		
LL1	17.9	1.1	0.7	**90.8**	**4.8**	5.2	0.0	0.5
LL2	16.6	1.0	0.7	**90.8**	**4.8**	4.8	1.7	0.7
LL3	8.9	0.5	0.4	**90.8**	**4.8**	2.6	0.0	0.9

Longitudinal seismic

	V (t)	ML (t.m)	MT (t.m)
DL	18.76	115	0.00
SIDL1	1.45	10.45	0.00
SIDL2	1.65	11.89	0.00
LL1	0.72	5.26	0.47
LL2	0.67	6.52	0.74
LL3	0.36	2.63	0.87

Tranverse seismic

	V (t)	ML (t.m)	MT (t.m)
DL	18.76	0.00	114.75
SIDL1	1.45	0.00	10.45
SIDL2	1.65	0.00	11.89
LL1	0.72	0.04	5.69
LL2	0.67	1.68	5.58
LL3	0.36	0.02	3.48

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For one Span Condition

	V (t)	ML (t.m)	MT (t.m)
DL	6.62	51.22	0.00
	6.62	0.00	51.22

Combination of Orthogonal Seismic Force

		V (t)	ML (t.m)	MT (t.m)	V (t)	ML (t.m)	MT (t.m)
DL		18.76	149.18	0.00	**18.8**	**149.2**	**0.0**
SIDL1		1.45	13.58	0.00	**1.5**	**13.6**	**0.0**
SIDL2		1.65	15.46	0.00	**1.7**	**15.5**	**0.0**
LL1		0.72	6.97	0.47	**0.7**	**7.0**	**0.5**
LL2		0.67	8.19	0.74	**0.7**	**8.2**	**0.7**
LL3		0.36	3.67	0.87	**0.4**	**3.7**	**0.9**
DL		18.76	0.00	149.18	**18.8**	**0.0**	**149.2**
SIDL1		1.45	0.00	13.58	**1.5**	**0.0**	**13.6**
SIDL2		1.65	0.00	15.46	**1.7**	**0.0**	**15.5**
LL1		0.72	0.04	7.27	**0.7**	**0.0**	**7.3**
LL2		0.67	1.68	7.54	**0.7**	**1.7**	**7.5**
LL3		0.36	0.02	4.27	**0.4**	**0.0**	**4.3**

	V (t)	ML (t.m)	MT (t.m)
DL+SIDL1+SIDL2+LL1	22.58	1527.79	0.47
DL+SIDL1+SIDL2+LL2	22.53	1529.01	0.74
DL+SIDL1+SIDL2+LL3	22.22	1524.49	0.87
DL+SIDL1+SIDL2+LL1	22.58	0.04	1528.09
DL+SIDL1+SIDL2+LL2	22.53	1.68	1528.36
DL+SIDL1+SIDL2+LL3	22.22	0.02	1525.09
DL+SIDL1+SIDL2+LL1	**23**	**1528**	**0**
DL+SIDL1+SIDL2+LL2	**23**	**1529**	**1**
DL+SIDL1+SIDL2+LL3	**23**	**2**	**1528**

1.07 Combination for design of foundation ULS & base pressure

Forces and moments are calculated at foundation bottom level

Following load combinations are considered As per Annex B of IRC:112-2011

1 Combination 1
2 Combination 2
3 Seismic Combination
4 Accidental Combination
5 Combination for base pressure check

		P (t)	ML (tm)	MT (tm)	HL(t)	HT(t)
1	DL Dead Load (due to superstructure+substructure+foundation)	3712.3	826.9	0.0	0.0	0.0
2	DL_SUP	326.8	826.9	0.0	0.0	0.0
3	PIER CAP	44.4	0.0	0.0	0.0	0.0
4	PIER	91.9	0.0	0.0	0.0	0.0
5	FDN	2329.7	0.0	0.0	0.0	0.0
6	SOIL	919.5	0.0	0.0	0.0	0.0
7	SIDL1 Super-imposed Dead Load due to wearing course	35.8			0.0	0.0
8	SIDL2 Super-imposed Dead Load other than wearing course	40.8			0.0	0.0
9	DL OSD	163.4	81.7	0.0	0.0	0.0
10	LL1 Live Load Maximum Reaction Case	89.5	1.0	58.2	0.0	0.0
11	LL2 Live Load Maximum Longitudinal Moment Case	83.0	41.5	91.7	0.0	0.0
12	LL3 Live Load Maximum Transverse Moment Case	44.7	0.5	107.4	0.0	0.0
13	LL1B Live Load Moment due to braking	2.9	77.3	0.0	12.3	0.0
14	LL2B Live Load Moment due to braking	2.6	70.4	0.0	11.2	0.0
15	LL3B Live Load Moment due to braking	0.0	0.0	0.0	0.0	0.0
16	W1 Wind vertically down without LL	35.9	91.3	933.6	2.7	25.2
17	W2 Wind vertically up without LL	-35.9	91.3	933.6	2.7	25.2
18	W3 Wind vertically down with LL	35.9	167.1	1745.0	4.4	45.7
19	W4 Wind vertically up with LL	-35.9	167.1	1745.0	4.4	45.7
20	S1 Seimic	22.6	1572.3	0.5	33.9	0.0
21	S2	22.6	0.0	1572.6	0.0	33.9
22	S3	-22.6	1572.3	0.5	33.9	0.0
23	S4	-22.6	0.0	1572.6	0.0	33.9
24	S5	6.6	520.5	0.0		
25	S6	-6.6	520.5	0.0		
26	C1 Colision load	0.0	0.0	0.0	0.0	0.0
27	C2 Colision load	0.0	0.0	0.0	0.0	0.0
28	WC1 Water current force	-279.0	19.7	301.5	3.2	42.7

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

Note:- In combination 1 & 2 wind is taken as accompanying load and LL is taken as the leading load.

	Combination 1				Combination 2				Seismic Combination				Combination for base pressure check		
	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)	P(t)	ML(tm)	MT(tm)
DL	1.35	5012	1116	0	1.00	3712	827	0	1.00	3712	827	0	3712	827	0
SIDL1	1.75	63	0	0	1.00	36	0	0	1.00	36	0	0	36	0	0
SIDL2	1.35	55	0	0	1.00	41	0	0	1.00	41	0	0	41	0	0
DL OSD	1.35	221	110	0	1.00	163	82	0	1.00	163	82	0	163	82	0
LL1	1.50	134	2	87	1.30	116	1	76	0.75	67	1	44	89	1	58
LL2	1.50	124	62	138	1.30	108	54	119	0.75	62	31	69	83	41	92
LL3	1.50	67	1	161	1.30	58	1	140	0.75	34	0	81	45	1	107
LL1B	1.50	4	116	0	1.30	4	100	0	0.75	2	58	0	3	77	0
LL2B	1.50	4	106	0	1.30	3	92	0	0.75	2	53	0	3	70	0
LL3B	1.50	0	0	0	1.30	0	0	0	0.75	0	0	0	0	0	0
W1	0.90	32	82	840	0.80	29	73	747	0.00	0	0	0	36	91	934
W2	0.90	-32	82	840	0.80	-29	73	747	0.00	0	0	0	-36	91	934
W3	0.90	32	150	1570	0.80	29	134	1396	0.00	0	0	0	36	167	1745
W4	0.90	-32	150	1570	0.80	-29	134	1396	0.00	0	0	0	-36	167	1745
S1	0.00	0	0	0	0.00	0	0	0	1.00	23	1572	0	23	1572	0
S2	0.00	0	0	0	0.00	0	0	0	1.00	23	0	1573	23	0	1573
S3	0.00	0	0	0	0.00	0	0	0	1.00	-23	1572	0	-23	1572	0
S4	0.00	0	0	0	0.00	0	0	0	1.00	-23	0	1573	-23	0	1573
S5	0.00	0	0	0	0.00	0	0	0	1.00	7	520	0	7	520	0
S6	0.00	0	0	0	0.00	0	0	0	1.00	-7	520	0	-7	520	0
WC1	1.00	-279	20	302	1.00	-279	20	302	1.00	-279	20	302	-279	20	302

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1 Combination 1

Load Case					Factored Forces/Moments			
		HL(t)	HT(t)		V	ML	MT	
					t	t-m	t-m	
Normal								
Case 1	DL+SIDL1+SIDL2				5129	1116	0	
Case 2	DL+SIDL1+SIDL2+LL1	18	0		5268	1234	87	
Case 3	DL+SIDL1+SIDL2+LL2	17	0		5258	1284	138	
Case 4	DL+SIDL1+SIDL2+LL3	0	0		5196	1117	161	
Downward Wind Condition								
Case 5	DL+SIDL1+SIDL2 + Wind	2	23		5162	1198	840	
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	2	23		5300	1384	1658	
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	4	41		5290	1435	1708	
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	4	41		5229	1267	1732	
Upward Wind Condition								
Case 9	DL+SIDL1+SIDL2 + Wind	2	23		5097	1198	840	
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	2	23		5236	1384	1658	
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	4	41		5225	1435	1708	
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	4	41		5164	1267	1732	
Span Dislodged Condition								
Case 13	DL+SIDL1+SIDL2				221	110	0	

5300	1384	1658	2159.6
5290	1435	1708	2230.5
5229	1267	1732	2145.9

Water current force

Case 14	DL+SIDL1+SIDL2 + WC1	3.2	42.7	4850	1136	302
Case 15	DL+SIDL1+SIDL2+LL1+ WC1	21.6	42.7	4989	1253	389
Case 16	DL+SIDL1+SIDL2+LL2+ WC1	20.0	42.7	4979	1304	439
Case 17	DL+SIDL1+SIDL2+LL3+ WC1	3.2	42.7	4917	1137	463
Downward Wind Condition		3.2	42.7	-279	20	302
Case 18	DL+SIDL1+SIDL2 + Wind + WC1	5.7	65.4	4883	1218	1142
Case 19	DL+SIDL1+SIDL2+LL1+ Wind+ WC1	5.7	65.4	5021	1404	1959
Case 20	DL+SIDL1+SIDL2+LL2+ Wind+ WC1	7.2	83.8	5011	1454	2010
Case 21	DL+SIDL1+SIDL2+LL3+ Wind+ WC1	7.2	83.8	4950	1287	2033
Upward Wind Condition		3.2	42.7	-279	20	302
Case 22	DL+SIDL1+SIDL2 + Wind + WC1	5.7	65.4	4818	1218	1142
Case 23	DL+SIDL1+SIDL2+LL1+ Wind+ WC1	5.7	65.4	4957	1404	1959
Case 24	DL+SIDL1+SIDL2+LL2+ Wind+ WC1	7.2	83.8	4946	1454	2010
Case 25	DL+SIDL1+SIDL2+LL3+ Wind+ WC1	7.2	83.8	4885	1287	2033
Span Dislodged Condition		3.2	42.7	-279	20	302
Case 26	DL+SIDL1+SIDL2+ WC1	3.2	42.7	-58	130	302

5021	1404	1959	2410.2
5011	1454	2010	2480.5
4950	1287	2033	2406.2

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	Job	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

2 Combination 2

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2			3789	827	0
Case 2	DL+SIDL1+SIDL2+LL1	16	0	3909	929	76
Case 3	DL+SIDL1+SIDL2+LL2	15	0	3900	972	119
Case 4	DL+SIDL1+SIDL2+LL3	0	0	3847	828	140
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	2	20	3818	900	747
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	2	20	3938	1062	1472
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	4	37	3929	1106	1515
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	4	37	3876	961	1536
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	2	20	3760	900	747
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	2	20	3880	1062	1472
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	4	37	3871	1106	1515
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	4	37	3818	961	1536
Span Dislodged Condition						
Case 13	DL+SIDL1+SIDL2	0	0	163	82	0

3938	1062	1472	1815
3929	1106	1515	1875.9
3876	961	1536	1811.6

Water current force						
Case 14	DL+SIDL1+SIDL2 + WC1	3	43	3510	847	302
Case 15	DL+SIDL1+SIDL2+LL1+ WC1	19	43	3630	948	377
Case 16	DL+SIDL1+SIDL2+LL2+ WC1	18	43	3621	992	421
Case 17	DL+SIDL1+SIDL2+LL3+ WC1	3	43	3568	847	441
Downward Wind Condition		3	43	-279	20	302
Case 18	DL+SIDL1+SIDL2 + Wind + WC1	5	63	3539	920	1048
Case 19	DL+SIDL1+SIDL2+LL1+ Wind+ WC1	5	63	3659	1082	1773
Case 20	DL+SIDL1+SIDL2+LL2+ Wind+ WC1	7	79	3650	1126	1817
Case 21	DL+SIDL1+SIDL2+LL3+ Wind+ WC1	7	79	3597	981	1837
Upward Wind Condition		3	43	-279	20	302
Case 22	DL+SIDL1+SIDL2 + Wind + WC1	5	63	3481	920	1048
Case 23	DL+SIDL1+SIDL2+LL1+ Wind+ WC1	5	63	3601	1082	1773
Case 24	DL+SIDL1+SIDL2+LL2+ Wind+ WC1	7	79	3592	1126	1817
Case 25	DL+SIDL1+SIDL2+LL3+ Wind+ WC1	7	79	3539	981	1837
Span Dislodged Condition		3	43	-279	20	302
Case 26	DL+SIDL1+SIDL2+ WC1	3	43	-116	101	302

3659	1082	1773	2077.1
3650	1126	1817	2137.2
3597	981	1837	2082.5

Case 3 Seismic Combination

Load Case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Downward seismic in longitudinal direction						
Case 1	DL+SIDL1+SIDL2+ seismic	33	0	3811	2399	0
Case 2	DL+SIDL1+SIDL2+LL1+ seismic	34	0	3881	2458	44
Case 3	DL+SIDL1+SIDL2+LL2+ seismic	34	0	3876	2483	69
Case 4	DL+SIDL1+SIDL2+LL3+ seismic	34	0	3845	2400	81
Downward seismic in transverse direction						
Case 5	DL+SIDL1+SIDL2	0	33	3811	827	1573
Case 6	DL+SIDL1+SIDL2+LL1+ seismic	0	34	3881	886	1616
Case 7	DL+SIDL1+SIDL2+LL2+ seismic	0	34	3876	911	1641
Case 8	DL+SIDL1+SIDL2+LL3+ seismic	0	34	3845	827	1653
Upward seismic in longitudinal direction						
Case 9	DL+SIDL1+SIDL2	33	0	3766	2399	0
Case 10	DL+SIDL1+SIDL2+LL1+ seismic	34	0	3836	2458	44
Case 11	DL+SIDL1+SIDL2+LL2+ seismic	34	0	3830	2483	69
Case 12	DL+SIDL1+SIDL2+LL3+ seismic	34	0	3800	2400	81
Upward seismic in transverse direction						
Case 13	DL+SIDL1+SIDL2	0	33	3766	827	1573
Case 14	DL+SIDL1+SIDL2+LL1+ seismic	0	34	3836	886	1616
Case 15	DL+SIDL1+SIDL2+LL2+ seismic	0	34	3830	911	1641
Case 16	DL+SIDL1+SIDL2+LL3+ seismic	0	34	3800	827	1653
Span Dislodged seismic in longitudinal direction						
Case 17	DL+SIDL1+SIDL2	18	0	157	602	0

3881	886	1616	1843
3830	2483	69	2484.1
3845	827	1653	1848.6

Water current force

Downward seismic in longitudinal direction						
Case 18	DL+SIDL1+SIDL2+WC1	36	43	3532	2419	302
Case 19	DL+SIDL1+SIDL2+LL1+ seismic+WC1	37	43	3602	2478	346
Case 20	DL+SIDL1+SIDL2+LL2+ seismic+WC1	37	43	3597	2503	371
Case 21	DL+SIDL1+SIDL2+LL3+ seismic+WC1	37	43	3566	2419	383
Downward seismic in transverse direction						
Case 22	DL+SIDL1+SIDL2+WC1	3	75	3532	847	1874
Case 23	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3	77	3602	905	1918
Case 24	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3	77	3597	931	1943
Case 25	DL+SIDL1+SIDL2+LL3+ seismic+WC1	3	77	3566	847	1955
Upward seismic in longitudinal direction						
Case 26	DL+SIDL1+SIDL2+WC1	36	43	3487	2419	302
Case 27	DL+SIDL1+SIDL2+LL1+ seismic+WC1	37	43	3557	2478	346
Case 28	DL+SIDL1+SIDL2+LL2+ seismic+WC1	37	43	3551	2503	371
Case 29	DL+SIDL1+SIDL2+LL3+ seismic+WC1	37	43	3521	2419	383
Upward seismic in transverse direction						
Case 30	DL+SIDL1+SIDL2+WC1	3	75	3487	847	1874
Case 31	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3	77	3557	905	1918
Case 32	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3	77	3551	931	1943
Case 33	DL+SIDL1+SIDL2+LL3+ seismic+WC1	3	77	3521	847	1955
Span Dislodged seismic in longitudinal direction						
Case 34	DL+SIDL1+SIDL2	21	43	-122	622	302

3602	2478	346	2501.6
3597	2503	371	2530.1
3566	847	1955	2130.3

4 Combination for base pressure check

Load Case		Factored Forces/Moments		
		V	ML	MT
		t	t-m	t-m
Normal				
Case 1	DL+SIDL1+SIDL2	3789	827	0
Case 2	DL+SIDL1+SIDL2+LL1	3881	905	58
Case 3	DL+SIDL1+SIDL2+LL2	3874	939	92
Case 4	DL+SIDL1+SIDL2+LL3	3834	827	107
Downward Wind Condition				
Case 5	DL+SIDL1+SIDL2 + Wind	3825	918	934
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	3917	1072	1803
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	3910	1106	1837
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	3870	994	1852
Upward Wind Condition				
Case 9	DL+SIDL1+SIDL2 + Wind	3753	918	934
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	3845	1072	1803
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	3839	1106	1837
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	3798	994	1852
Span Dislodged Condition				
Case 13	DL+SIDL1+SIDL2	163	82	0
Downward seismic in longitudinal direction				
Case 14	DL+SIDL1+SIDL2	3811	2399	0
Case 15	DL+SIDL1+SIDL2 + seismic	3904	2477	59
Case 16	DL+SIDL1+SIDL2+LL1+ seismic	3897	2511	92
Case 17	DL+SIDL1+SIDL2+LL2+ seismic	3856	2400	108
Downward seismic in transverse direction				
Case 18	DL+SIDL1+SIDL2	3811	827	1573
Case 19	DL+SIDL1+SIDL2 + seismic	3904	905	1631
Case 20	DL+SIDL1+SIDL2+LL1+ seismic	3897	939	1664
Case 21	DL+SIDL1+SIDL2+LL2+ seismic	3856	827	1680
Upward seismic in longitudinal direction				
Case 22	DL+SIDL1+SIDL2	3766	2399	0
Case 23	DL+SIDL1+SIDL2 + seismic	3859	2477	59
Case 24	DL+SIDL1+SIDL2+LL1+ seismic	3852	2511	92
Case 25	DL+SIDL1+SIDL2+LL2+ seismic	3811	2400	108
Upward seismic in transverse direction				
Case 26	DL+SIDL1+SIDL2	3766	827	1573
Case 27	DL+SIDL1+SIDL2 + seismic	3859	905	1631
Case 28	DL+SIDL1+SIDL2+LL1+ seismic	3852	939	1664
Case 29	DL+SIDL1+SIDL2+LL2+ seismic	3811	827	1680
Span Dislodged Condition seismic in longitudinal direction				
Case 30	DL+SIDL1+SIDL2	157	602	0

	water current force			
Case 31	DL+SIDL1+SIDL2+WC1	3510	847	302
Case 32	DL+SIDL1+SIDL2+LL1+WC1	3602	925	360
Case 33	DL+SIDL1+SIDL2+LL2+WC1	3595	959	393
Case 34	DL+SIDL1+SIDL2+LL3+WC1	3555	847	409
Downward Wind Condition		-279	20	302
Case 35	DL+SIDL1+SIDL2 + Wind +WC1	3546	938	1235
Case 36	DL+SIDL1+SIDL2+LL1+ Wind+WC1	3638	1092	2105
Case 37	DL+SIDL1+SIDL2+LL2+ Wind+WC1	3631	1126	2138
Case 38	DL+SIDL1+SIDL2+LL3+ Wind+WC1	3591	1014	2154
Upward Wind Condition		-279	20	302
Case 39	DL+SIDL1+SIDL2 + Wind +WC1	3474	938	1235
Case 40	DL+SIDL1+SIDL2+LL1+ Wind+WC1	3566	1092	2105
Case 41	DL+SIDL1+SIDL2+LL2+ Wind+WC1	3560	1126	2138
Case 42	DL+SIDL1+SIDL2+LL3+ Wind+WC1	3519	1014	2154
Span Dislodged Condition		-279	20	302
Case 43	DL+SIDL1+SIDL2+WC1	-116	101	302
Downward seismic in longitudinal direction		-279	20	302
Case 44	DL+SIDL1+SIDL2+WC1	3532	2419	302
Case 45	DL+SIDL1+SIDL2 + seismic +WC1	3625	2497	360
Case 46	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3618	2531	394
Case 47	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3577	2419	409
Downward seismic in transverse direction		-279	20	302
Case 48	DL+SIDL1+SIDL2+WC1	3532	847	1874
Case 49	DL+SIDL1+SIDL2 + seismic +WC1	3625	925	1932
Case 50	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3618	959	1966
Case 51	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3577	847	1982
Upward seismic in longitudinal direction		-279	20	302
Case 52	DL+SIDL1+SIDL2+WC1	3487	2419	302
Case 53	DL+SIDL1+SIDL2 + seismic +WC1	3580	2497	360
Case 54	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3573	2531	394
Case 55	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3532	2419	409
Upward seismic in transverse direction		-279	20	302
Case 56	DL+SIDL1+SIDL2+WC1	3487	847	1874
Case 57	DL+SIDL1+SIDL2 + seismic +WC1	3580	925	1932
Case 58	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3573	959	1966
Case 59	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3532	847	1982
Span Dislodged Condition seismic in longitudinal direction		-279	20	302
Case 60	DL+SIDL1+SIDL2+WC1	-122	622	302

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
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3917	1072	1803	2097.9	
3618	2531	394	2561.2	
3591	1014	2154	2380.7	

1 Normal				
Load Case	Factored Forces/Moments			
	V	ML	MT	
	t	t-m	t-m	
Case 1 Normal max. Vertical load	3881	905	58	907.05
Case 2 Normal max. longitudinal moment	3874	939	92	943.31
Case 3 Normal max. transverse moment	3834	827	107	834.36
2 Seismic / wind				
Load Case	Factored Forces/Moments			
	V	ML	MT	
	t	t-m	t-m	
Case 1 Seismic/wind max. Vertical load	3917	1072	1803	2097.9
Case 2 Seismic/wind max. longitudinal moment	3618	2531	394	2561.2
Case 3 Seismic/wind max. transverse moment	3591	1014	2154	2380.7

Max. Moment in resultant direction = **2561** tm

Net Resisting moment from active and passive earth pressure **42369** tm **OK**

Base Pressure

Net Bearing capacity of soil (t/sqm) = **56.5** Ref: Soil report

A = 44 m^2

	Concrete t/m^3	Soil t/m^3
Dry Density	**2.5**	**2**
Wet Density	**2.0**	**1**

max allowable bearing pressure (Normal) 88.5 t/m^2

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
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Load Case						With Correction Factor		Remarks
	P/A			Max	Min	Max	Min	
				t/m^2	t/m^2	t/m^2	t/m^2	
4	**Combination for base pressure check (normal)**							
Case 1	88			87.9	87.9	87.9	87.9	SAFE
Case 2	88			87.7	87.7	87.7	87.7	SAFE
Case 3	87			86.8	86.8	86.8	86.8	SAFE
5	**Combination for base pressure check (Seismic/Wind)**							
Case 1	89			88.7	88.7	70.9	70.9	SAFE
Case 2	82			81.9	81.9	65.5	65.5	SAFE
Case 3	81			81.3	81.3	65.0	65.0	SAFE

Load on well cap	max. load			655	area of pier =		11.14	m2
Load intensity		59	t/m2	self wt. of cap =		3.75	t/m2	total = 62.50 t/m2
Effective dia of steining		5.40	m	shear perimeter		=	38.584	t/m
Sagging moment at bottom			wl^2/30	=	60.8	tm		
Hogging moment at top			wl^2/60	=	30.4	tm		

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
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1.08 Verification of structural strength for Foundation

ULTIMATE LIMIT STATE

Grade of Concrete

As per clause 6.4.2.8, IRC:112-2011

f_{ck}	=	35	N/mm^2	
f_{cd}	=	15.63	N/mm^2	Combination (1)
f_{cd}	=	15.63	N/mm^2	Seismic Combi.
f_{cd}	=	15.63	N/mm^2	Combination (2)
E_c	=	32000	MPa	

Grade of steel

f_Y	=	500	N/mm^2	
f_{Yd}	=	435	N/mm^2	Combination (1)
f_{Yd}	=	435	N/mm^2	Seismic Combi.
f_{Yd}	=	435	N/mm^2	Combination (2)

Refer Fig. 6.2 of IRC:112-2011

For steel reinforcement, simplified bilinear diagram is used

Minimum strain in steel reinforcement = 0.87 fy / Es

Es = 2.0E+05 MPa

Cu = fcd*b*(3/7xu,lim + 2/3*4/7xu,lim)

= 17/21*fcd*b*xu

= 0.8095 *fcd*b*xu

Refer Chaper 5 of Reinfroced Concrete Limit State Design by Ashok K. Jain

cg of compression block from top = 0.416 xu

Tu = fyd*Ast

$R_{lim} = M_{u,Lim}/bd^2$ = 0.8095fcd*(xu $_{lim}$/d)*(1-0.416*xu $_{lim}$/d)

	Basic	Accidental	Seismic
$xu_{,lim}/d$	0.62	0.62	0.62
$R_{lim} = M_{u,Lim}/bd^2$	5.80	5.80	5.80

Here R_{lim} is in MPa

Calculation of Reinforcement

Width of section b = 1000 mm

Depth of section D = 1500 mm

Clear cover = 75

Moment on the section	**BM at support**			**BM at mid**		
	Combination (1)			Combination (1)		
Actual moment (KNm)	608			304		
b	1000			1000		
D	1500			1500		
c	75			75		
d	1400.5			1400.5		
f_{cd}	15.63			15.63		
f_{Yd}	435			435		
$xu_{,lim}/d$	0.62			0.62		
$R_{sls} = M_{u,sls}/bd^2$	5.80			5.80		
$M_{u,Lim}$ (KNm)	11382			11382		
	OK			**OK**		
Ast Req.	1008			501		
Dia of bar (main tension) (mm)	25			25		
Spacing (mm)	175			175		
+ dia of bar (main tension) (mm)	0			0		
Spacing (mm)	175			175		
Ast provided (sq mm)	2805			2805		

Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.		Designed by:	-
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Dia of bar(main compresion at top mm)	25			25		
Spacing (mm)	175			175		
Area of main compresion (mm^2)	2805			2805		
f_{ctm}	2.8			2.8		
f_{yk}	435			435		
cl. 16.6.1 (2) of IRC :112-2011						
$A_{S.min} = 0.26\ f_{ctm}\ b_t\ d / f_{yk} >= 0.0013\ b_t\ d$	2345			2345		
A_{ct}	1403631			1403631		
$f_{ct,eff}$	2.9			2.9		
$k_c = 0.4\ \{ 1 - \sigma_c / (k_1\ f_{ct,eff}\ h/h^*) \} <= 1$	0.4			0.4		
For Bending or bending combined with axial force						
k	0.65			0.65		
σ_s	435			435		
cl. 12.3.6 (4) of IRC :112-2011						
$A_{S.min} = k_c\ k\ f_{ct,eff}\ A_{ct} / \sigma_s$	2434			2434		
	OK			**OK**		
As.max = 0.025 Ac (main tension)	37500			37500		
cl. 16.5.1.1 (2) of IRC :112-2011	**OK**			**OK**		
As.max = 0.04 Ac (tension + compresion)	60000			60000		
x (mm)	96			96		
x/d	0.069			0.069		
	OK			**OK**		
z (mm)	1360			1360		
MR (KNm)	1659			1659		
	OK			**OK**		
Shear on the section						
Actual shear at effective depth V_{Ed} (KN)	386			386		
Total depth at effective depth from face of sup	1500			1500		
Effective depth at effective depth from face of	1388			1388		
Actual shear stress (N/mm2)	0.309			0.309		
Max shear capacity, 0.135 fck(1-fck/310)	4.2			4.2		
	OK			**OK**		
Min shear capacity, 0.0924 fck(1-fck/310)	2.9			2.9		
$\Theta = 0.5 \times \sin^{-1}$ (Applied shear stress / 0.135/fck/(1-fck/310))	2.1			2.1		
Min angle of inclination, Θ (deg)	21.8			21.8		
cl. 10.3.2(2) Eq. 10.2 of IRC :112-2011						
K = 1+Sqrt(200/d) <= 2.0	1.380			1.380		
cl. 10.3.2(2) Eq. 10.3 of IRC :112-2011						
$\nu_{min} = 0.031\ K^{3/2}\ fck^{1/2}$	0.297			0.297		
cl. 10.3.1 of IRC :112-2011						
$\rho 1 = A_{sl}/(b_w\ d) <= 0.02$	0.002			0.002		
	OK			**OK**		
$0.12\ K\ (80\ \rho 1\ f_{ck})^{0.33}$	0.293			0.293		
Axial compressive force N_{Ed} (KN)	0.0			3.0		
$\sigma_{cp} = N_{Ed} / A_c <= 0.2\ f_{cd}$	0.0			0.0		
cl. 10.3.2(2) Eq. 10.1 of IRC :112-2011						
$V_{Rd.c} = [0.12K(80\rho 1\ f_{ck})^{0.33} + 0.15\sigma_{cp}]b_w d <=$	407			408		
	OK			**OK**		

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

1.11 Combination for design of pier (ULS)

Forces and moments are calculated at foundation top level

Following load combinations are co As per Annex B of IRC:112-2011

1 Basic Combination
2 Accidental Combination
3 Seismic Combination

		P (t)	ML (tm)	MT (tm)	HL(t)	HT(t)
1	DL Dead Load (due to superstructure+substructure)	463	0	0	0	0
2	DL_SUP	327	0	0	0	0
3	PIER CAP	44	0	0	0	0
4	PIER	92	0	0	0	0
5	SIDL1 Super-imposed Dead Load due to wearing course	36			0	0
6	SIDL2 Super-imposed Dead Load other than wearing cours	41			0	0
7	DL OSD	163	82	0	0	0
8	SIDL1 OSD	18	9	0	0	0
9	SIDL2 OSD	20	10	0	0	0
10	LL1 Live Load Maximum Reaction Case	89	1	58	0	0
11	LL2 Live Load Maximum Longitudinal Moment Case	83	41	92	0	0
12	LL3 Live Load Maximum Transverse Moment Case	45	1	107	0	0
13	LL1 Live Load Moment due to braking	3	59	0	12.26	0
14	LL2 Live Load Moment due to braking	3	54	0	11.18	0
15	LL3 Live Load Moment due to braking	0	0	0	0.00	0
16	W1 Wind vertically down without LL	36	16	140	2.72	25.18
17	W2 Wind vertically up without LL	-36	16	140	2.72	25.18
18	W3 Wind vertically down with LL	36	27	306	4.44	45.69
19	W4 Wind vertically up with LL	-36	27	306	4.44	45.69
20	S1 Seimic	23	1529	1	33.9	0
21	S2	23	2	1528	0.0	33.9
22	S3	-23	1529	1	33.9	0
23	S4	-23	2	1528	0.0	33.9
26	WC1	-37	12	218	2.5	37.1

	Basic Combination								Seismic Combination			
	PSF	P (t)	ML (tm)	MT (tm)					PSF	P (t)	ML (tm)	MT (tm)
DL	1	463	0	0					1	463	0	0
SIDL1	1	36	0	0					1	36	0	0
SIDL2	1	41	0	0					1	41	0	0
DL OSD	1	163	82	0					1	163	82	0
SIDL1 OSD	1	18	9	0					1	18	9	0
SIDL2 OSD	1	20	10	0					1	20	10	0
LL1	1	89	1	58					1	89	1	58
LL2	1	83	41	92					1	83	41	92
LL3	1	45	1	107					1	45	1	107
LL1	1	3	59	0					1	3	59	0
LL2	1	3	54	0					1	3	54	0
LL3	1	0	0	0					1	0	0	0
W1	1	36	16	140					1	36	16	140
W2	1	-36	16	140					1	-36	16	140
W3	1	36	27	306					1	36	27	306
W4	1	-36	27	306					1	-36	27	306
S1	0	0	0	0					1	23	1529	1
S2	0	0	0	0					1	23	2	1528
S3	0	0	0	0					1	-23	1529	1
S4	0	0	0	0					1	-23	2	1528
WC1	1	-36.77	12.194	217.92					1	-36.767	12.194	217.92

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

1 Basic Combination

LWL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2			540	0	0
Case 2	DL+SIDL1+SIDL2+LL1	12	0	632	60	58
Case 3	DL+SIDL1+SIDL2+LL2	11	0	625	95	92
Case 4	DL+SIDL1+SIDL2+LL3	0	0	584	1	107
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	3	25	576	16	140
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	3	25	668	87	364
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	4	46	661	122	398
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	4	46	620	28	413
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	3	25	504	16	140
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	3	25	596	87	364
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	4	46	589	122	398
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	4	46	549	28	413
Span Dislodged Condition						
Case 10	DL+SIDL1+SIDL2	0	0	202	101	0

668	87	364
661	122	398
620	28	413

HFL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2 +WC1	3	37	502.9	12.2	217.9
Case 2	DL+SIDL1+SIDL2+LL1 +WC1	15	37	595.3	72.1	276.1
Case 3	DL+SIDL1+SIDL2+LL2+WC1	14	37	588.5	107.4	309.6
Case 4	DL+SIDL1+SIDL2+LL3+WC1	3	37	547.7	12.7	325.3
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind +WC1	5	62	538.8	27.8	358.2
Case 6	DL+SIDL1+SIDL2+LL1+ Wind+WC1	5	62	631.2	99.2	581.9
Case 7	DL+SIDL1+SIDL2+LL2+ Wind+WC1	7	83	624.4	134.5	615.5
Case 8	DL+SIDL1+SIDL2+LL3+ Wind+WC1	7	83	583.6	39.9	631.2
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind +WC1	5	62	467.0	27.8	358.2
Case 10	DL+SIDL1+SIDL2+LL1+ Wind+WC1	5	62	559.4	99.2	581.9
Case 11	DL+SIDL1+SIDL2+LL2+ Wind+WC1	7	83	552.6	134.5	615.5
Case 12	DL+SIDL1+SIDL2+LL3+ Wind+WC1	7	83	511.8	39.9	631.2
Span Dislodged Condition						
Case 10	DL+SIDL1+SIDL2+WC1	3	37	164.9	113.0	217.9

631	99	582
624	135	615
584	40	631

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

3 Seismic Combination

LWL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Downward seismic in longitudinal direction						
Case 1	DL+SIDL1+SIDL2	33	0	562	1529	1
Case 2	DL+SIDL1+SIDL2+LL1+ seismic	34	0	655	1589	59
Case 3	DL+SIDL1+SIDL2+LL2+ seismic	34	0	648	1624	92
Case 4	DL+SIDL1+SIDL2+LL3+ seismic	34	0	607	1530	108
Downward seismic in transverse direction						
Case 5	DL+SIDL1+SIDL2	0	33	562	2	1528
Case 6	DL+SIDL1+SIDL2+LL1+ seismic	0	34	655	62	1587
Case 7	DL+SIDL1+SIDL2+LL2+ seismic	0	34	648	97	1620
Case 8	DL+SIDL1+SIDL2+LL3+ seismic	0	34	607	2	1636
Upward seismic in longitudinal direction						
Case 9	DL+SIDL1+SIDL2	33	0	517	1529	1
Case 10	DL+SIDL1+SIDL2+LL1+ seismic	34	0	610	1589	59
Case 11	DL+SIDL1+SIDL2+LL2+ seismic	34	0	603	1624	92
Case 12	DL+SIDL1+SIDL2+LL3+ seismic	34	0	562	1530	108
Upward seismic in transverse direction						
Case 13	DL+SIDL1+SIDL2	0	33	517	2	1528
Case 14	DL+SIDL1+SIDL2+LL1+ seismic	0	34	610	62	1587
Case 15	DL+SIDL1+SIDL2+LL2+ seismic	0	34	603	97	1620
Case 16	DL+SIDL1+SIDL2+LL3+ seismic	0	34	562	2	1636
Span Dislodged seismic in longitudinal direction						
Case 17	DL+SIDL1+SIDL2	20.533	0	179	1630	1

655	1589	59
179	1630	1
607	2	1636

HFL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2+WC1	35	37	525.5	1541.2	218.7
Case 2	DL+SIDL1+SIDL2+LL1+ seismic+WC1	36	37	617.8	1601.1	276.8
Case 3	DL+SIDL1+SIDL2+LL2+ seismic+WC1	36	37	611.0	1636.4	310.4
Case 4	DL+SIDL1+SIDL2+LL3+ seismic+WC1	36	37	570.2	1541.7	326.1
Downward seismic in transverse direction						
Case 5	DL+SIDL1+SIDL2+WC1	3	70	525.5	13.9	1746.3
Case 6	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3	71	617.8	73.8	1804.4
Case 7	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3	71	611.0	109.0	1838.0
Case 8	DL+SIDL1+SIDL2+LL3+ seismic+WC1	3	71	570.2	14.4	1853.7
Upward seismic in longitudinal direction						
Case 9	DL+SIDL1+SIDL2+WC1	35	37	480.4	1541.2	218.7
Case 10	DL+SIDL1+SIDL2+LL1+ seismic+WC1	36	37	572.7	1601.1	276.8
Case 11	DL+SIDL1+SIDL2+LL2+ seismic+WC1	36	37	565.9	1636.4	310.4
Case 12	DL+SIDL1+SIDL2+LL3+ seismic+WC1	36	37	525.1	1541.7	326.1
Upward seismic in transverse direction						
Case 13	DL+SIDL1+SIDL2+WC1	3	70	480.4	13.9	1746.3
Case 14	DL+SIDL1+SIDL2+LL1+ seismic+WC1	3	71	572.7	73.8	1804.4
Case 15	DL+SIDL1+SIDL2+LL2+ seismic+WC1	3	71	565.9	109.0	1838.0
Case 16	DL+SIDL1+SIDL2+LL3+ seismic+WC1	3	71	525.1	14.4	1853.7
Span Dislodged seismic in longitudinal direction						
Case 17	DL+SIDL1+SIDL2+WC1	23	37	142.4	1642.1	218.7

618	1601	277
142	1642	219
570	14	1854

1.12 Combination for design of pier (SLS)

Forces and moments are calculated at foundation top level

Following load combinations are co As per Annex B of IRC:112-2011

1 Rare Combination
2 Frequent Combination
3 Quasi-Perma. Combination

		P (t)	ML (tm)	MT (tm)	HL(t)	HT(t)
1	DL Dead Load (due to superstructure+substructure)	463	0	0	0	0
2	DL_SUP	327	0	0	0	0
3	PIER CAP	44	0	0	0	0
4	PIER	92	0	0	0	0
5	SIDL1 Super-imposed Dead Load due to wearing course	36			0	0
6	SIDL2 Super-imposed Dead Load other than wearing course	41			0	0
7	DL OSD	163	82	0	0	0
8	SIDL1 OSD	18	9	0	0	0
9	SIDL2 OSD	20	10	0	0	0
10	LL1 Live Load Maximum Reaction Case	89	1	58	0	0
11	LL2 Live Load Maximum Longitudinal Moment Case	83	41	92	0	0
12	LL3 Live Load Maximum Transverse Moment Case	45	1	107	0	0
13	LL1 Live Load Moment due to braking	3	59	0	12.26	0
14	LL2 Live Load Moment due to braking	3	54	0	11.18	0
15	LL3 Live Load Moment due to braking	0	0	0	0.00	0
16	W1 Wind vertically down without LL	36	16	140	2.72	25.18
17	W2 Wind vertically up without LL	-36	16	140	2.72	25.18
18	W3 Wind vertically down with LL	36	27	306	4.44	45.69
19	W4 Wind vertically up with LL	-36	27	306	4.44	45.69
20	S1 Seimic	23	1529	1	33.9	0
21	S2	23	2	1528	0.0	33.9
22	S3	-23	1529	1	33.9	0
23	S4	-23	2	1528	0.0	33.9
24	WC1	-37	12	218	2.5	37.1

	Rare Combination				Frequent Combination				Quasi-Perma. Combination			
	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)	PSF	P (t)	ML (tm)	MT (tm)
DL	1	463	0	0	1	463	0	0	1	463	0	0
SIDL1	1	36	0	0	1	36	0	0	1	36	0	0
SIDL2	1	41	0	0	1	41	0	0	1	41	0	0
DL OSD	1	163	82	0	1	163	82	0	1	163	82	0
SIDL1 OSD	1	18	9	0	1	18	9	0	1	18	9	0
SIDL2 OSD	1	20	10	0	1	20	10	0	1	20	10	0
LL1	1	89	1	58	1	89	1	58	1	89	1	58
LL2	1	83	41	92	1	83	41	92	1	83	41	92
LL3	1	45	1	107	1	45	1	107	1	45	1	107
LL1	1	3	59	0	1	3	59	0	1	3	59	0
LL2	1	3	54	0	1	3	54	0	1	3	54	0
LL3	1	0	0	0	1	0	0	0	1	0	0	0
W1	1	36	16	140	1	36	16	140	1	36	16	140
W2	1	-36	16	140	1	-36	16	140	1	-36	16	140
W3	1	36	27	306	1	36	27	306	1	36	27	306
W4	1	-36	27	306	1	-36	27	306	1	-36	27	306
S1	0	0	0	0	0	0	0	0	1	23	1529	1
S2	0	0	0	0	0	0	0	0	1	23	2	1528
S3	0	0	0	0	0	0	0	0	1	-23	1529	1
S4	0	0	0	0	0	0	0	0	1	-23	2	1528
WC1	1	-37	12	218	1	-37	12	218	1	-37	12	218

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

1 Rare Combination

LWL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2			540	0	0
Case 2	DL+SIDL1+SIDL2+LL1	12	0	632	60	58
Case 3	DL+SIDL1+SIDL2+LL2	11	0	625	95	92
Case 4	DL+SIDL1+SIDL2+LL3	0	0	584	1	107
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	3	25	576	16	140
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	3	25	668	87	364
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	4	46	661	122	398
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	4	46	620	28	413
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	3	25	504	16	140
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	3	25	596	87	364
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	4	46	589	122	398
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	4	46	549	28	413
Span Dislodged Condition						
Case 10	DL+SIDL1+SIDL2	0	0	202	101	0
				P (t)	ML(tm)	MT(tm)
				668	87	364
				661	122	398
				620	28	413

HFL case		HL(t)	HT(t)	V	ML	MT
Normal				t	t-m	t-m
Case 1	DL+SIDL1+SIDL2 +WC1	2.51	37.10	502.9	12.2	217.9
Case 2	DL+SIDL1+SIDL2+LL1 +WC1	14.77	37.10	595.3	72.1	276.1
Case 3	DL+SIDL1+SIDL2+LL2+WC1	13.69	37.10	588.5	107.4	309.6
Case 4	DL+SIDL1+SIDL2+LL3+WC1	2.51	37.10	547.7	12.7	325.3
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind +WC1	5.23	62.29	538.8	27.8	358.2
Case 6	DL+SIDL1+SIDL2+LL1+ Wind+WC1	5.23	62.29	631.2	99.2	581.9
Case 7	DL+SIDL1+SIDL2+LL2+ Wind+WC1	6.95	82.79	624.4	134.5	615.5
Case 8	DL+SIDL1+SIDL2+LL3+ Wind+WC1	6.95	82.79	583.6	39.9	631.2
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind +WC1	5.23	62.29	467.0	27.8	358.2
Case 10	DL+SIDL1+SIDL2+LL1+ Wind+WC1	5.23	62.29	559.4	99.2	581.9
Case 11	DL+SIDL1+SIDL2+LL2+ Wind+WC1	6.95	82.79	552.6	134.5	615.5
Case 12	DL+SIDL1+SIDL2+LL3+ Wind+WC1	6.95	82.79	511.8	39.9	631.2
Span Dislodged Condition						
Case 10	DL+SIDL1+SIDL2+WC1	2.51	37.10	164.9	113.0	217.9
				P (t)	ML(tm)	MT(tm)
				631	99	582
				624	135	615
				584	40	631

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

2 Frequent Combination

LWL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Normal						
Case 1	DL+SIDL1+SIDL2			540	0	0
Case 2	DL+SIDL1+SIDL2+LL1	12	0	632	60	58
Case 3	DL+SIDL1+SIDL2+LL2	11	0	625	95	92
Case 4	DL+SIDL1+SIDL2+LL3	0	0	584	1	107
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind	3	25	576	16	140
Case 6	DL+SIDL1+SIDL2+LL1+ Wind	3	25	668	87	364
Case 7	DL+SIDL1+SIDL2+LL2+ Wind	4	46	661	122	398
Case 8	DL+SIDL1+SIDL2+LL3+ Wind	4	46	620	28	413
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind	3	25	504	16	140
Case 10	DL+SIDL1+SIDL2+LL1+ Wind	3	25	596	87	364
Case 11	DL+SIDL1+SIDL2+LL2+ Wind	4	46	589	122	398
Case 12	DL+SIDL1+SIDL2+LL3+ Wind	4	46	549	28	413
Span Dislodged Condition						
Case 10	DL+SIDL1+SIDL2	0	0	202	101	0
				P (t)	ML(tm)	MT(tm)
				668	87	364
				661	122	398
				620	28	413

HFL case		HL(t)	HT(t)	V	ML	MT
Normal				t	t-m	t-m
Case 1	DL+SIDL1+SIDL2 +WC1	2.51	37.10	502.9	12.2	217.9
Case 2	DL+SIDL1+SIDL2+LL1 +WC1	14.77	37.10	595.3	72.1	276.1
Case 3	DL+SIDL1+SIDL2+LL2+WC1	13.69	37.10	588.5	107.4	309.6
Case 4	DL+SIDL1+SIDL2+LL3+WC1	2.51	37.10	547.7	12.7	325.3
Downward Wind Condition						
Case 5	DL+SIDL1+SIDL2 + Wind +WC1	5.23	62.29	538.8	27.8	358.2
Case 6	DL+SIDL1+SIDL2+LL1+ Wind+WC1	5.23	62.29	631.2	99.2	581.9
Case 7	DL+SIDL1+SIDL2+LL2+ Wind+WC1	6.95	82.79	624.4	134.5	615.5
Case 8	DL+SIDL1+SIDL2+LL3+ Wind+WC1	6.95	82.79	583.6	39.9	631.2
Upward Wind Condition						
Case 9	DL+SIDL1+SIDL2 + Wind +WC1	5.23	62.29	467.0	27.8	358.2
Case 10	DL+SIDL1+SIDL2+LL1+ Wind+WC1	5.23	62.29	559.4	99.2	581.9
Case 11	DL+SIDL1+SIDL2+LL2+ Wind+WC1	6.95	82.79	552.6	134.5	615.5
Case 12	DL+SIDL1+SIDL2+LL3+ Wind+WC1	6.95	82.79	511.8	39.9	631.2
Span Dislodged Condition						
Case 10	DL+SIDL1+SIDL2+WC1	2.51	37.10	164.9	113.0	217.9
				P (t)	ML(tm)	MT(tm)
				631	99	582
				624	135	615
				584	40	631

3 Quasi-Perma. Combination

LWL case				Factored Forces/Moments		
		HL(t)	HT(t)	V	ML	MT
				t	t-m	t-m
Downward seismic in longitudinal direction						
Case 1	DL+SIDL1+SIDL2	33	0	562	1529	1
Case 2	DL+SIDL1+SIDL2+LL1+ seismic	34	0	655	1589	59
Case 3	DL+SIDL1+SIDL2+LL2+ seismic	34	0	648	1624	92
Case 4	DL+SIDL1+SIDL2+LL3+ seismic	34	0	607	1530	108
Downward seismic in transverse direction						
Case 5	DL+SIDL1+SIDL2	0	33	562	2	1528
Case 6	DL+SIDL1+SIDL2+LL1+ seismic	0	34	655	62	1587
Case 7	DL+SIDL1+SIDL2+LL2+ seismic	0	34	648	97	1620
Case 8	DL+SIDL1+SIDL2+LL3+ seismic	0	34	607	2	1636
Upward seismic in longitudinal direction						
Case 9	DL+SIDL1+SIDL2	33	0	517	1529	1
Case 10	DL+SIDL1+SIDL2+LL1+ seismic	34	0	610	1589	59
Case 11	DL+SIDL1+SIDL2+LL2+ seismic	34	0	603	1624	92
Case 12	DL+SIDL1+SIDL2+LL3+ seismic	34	0	562	1530	108
Upward seismic in transverse direction						
Case 13	DL+SIDL1+SIDL2	0	33	517	2	1528
Case 14	DL+SIDL1+SIDL2+LL1+ seismic	0	34	610	62	1587
Case 15	DL+SIDL1+SIDL2+LL2+ seismic	0	34	603	97	1620
Case 16	DL+SIDL1+SIDL2+LL3+ seismic	0	34	562	2	1636
Span Dislodged seismic in longitudinal direction						
Case 17	DL+SIDL1+SIDL2	20.533	0	179	1630	1
				P (t)	ML(tm)	MT(tm)
				655	1589	59
				179	1630	1
				607	2	1636

HFL case		HL(t)	HT(t)	V	ML	MT
Normal				t	t-m	t-m
Case 1	DL+SIDL1+SIDL2+WC1	35.30	37.10	525.5	1541.2	218.7
Case 2	DL+SIDL1+SIDL2+LL1+ seismic+WC1	36.38	37.10	617.8	1601.1	276.8
Case 3	DL+SIDL1+SIDL2+LL2+ seismic+WC1	36.38	37.10	611.0	1636.4	310.4
Case 4	DL+SIDL1+SIDL2+LL3+ seismic+WC1	36.38	37.10	570.2	1541.7	326.1
Downward seismic in transverse direction						
Case 5	DL+SIDL1+SIDL2+WC1	2.51	69.89	525.5	13.9	1746.3
Case 6	DL+SIDL1+SIDL2+LL1+ seismic+WC1	2.51	70.98	617.8	73.8	1804.4
Case 7	DL+SIDL1+SIDL2+LL2+ seismic+WC1	2.51	70.98	611.0	109.0	1838.0
Case 8	DL+SIDL1+SIDL2+LL3+ seismic+WC1	2.51	70.98	570.2	14.4	1853.7
Upward seismic in longitudinal direction						
Case 9	DL+SIDL1+SIDL2+WC1	35.30	37.10	480.4	1541.2	218.7
Case 10	DL+SIDL1+SIDL2+LL1+ seismic+WC1	36.38	37.10	572.7	1601.1	276.8
Case 11	DL+SIDL1+SIDL2+LL2+ seismic+WC1	36.38	37.10	565.9	1636.4	310.4
Case 12	DL+SIDL1+SIDL2+LL3+ seismic+WC1	36.38	37.10	525.1	1541.7	326.1
Upward seismic in transverse direction						
Case 13	DL+SIDL1+SIDL2+WC1	2.51	69.89	480.4	13.9	1746.3
Case 14	DL+SIDL1+SIDL2+LL1+ seismic+WC1	2.51	70.98	572.7	73.8	1804.4
Case 15	DL+SIDL1+SIDL2+LL2+ seismic+WC1	2.51	70.98	565.9	109.0	1838.0
Case 16	DL+SIDL1+SIDL2+LL3+ seismic+WC1	2.51	70.98	525.1	14.4	1853.7
Span Dislodged seismic in longitudinal direction						
Case 17	DL+SIDL1+SIDL2+WC1	23.04	37.10	142.4	1642.1	218.7
				P (t)	ML(tm)	MT(tm)
				618	1601	277
				142	1642	219
				570	14	1854

Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJ
Client	MP PWD
Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1

STRUCTRAL DESIGN OF WELL M 35

DESIGN DATA :-

OUTER DIAMETER OF PIER SHAFT = 7.50 M

INNER DIAMETER OF PIER SHAFT = 4.7 M

GROSS AREA OF SECTION = π /4 X (7.5 ^ 2 - 4.7 ^ 2) = 26.829 M^2

GROSS M. I. OF SECTION = π / 64 X (7.5 ^ 4 - 4.7 ^ 4) = 131.362 M^4

RADIUS OF GYRATION r = $\sqrt{\frac{131.362}{26.829}}$ = 2.213

EFFECTIVE LENGTH (BOTH SPAN IN POSITION)

= 86 - 54.50 + 6.30 = 37.80

EFFECTIVE LENGTH (LEFT SPAN NOT IN POSITION)

= 86 - 54.5 + 23.63 = 55.13

SLENDERNESS RATIO (BOTH SPAN IN POSITION)

= 37.80 / 2.21 = 17.08 < 50 SHORT COLUMN

SLENDERNESS RATIO (LEFT SPAN NOT IN POSITION)

= 55.125 / 2.21274716 = 24.91 < 50 SHORT COLUMN

PERMISSIBE STRESS IN M 35 CONCRETE

σ c allowable = 116.7 Kg/cm^2

σ c o allowable = 87.5 Kg/cm^2

Permisible basic tensile stress = -28 Kg/cm^2

Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.
Client	MP PWD
Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1

Design of steining

Longitudinal Reinforcement

Minimum steel required on outer face (as per cl. 708.3.5 of IRC:78-2000)

0.12 % of gross x-sectional area = $\frac{0.12}{100}$ x (7500 ^2 - 4700 ^2) x 3.14 / 4

= **32195** mm^2

Ast provided near outer face of steining = **136** bars of dia. **16** mm (Ast provided = 27344 mm^2) 173

Ast provided near inner face of steining = **136** bars of dia. **16** mm (Ast provided = 27344 mm^2) 109

Total steel provided	=	27344	+	27344	=	**54689 mm^2**	**Hence O.K.**

Transverse Hoop Reinforcement

Hoop Reinforcement in well steining /m = $\frac{0.04}{100}$ x Volume of Well Steining /metre height

= $\frac{0.04}{100}$ x $\frac{\pi}{4}$ x (7.50 ^2 - 4.70 ^2)

= 0.0107 m^3/m = **10731681 mm^3/m**

Adopt 10 dia stirrups @ **200 mm c/c on outer face of steining.**

Ast provided = $\frac{\pi}{4}$ x 10.00 ^2 x $\frac{\pi}{200}$ x 7350.00 x **1000**

= **9067699 mm^3/m**

Adopt 10 dia stirrups @ **200 mm c/c on inner face of steining.**

Ast provided = $\frac{\pi}{4}$ x 10.00 ^2 x $\frac{\pi}{200}$ x 4850.00 x **1000**

= **5983448 mm^3/m**

Total Volume of Transverse Reinforcement provided = 9067699 + 5983448 = **15051147** > **10731681 mm^3/m**

Hence O.K.

Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.
Client	MP PWD
Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1

COMPUTAION OF TRANSFORMED SECTION PROPERTIES

Min. steel required on outer face (as per cl. 708.3.5 of IRC:78-2014) C/C DIST. OF BAR Ast PERCENTAGE

PROVIDE 136 Nos. OF 16 mm DIA. AT OUTER FACE HYSD BARS 173 273.44 50

136 Nos. OF 16 mm DIA. AT INNER FACE HYSD BARS 109 273.44 50

STEEL AREA = 546.89 CM^2 EFFETIVE COVER c = 7.5 CM

% OF STEEL = 0.204 % RADIUS OF PIER R = 375 CM

R -c = 367.5 CM

d = 735 CM

MODULAR RATIO = 10

AREA OF CONCRETE Ac = 268292.01 - 546.89 = 267745.12 CM^2

TRANSFORMED AREA = Ac + m. As = 267745.12 + 5468.88 = 273214.01 CM^2

TRANFORMED M. I. = Ig + (m - 1) * As * (R - c) ^2

= 13136247667.7 + 664746328.0

= 13800993995.7

TRANSFORMED Z = I / R = 36802651 CM^3

CHECK FOR CRACKED / UNCRACKED SECTION

			SEISMIC CONDITION	
	LOADS & MOMENTS			
1	P	T	3880.72	3596.64
2	M	T.m	1842.99	2530.14
	ACTUAL STRESS	Kg/cm^2		
3	σ co cal (P / A)		14.20	13.16
4	σ cbc,cal (M / Z)		5.01	6.87
	PERMISSIBLE STRESS	Kg/cm^2		
5	σ co		116.81	116.81
6	σ cbc		155.75	155.75
	CHECK FOR MIN. STEEL AREA			
7	CONCRETE AREA REQ. FOR DIRECT STRESS		33222.53	30790.47
8	0.8 % OF AREA REQ.=0.008 X (7)		265.78	246.32
9	0.3 % OF GROSS AREA REQ.		804.88	804.88
10	AREA OF STEEL PROVIDED		546.89	546.89
	CHECK FOR SAFETY OF SECTION			
11	$\frac{\sigma \text{ co cal}}{\sigma \text{ co}} + \frac{\sigma \text{ cbc,cal}}{\sigma \text{ cbc}}$		0.15	0.16
12	CHECK FOR CRACKED / UNCRACKED SECTION			
	σ co cal - σ cbc,cal		9.20	6.29
13	PERMISSIBLE TENSILE STRESS IN CONCRETE		-28	-28
14	SECTION IS TO BE DESIGN AS A CRACKED		UNCRACKED	UNCRACKED

Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST -
Client	MP PWD
Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1

Design of Well Curb

Volume of well curb	=	34.33	m^3								
Outer dia of the well	=	7.50	m								
Inner dia of the well	=	4.70	m	6.100							
Thickness of the steining	=	1.40	m								
Depth of well curb	=	2.45	m								
Perimeter		19.16	m								
As per Cl:708.7.3 of IRC:78-2014											
Min. reinforcement in terms of wt.	=	**72**	kg / m^3								
Clear cover = 75 mm											
Reinforcement required	=	72	x	34.333	=	2472	kg			**129**	**Kg/m**
Provide **33** Nos.		**20**	dia bar around the periphery.					33.00	2.47	81.48	**Kg/m**
5 Nos. of stirrups per meter		**12**	dia in well curb @	200	mm c/c around the periphery.			5.53	0.89	24.56	**Kg/m**
11 Nos. of links		**10**	dia in one set.	**5**	sets per meter			0.96	0.62	32.59	**Kg/m**
										138.63	**Kg/m** **OK**

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

1.13 Appendex 1

$$T = 2 \times \sqrt{\frac{D}{1000*F}}$$

T = Time Period

D = Appropriate Dead Load & Live Load

F = Horizontal force to be applied at superstructure mass to produce 1 mm deflection at top of pier.

						DL +SIDL		
			Longi. Direction			**Transverse Direction**		
	d	=	$F*L^3/3EI$					
Height of pier,piercap	h	=	4.80	m				
	F	=	***1988.7***	KN		***28887***	KN	
	D	=	4631	KN				
Dia. of the Pier		=	2.00	m				
Length of the Pier		=	6.00	m				
Area of Pier	Ap	=	11.1416	m		11.142	m	
Inertia	Ip	=	4	m^4		36.000	m^4	
E of conc	E	=	3.0E+07	KN/m^2				
CG of Superstructure	H2	=	6.098	m		4.80		
Piercap Top Level	H1	=	4.80	m				
Pier Top Level		=	90.30					
At top of pier	d	=	1.00	mm		1.00	mm	
	T	=	0.1	sec		0.03	sec	
	Ah	=	(Z/2)*(Sa/g)*(I/R)					
Seismic Zone	II							
	Z	=	0.1					
	I	=	1.2					
(RCC Column with out Ductile Detailing)	R	=	3.3					
For founding on medium soil	II	Type of soil						
	Sa/g	=	2.50			2.50		
	Ah	**=**	**0.045**			**0.045**		

PU- MU INTERACTION CURVE
FOR DESIGN OF PIER SHAFT

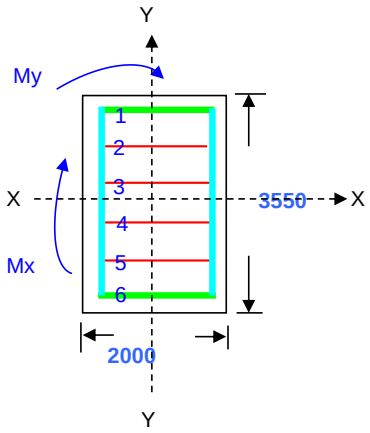
	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

Checking of Stress in Wall Type Pier Shaft

Given Data :

Pier Size:

Width (B)	=	2000.00	mm
Depth (D)	=	3550.00	mm
Area (Ac)	=	7.10E+06	mm^2
Length (Lo)	=	**3.300**	m



Section Reinforcement:

Dia. Of Bars	=	25	mm
Nos. of Bars Along Width (B)	=	22	Nos.
Nos. of Bars Along Depth (D)	=	26	Nos.
Total Nos. of Bars	=	96	Nos.
Total Ast	=	47123.89	mm^2
% of Reinforcement (Pt)	=	0.66	%
Clear Cover	=	50	mm
Dia of Tie bar	=	12	mm

D/B = 1.78 Wall Type Pier

Horizontal Reinforcement for wall type Pier as per IRC: 112-2011 , Clause 16.3.2

maximum of following	=	0.2500	x	47123.89	=	11780.97	
	=	0.001	x	7.10E+06	=	7.10E+03	
Maximum Horizontal Reinf.	=	11781	mm^2				
Min dia of bar	=	0.250	x	25	=	6.25	mm
	or	8	mm				
Maximum Spacing between bars	<=	300	mm c/c				
Provide 20 L	12 dia	@	250 c/c	=	9048	mm^2	Link
Provide 2 L	16 dia	@	125 c/c	=	3217	mm^2	
					12265		**OK**

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

Material Properties:

Concrete Grade	=	M35		Factor of Safety For Concrete	=	1.5		
Steel Grade	=	Fe500	D	Factor of Safety For Steel	=	1.15		
Characteristic Compressive Strength of Concrete at 28 days,fck	=	35.00	N/mm^2	Maximum Strain in Concrete,ε_{cu}	=	0.0035		
Design strength of concrete, fcd=0.446fck	=	15.63	N/mm^2	Yield Strain in Concrete, ε_{cd}	=	0.002		
Yield Strength of Reinforcement Steel ,fy	=	500.00	N/mm^2	Yield Strain in Steel,$\varepsilon_{st}(\varepsilon_{yd})$	=	0.002174		
Design Yield Strength of Reinforcement Steel ,fyd	=	434.78	N/mm^2	Max. Strain in steel, ε_{stu}	=	0.004174		
Modulus of Elasticity of Concrete (E_c)	=	29580.40	N/mm^2	Modulus of Elasticity of Steel (Es)	=	200000	N/mm^2	

Summary of Design Forces at Pier Shaft Bottom :

Load Cases	Axial Load (kN)	Initial Moment (kNm)		Final Design Moment (kNm)	
	Pu or N_{Ed}	Mx(MT)	My(ML)	Mux	Muy
Normal Case 1	**4574**	**3571**	**854**	3570.94	853.99
Normal Case 2	**4146**	**6038**	**1320**	6037.601	1319.77
Normal Case 3	**3746**	**6192**	**391**	6191.626	391.22
Seismic Case 4	**4363**	**578**	**9915**	590.3955	9920.50
Seismic Case 5	**1477**	**2145**	**10436**	2149.412	10438.17
Seismic Case 6	**3535**	**12512**	**141**	12522.21	145.96

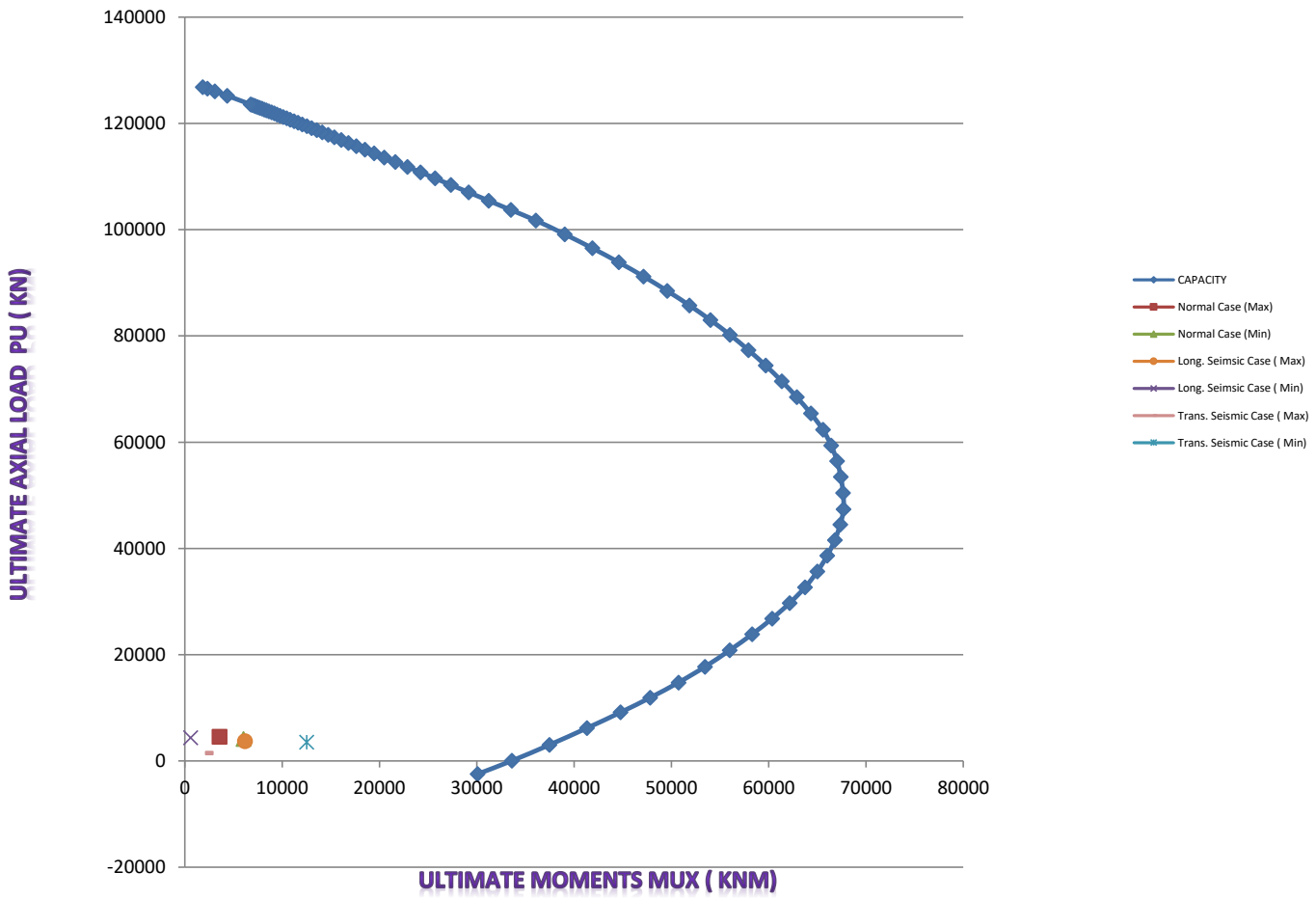
Summery of Results:

Load Cases	Axial Load (kN)	Final Design Moment (kNm)		Permissible Moment (kNm)		Max. Tensile Stress in Steel (N/mm^2)		$\left(\frac{Mux}{Mux1}\right)^{\alpha}+\left(\frac{Muy}{Muy1}\right)^{\alpha}\le 1$	Section Status
	Pu or N_{Ed}	Mux	Muy	Mux1	Muy1	Fsx	Fsy		
Normal Case 1	4574	3571	854	39358.34	22336.13	434.8	434.8	0.13	**SAFE**
Normal Case 2	4146	6038	1320	38836.64	22038.36	434.8	434.8	0.22	**SAFE**
Normal Case 3	3746	6192	391	38347.93	21741.69	434.8	434.8	0.18	**SAFE**
Seismic Case 4	4363	590	9920	39100.62	22190.55	434.8	434.8	0.46	**SAFE**
Seismic Case 5	1477	2149	10438	35471.96	20060.25	434.8	434.8	0.58	**SAFE**
Seismic Case 6	3535	12522	146	38090.21	21585.25	434.8	434.8	0.34	**SAFE**

	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF SUBSTRUCTURE & WELL FOUNDATION OF PIER P1	Date & Rev.	09/01/2024

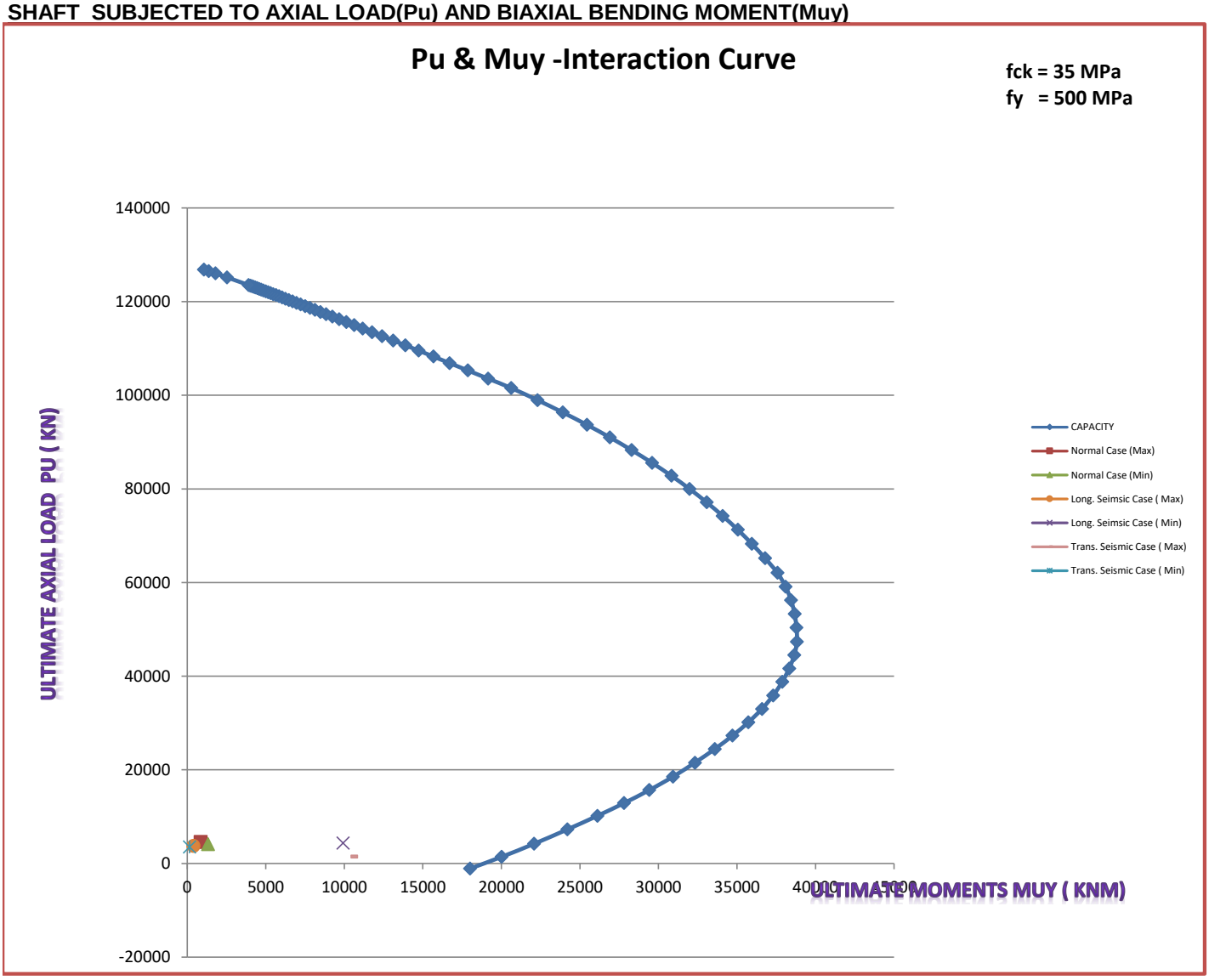
Pu & Mux -Interaction Curve

fck = 35 MPa
fy = 500 MPa



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SHAFT SUBJECTED TO AXIAL LOAD(Pu) AND BIAXIAL BENDING MOMENT(Muy)



CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

ANNEXURE :- DESIGN OF STOPPERS

-	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF STOPPERS	Date & Rev.	09-01-2024

Design of Transverse Stoppers

Material Properties:

Grade of Concrete	=	M 35	
fck	=	35 N/mm^2	- Refer Table No 6.5 of IRC : 112-2011
fctm	=	2.77 N/mm^2	-Refer Table 6.5 of IRC:112-2011
Ec	=	29580 N/mm^2	(5000 x sqrt (fck)
Grade of Steel	=	Fe 500	(HYSD Bars)
fy or fyk	=	500.00 N/mm^2	- Refer Table No 18.1 of IRC : 112-2011
fyd	=	434.78 N/mm^2	(fy/1.15) - Refer Fig 6.4 of IRC : 112-2011
Es	=	200000 N/mm^2	- Refer Fig 6.3.5 of IRC : 112-2011

Design Loads

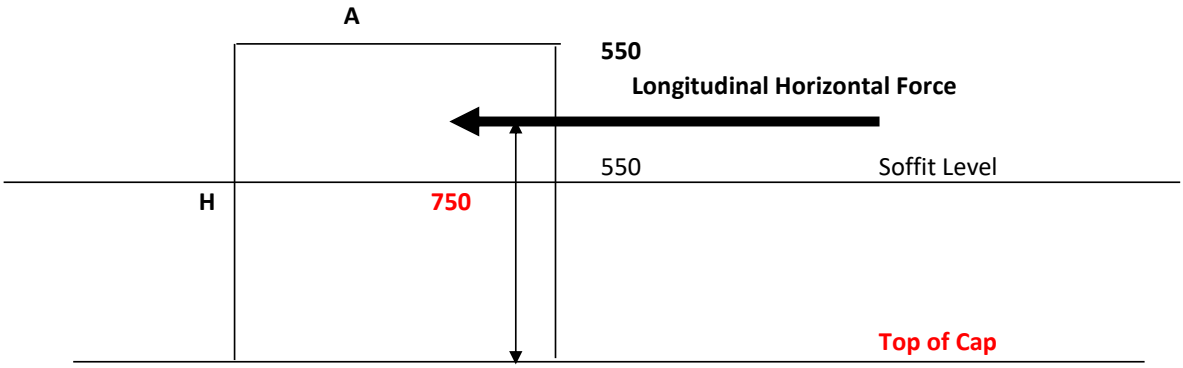
Nos. of Stopper = 1

From water 35 t **From seismic 16 t** **Vu** = **339.2 kN**

WATER CURRENT/SEISMIC FORCE IN TRANVERSE DIRECTION ON SUPER STRUCTURE = 339.2 KN REFER PIER DESIGN CALCULAT

Shape and Dimension of Stopper

Plan Dimension		Height
A	**B**	**H**
(in mm)	(in mm)	(in mm)
500	2500	1300



Max. Height of Pedestal with bearing = 500 mm

Dist. of Point of Application of Force **a** = 750 mm

Adopting **25.00** mm dia Main bars and **12.00** mm dia for Stirrups

						dia. of bar/2	
Effective Depth	**d'**	=	500	-50	0.0	-12.50	-12.00

= **425.500 mm**

-	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF STOPPERS	Date & Rev.	09-01-2024

Ratio a/d' = 1.763 > 1

Hence, Restrainer has to be Design as a Cantilever

Cantilever Moment = 339.199 x 0.75
= 254.40 kN-m

Check For Depth of Stopper :

Mult = 0.167 x fck x b x d^2 = 254.40 kNm

Effctive Depth Required = SQRT(254.40 x 1000000) / (0.167 x 35 x 2500)

= 131.95 mm < 425.50 SAFE

R= Mu/(b d^2) = 0.56

Minimum Longitudinal Reinforcement : - Refer Eq. 16.5.1.1 & 16.6.1.1 of IRC: 112-2011

As. Min = 0.26 x (fctm / fyk) x b . d

OR = 0.0015 x b . d -Refer Clause 16.9 of IRC:112-2011'

Whichever is higher

b = 2500.00 mm

d = 425.50 mm

Ast min = **1595.63** mm^2

Area of Steel Required:

$\frac{pt}{100} = \frac{Ast_{req}}{b\,d} = \frac{fck}{2\,fy}\{1 - \sqrt{1 - 4.598\,R/fck}\}$

= 0.001

Ast_{req} = 1400.88963 mm^2

Ast required = max(Ast_{min}, Ast_{req}) = **1595.63** mm^2

Provide 24 nos. 25.00 mm dia bars (Provided 11781 mm^2)

Spacing of Bar **100 mm c/c** OK

Percentage of steel (Pt) = 1.11 %

Moment of Resistance of Section corresponding to Provided Ast

$Mu = 0.87\,f_y\,A_{st}\,d\,[1 - A_{st}\,f_y / b\,d\,f_{ck}]$

= 1835.57 kNm > 254.40 kNm

SAFE

Design Shear Strength of Concrete,(τ_c) without Shear Reinforcement:

-	Project	CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.	Designed by:	-
	Client	MP PWD	Checked by:	-
	Job Name	DESIGN OF STOPPERS	Date & Rev.	09-01-2024

As per Clause 10.3.2 of IRC:112-2011,

Design shear resistance of the member without shear reinforcement is given by:

$$V_{Rd,c} = \left[0.12K(80\rho_1 f_{ck})^{0.33} + 0.15\sigma_{cp}\right]b_w\, d \qquad \textbf{eq.1}$$

Subjected to minimum of = $V_{Rd,c} = \left(V_{\min} + 0.15\,\sigma_{cp}\right)b_w\, d$ **eq.2**

where,

K= 1 + SQRT(200/d) ≤ 2.0 , where d is depth in mm

K= 1.69

vmin= $0.031\ K^{3/2}\ fck^{1/2}$, fck = 35.00 N/mm^2

Hence **vmin= 0.4014 N/mm^2**

σ_{cp} = Concrete compressive stress in concrete at centroidal axis in the direction of axial load or prestressing

σ_{cp} = N_{Ed}/A_c < 0.2 f_{cd} where , f_{cd} = 0.67 fck/1.5

σ_{cp} = **0.00** N/mm^2

Hence,

τ_c = $V_{Rd,c}/(b_w.d)$ = V_{min} + 0.15 σ_{cp} = 0.4014 N/mm^2 **From eq.1**

$\rho 1$ = Steel Ratio = Asl/(bw . d) ≤ 0.02 Hence $\rho 1$ = 0.0111

τ_c = $V_{Rd,c}/(b_w.d)$ = 0.6355 N/mm^2 **From eq.2**

Max of eq.1 & eq.2

τ_c = $V_{Rd,c}/(b_w.d)$ = 0.6355 N/mm^2 Corresponds to steel ratio = 1.107% & M35 Grade of Concrete

Shear stress(Tv) = Ved/(bw*z) = $V_{Ed}/(b*0.9d)$, Where V_{Ed} = Normal Shear Force at Pier Face

Tv = 339.20 x 1000 = **0.35** N/mm^2

2500.00 x 382.95 < 0.635 MPa

AsTv is lesser than Tc Hence No Shear Reinforcement is need to be provided.

Design parameters :- Elastomeric bearing pad for braket

CONSTRUCTION OF SUBMERSIBLE BRIDGE ACROSS KUNWARI RIVER ON SUJANGHARHI ROAD DIST - MORENA.

force due to water / seismic in transverse direction		**339.2 kN**	REF. PIER DESIGN			
Design LOAD, Vmax	**=**	**169.6 kN**	EACH SUPPORT	Design displacement, e_l(long.)	=	0 mm
Design horizontal load, Hmax (long.)	=	33.9199 kN		Design displacement, e_l(lat.)	=	0 mm
Design horizontal load, Hmax (lat.)	=	33.9199 kN		Bulk modulus, E_b	=	2000 N/mm2
Maximum design strain, εtd (Cl. 5.2.3)	=	7.0		Yield stress of steel, f_y	=	250 N/mm2
Shear modulus, G	=	0.9 N/mm2		Partial safety factor, γ_m	=	1.0
Stress correction factor, Kp	=	1.3		Factor for tensile stresses, K_h	=	1.0
Design rotation, α	=	0		Type loading factor, K_L	=	1.0
Design vertical load, Vmin	**=**	**42.4 kN**				

Bearing parameters :-

Length, a	=	275 mm		No. of elastomer layer, n	=	1.0
Width, b	=	160 mm		No. of steel laminate, n+1	=	2.0
Total height, T	**=**	**22 mm**		Side cover, c	=	6 mm
Thickness of individual elastomer layer, t_i or t_e	=	8 mm		Edge cover	=	4 mm
Thickness of steel laminate, t_s	=	3 mm		Effective area, Ae = (a-2c)x(b-2c)	=	38924 mm2
				Reduced area, Ar	=	36853 mm2
Total elastomer thickness, T_e	=	16 mm		Ae*(1-(v_{xd}/(a-2c))-(v_{yd}/(b-2c)))		
				K_{rd}	=	3.0

Design Check :- (Cl No. 5.2.3.2)

Compressive strain

Shape factor, S=0.5x((a-2c)x(b-2c))/((a-2c)+(b-2c))xti	=	**5.92**				
Compressive strain, $\varepsilon_{c,d}$ =1.5 x F_{zd} / G x A_r x S	=	1.30				

Shear Strain :- (Cl No.5.2.3.5)

Shear displacement, long $V_{x,d}$	=	14.0		{ ($H*T_e$)/(A_e*G)} + el		
Shear displacement, lat $V_{y,d}$	=	0.0				
Shear displacement ,$V_{xy,d}$	=	14.0				
Strain,εqd=vxyd/Te	=	0.64	<	1.00	**HENCE OK**	

Strain due to angular rotation :- (Cl No. 5.2.3.4)

Strain,εαd=(α*((a-2c)^2+(b-2c)^2)*ti)/(2*Σti^3)	=	0.00				

Total Design strain (Cl No.5.2.3)

Total strain, εtd=K_L(εcd+εqd+εαd)	=	1.93	<	7.00	**HENCE OK**	

Thickness of steel laminate (ts) (Cl.No.5.2.3.5)

Allowable thickness of steel laminate	>	2 mm				
Required thickness, ts=(K_p*V_{max}*(t_1+t_2)*K_h x γ_m)/(f_yxA_r)	=	0.38 mm		**HENCE OK**		
Thickness provided	=	3 mm		**Thickness given > Required thickness**		

Buckling Stability :- (Cl. No.5.2.3.6)

Buckling Stability, 2*(a-2c)*GxS/3*Te	=	58.4 N/mm2		**HENCE OK**		
Max Compressive stress, σ_{max}=(V_{max}/A_r)	=	4.6 N/mm2		**Buckling Stability > σmax**		

Vertical Deflection (Cl. No.5.2.3.7)

Total Vertical defl., ΣV_{zd}=σ_{max}x$t_i$xnx(1/5xGxS_2+1/E_b)	=	0.25 mm				

Rotational deflection

Rotational deflection, ((a-2C)xα)+(b-2C)xα))/K_{rd}	=	0.00 mm	<	0.25	**HENCE OK**	(Cl. No.5.2.3.6)

Friction Check

Min compressive stress, σ_{min}=(V_{min}/A_r)	=	1.15	<	3.00	**HENCE OK**	(Cl. No.5.2.3.6)
Coefficient of friction, μ=(0.1+1.5 x k_f/σ_{min})	=	0.88				
Resultant Horizontal force, Hr	=	33.9 kN				
Friction force,Fr = μxVmin	=	37.4 kN	>	33.92	**HENCE OK**	

S. Nos.	DEFLECTION SHOWN IN THESE TABLE AT L/4, L/2 & 3L/4	L/4	L/2	3L/4
1	MAX. DEFLECTION AS PER DESIGN LIVE LOADING 70RW (MM)	5.65	7.61	5.07
2	MAX. DEFLECTION AS PER DESIGN LIVE LOADING 2A (MM)	5.40	7.33	5.08
3	MAX. DEFLECTION AFTER TEST LIVE LOADING STAGE 1 (MM)	3.17	4.48	3.19
4	MAX. DEFLECTION AFTER TEST LIVE LOADING STAGE 2 (MM)	6.34	8.96	6.38
	MAX. PERMISIBLE DEFLECTION SPAN/800 (MM)	**17.5**	**17.5**	**17.5**

S. Nos.	REACTION AT SUPPORT	LEFT SUPPORT (T)	RIGHT SUPPORT (T)	Total (T)
1	MAX. REACTION AS PER DESIGN LIVE LOADING 70R	66.3	33.6	100.0
2	MAX. REACTION AS PER DESIGN LIVE LOADING 2A	69.7	41.0	110.8
3	MAX. REACTION AFTER TEST LIVE LOADING STAGE 1	27.3	28.7	56.0
4	MAX. REACTION AFTER TEST LIVE LOADING STAGE 2	54.6	57.4	112.0

```
STAAD SPACE
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 0.2 0 0; 3 6.2 0 0; 4 12.2 0 0; 5 18.2 0 0; 6 24.2 0 0; 7 24.4 0 0;
************************************************
*        STAAD.Pro Generated Comment        *
************************************************
*1 1 2 6
************************************************
MEMBER INCIDENCES
1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 6 7;
**************************************
DEFINE MATERIAL START
ISOTROPIC CONCRETE1
E 3e+007
POISSON 0.17
DENSITY 25
ALPHA 1.17e-005
DAMP 0.05
END DEFINE MATERIAL
UNIT METER MTON
MEMBER PROPERTY AMERICAN
1 TO 6 PRIS YD 1.2 ZD 8.4 YB 0.925 ZB 5.4
*****************************************
UNIT METER KN
CONSTANTS
MATERIAL CONCRETE1 ALL
SUPPORTS
2 6 PINNED
UNIT METER MTON
DEFINE MOVING LOAD
*70R Wheeled (13.4m)
TYPE 1 LOAD 8 12 12 17 17 17 17
DIST 3.96 1.52 2.13 1.37 3.05 1.37
*Class A (18.8m)
TYPE 2 LOAD 2.7 2.7 11.4 11.4 6.8 6.8 6.8 6.8
DIST 1.1 3.2 1.2 4.3 3 3 3
**TEST LOAD
TYPE 3 LOAD 7 10.5 10.5
DIST 3.52 1.48
TYPE 4 LOAD 10.5 10.5 7
DIST 1.48 3.52
*****************************
*SYM. 70R Wheeled
LOAD GENERATION 1
TYPE 1 0 0 0 XINC 0.1
**2 lane Class A
***********
LOAD GENERATION 1
TYPE 2 0 0 0 XINC 0.1
TYPE 2 0 0 0 XINC 0.1
**TEST LOAD
LOAD GENERATION 1
TYPE 3 4.835 0 0 XINC 0.1
```

```
TYPE 4 15.165 0 0 XINC 0.1
************
*******
LOAD GENERATION 1
TYPE 3 4.835 0 0 XINC 0.1
TYPE 3 4.835 0 0 XINC 0.1
***********
TYPE 4 15.165 0 0 XINC 0.1
TYPE 4 15.165 0 0 XINC 0.1
**********
PERFORM ANALYSIS PRINT STATICS CHECK
FINISH
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